
Low-dimensional topology of deep neural networks

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Abstract

We study layered models, including feedforward networks and transformers, by limiting each layer to a width of $d = 3$ neurons, i.e., a representation space of $\mathbb{R}^3$. This allows us to examine how a neural network changes low-dimensional topological invariants like links and knots, as well as more sophisticated measures like Milnor’s μ -invariant, through the layers. Note that one may simplify or even trivialize just about any topological structure by simply increasing dimension; for example, any knot is equivalent to an unknot in $\mathbb{R}^4$. By restricting to $\mathbb{R}^3$, we not only isolate the effects of activation and depth from that of width, we work in a space that lends itself to easy visualization. We provide full mathematical proofs and empirical experiments to justify the following insights: When measured by their power to effect topological changes, ResNets are as powerful as transformers; both are strictly more powerful than feedforward neural networks, which are in turn more powerful than invertible models like flow-based models; but using a non-monotone activation would allow the feedforward networks to become as powerful as ResNets and transformers. These results suggest that low-dimensional topology can be an important tool to guide future designs of AI architectures. We then generalize our results from $d = 3$ to arbitrary $d > 3$.

1. Introduction

The last 50 years in topology has arguably been the half-century of *low-dimensional topology*, with many astounding groundbreaking advances in the topology of 3-manifolds (Wall, 1984; Birman, 1991; Lott, 2007) and 4-manifolds (Atiyah, 1987; Milnor, 1987). We will show that the innovative topological invariants developed to study low-

dimensional (i.e., 3- or 4-dimensional) manifolds may be put to good use in the analysis and design of modern AI models.

We begin with a simple example. Consider two interlocked rings in $\mathbb{R}^3$ —the Hopf link. Can a neural network learn to classify points on these rings into their respective classes? This deceptively simple question reveals deep connections between topology and machine learning. Under the manifold hypothesis (Bengio et al., 2013), real-world data concentrates near low-dimensional manifolds embedded in high-dimensional space. When class manifolds are topologically entangled—linked like chain links or knotted like a pretzel—we show that classification *requires* geometrically “untangling” them. Such topological complexity is not merely theoretical: approximately 1–2% of proteins contain genuine knots (Rubach et al., 2024), and the first natural RNA knot was recently discovered (Niemyska et al., 2024). Similarly, we show that learning compact representations of a knotted manifold *requires* “unknotting” it. These are classical problems in low-dimensional topology, and we show they impose fundamental constraints on neural network architectures.

Existing neural network theory largely ignores such topological structure. Universal approximation theorems (Cybenko, 1989; Hornik et al., 1989) guarantee that sufficiently wide networks can approximate any function, but say nothing about *which* architectures succeed on *which* data. Depth-width tradeoffs (Telgarsky, 2016; Eldan & Shamir, 2016) and approximation rates (Barron, 1993; Yarotsky, 2017) characterize function complexity, not data complexity. Topological data analysis (Carlsson, 2009) studies intrinsic data shape but not how embeddings in ambient space constrain learning. Meanwhile, practitioners observe that certain architectural choices—skip connections, non-monotonic activations, attention—consistently outperform alternatives, yet vision-language models still struggle with 3D spatial reasoning (Zha et al., 2025). The reasons are complex, involving training stability, information flow, and incremental layer-wise transformations, and remain incompletely understood; we provide a topological perspective.

When does topological structure create fundamental barriers for neural networks, and which mechanisms overcome them? The Hopf link (Figure 1) provides a canonical ex-

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ample: two circles interlock with linking number $lk = 1$, measuring how many times one curve winds through the other. Linear separability requires $lk = 0$ —if a hyperplane separates two curves, projecting along its normal yields disjoint planar curves with no crossings. We prove that width-3 feedforward networks with ReLU activations *preserve* linking numbers: invertible affine layers preserve lk as homeomorphisms, and monotonic activations preserve lk via straight-line homotopies. Since the Hopf link has $lk = 1 \neq 0$, no such network can achieve linear separability, regardless of depth. Similarly, decision regions cannot be made disjoint and convex, so logit-based classifiers cannot achieve perfect accuracy (Appendix A.2).

How do modern architectures escape? The key is *folding*: coordinate-wise non-monotonic transformations like the absolute value $|x|$. A ResNet block can synthesize $|x| = x + 2 \cdot \text{ReLU}(-x)$ using only ReLU and skip connections. Non-monotonic activations (GELU, Swish) fold directly. Attention mechanisms create input-dependent convex combinations that locally approximate folding. These mechanisms break the homotopy argument underlying our impossibility theorems, enabling topological transformations that monotonic networks cannot perform.

Our contributions. We establish a rigorous framework connecting low-dimensional topology to neural network expressivity and learnability:

1. **Topology–ML correspondences.** Classification of linked data *requires* unlinking, and injective embeddings of knotted data *require* unknotting (Section 3, Appendices A.2, F).
2. **Topological expressivity analysis.** Width- n feedforward networks with monotonic activations cannot separate linked or knotted manifolds in $\mathbb{R}^n$ (Sections 3–5, Appendix C). Skip connections, attention, non-monotonic activations, and width expansion enable folding (Section 6, Appendix I). Table 1 summarizes.
3. **Learnability beyond expressivity.** 100-seed experiments on synthetic Hopf links and CIFAR-10 establish that expressivity gaps manifest as accuracy ceilings, with folding architectures achieving systematic advantages (Section 7).
4. **Linking detection algorithm.** An executable algorithm detects linking in point clouds via spatial graphs, cycle bases, and Gauss integrals; applied to CIFAR-10, linking consistency predicts pairwise classification difficulty (Section 7.6, Appendices K, L.8).
5. **Theoretical applications.** A new proof that universal approximation requires width $\geq n + 1$ for monotonic activations, removing Lipschitz assumptions from prior work (Hanin & Sellke, 2017; Johnson, 2019; Rochau

et al., 2024) (Appendix H).

Table 1 previews our main expressivity results: which architectures can perform which topological transformations under width constraints. The paper develops the theory (Sections 3–6), validates predictions experimentally (Section 7), and demonstrates real-world relevance via CIFAR-10 linking detection (Section 7.6).

Table 1. Topological expressivity of width- n architectures. $\times$ = cannot perform; $\checkmark$ = can perform. **AH**: Ambient homeomorphism (flow-based models, Neural ODEs; Appendix B). **FM**: Feedforward monotonic (ReLU, sigmoid, tanh). **AE**: Autoencoder with width- n bottleneck. **FN**: Feedforward non-monotonic (GELU, Swish). **R**: ResNet. **T**: Transformer.

Topological Transformation	AH	FM	AE	FN	R	T
Deform shapes (no topological change)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Merge connected components	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Fill up holes	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Unlink 2-component link	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
Unlink multi-component link	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
Unknot nontrivial knot	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$

2. Related Work

Topological data analysis extracts intrinsic shape features (holes, connected components) via persistent homology (Carlsson, 2009; Edelsbrunner & Harer, 2010; Cang & Wei, 2017). Naitzat et al. (2020) study how deep networks change Betti numbers of data manifolds; we study how narrow networks *preserve* linking numbers—an extrinsic invariant depending on how manifolds are embedded in ambient space, not just on the manifolds themselves. This extrinsic perspective connects to early intuitions about neural networks “untangling” data (Olah, 2014). Recent work identifies other topological limitations: higher-order message-passing cannot compute homology (Eitan et al., 2025), E(3)-invariant GNNs cannot distinguish enantiomers (Dumitrescu et al., 2025), and topological obstructions constrain generative models (Esmaeili et al., 2023). These concern what networks can *detect*; we study what they can *transform*—a question also relevant to world models that must represent geometric structure (LeCun, 2022). Width bounds for universal approximation (Hanin & Sellke, 2017; Johnson, 2019; Rochau et al., 2024) show width- n networks cannot approximate all functions; our bounds differ: arbitrarily deep width- n networks cannot separate linked data. Modern molecular architectures lift $\mathbb{R}^3$ coordinates to 64–384 dimensional features (Schütt et al., 2017; Satorras et al., 2021; Jumper et al., 2021); our theory offers a topological rationale for this practice.

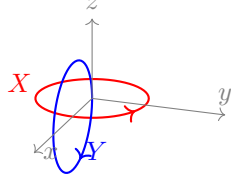


Figure 1. The Hopf link: two circles linked in $\mathbb{R}^3$. See Figure 4 for unlinking via non-monotonic activations.

3. The Hopf Link: An Intuitive Example

Consider two interlinked circles in 3D space forming the Hopf link (for topological background, see Rolfsen, 1976; Hatcher, 2002). Can a width-3 ReLU network classify points on these circles into separate classes? Here a width- n feedforward network is $F = A_L \circ \sigma \circ \dots \circ \sigma \circ A_1$ where each $A_i(x) = W_i x + b_i$ is affine with $W_i \in \mathbb{R}^{n \times n}$ and σ is coordinate-wise (e.g., ReLU). Our results apply to any architecture with intermediate width $\leq n$, including CNNs and bottleneck autoencoders (Appendix A.1).

Example 3.1 (The Hopf link). The Hopf link consists of two interlinked circles in $\mathbb{R}^3$ given parametrically by:

$$X(t) = (\cos(t), \sin(t), 0) \quad (1)$$

$$Y(s) = (1 + \cos(s), 0, \sin(s)) \quad (2)$$

for $t, s \in [0, 2\pi]$. These two circles are topologically linked with $\text{lk}(X, Y) = 1$, meaning they cannot be separated without cutting one of the curves.

This seemingly simple classification task reveals deep topological constraints that narrow neural networks cannot overcome.

3.1. Linking Numbers

The core of our argument is a topological invariant that quantifies how two closed curves are intertwined in 3D space. Complete topological background appears in Appendix C.

Definition 3.2 (Linking number). Let $X, Y \subset \mathbb{R}^3$ be two disjoint, oriented, simple closed curves. The linking number $\text{lk}(X, Y)$ is defined by the Gauss integral:

$$\text{lk}(X, Y) = \frac{1}{4\pi} \oint_X \oint_Y \frac{(x - y) \cdot (dx \times dy)}{|x - y|^3}$$

This measures the degree of the Gauss map $G : X \times Y \rightarrow S^2$ given by $G(x, y) = \frac{x - y}{|x - y|}$. The linking number is always an integer. For the Hopf link, $\text{lk}(X, Y) = \pm 1$.

Equivalent Combinatorial Formula: For any regular projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$:

$$\text{lk}(X, Y) = \frac{1}{2} \sum_{p \in \pi(X) \cap \pi(Y)} \epsilon_p$$

where $\epsilon_p = \pm 1$ is the sign of each crossing.

3.2. Linear Separability Requires Zero Linking Number

Linear separability is the goal of representation learning for classification: the final layer (classifier head) applies a linear transformation to features, so perfect classification requires that class representations become linearly separable. This connection is made precise in Appendix A.2. The task imposes a strict topological constraint:

Lemma 3.3 (Linear separability implies $\text{lk} = 0$). *If two disjoint simple closed curves $X, Y \subset \mathbb{R}^3$ are linearly separable, then $\text{lk}(X, Y) = 0$.*

Proof. Linear separability implies a hyperplane H separates X and Y into disjoint half-spaces. Projecting along the normal direction yields $\pi(X) \cap \pi(Y) = \emptyset$ in the plane. By the combinatorial formula, $\text{lk}(X, Y) = 0$. $\square$

This exposes the conflict: to separate the Hopf link, a network must change the linking number from ± 1 to 0.

3.3. Network Operations and Linking Number Preservation

We analyze each layer type in a width-3 ReLU network. The key technical lemmas are stated here; complete proofs appear in Appendix C.

Definition 3.4 (Homotopy). A homotopy between two continuous maps $f, g : X \rightarrow Z$ is a continuous function $H : X \times [0, 1] \rightarrow Z$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$. Intuitively, H provides a continuous “interpolation” from f to g , deforming $f(X)$ to $g(X)$ within Z . The straight-line homotopy $H_t(x) = (1 - t)f(x) + tg(x)$ is the simplest example when $Z = \mathbb{R}^n$.

Lemma 3.5 (Homotopy invariance). *If X' is homotopic to X via a homotopy avoiding Y , then $\text{lk}(X', Y) = \text{lk}(X, Y)$.*

Proof sketch. The function $t \mapsto \text{lk}(X_t, Y)$ is continuous and integer-valued on $[0, 1]$, hence constant. $\square$

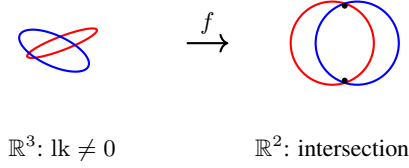
Lemma 3.6 (Rank-deficient intersection). *Let $X, Y \subset \mathbb{R}^3$ be disjoint curves with $\text{lk}(X, Y) \neq 0$. For any rank-deficient affine $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, we have $f(X) \cap f(Y) \neq \emptyset$.*

Proof sketch. Reduce via SVD to projection. If projected curves were disjoint, the combinatorial formula gives $\text{lk} = 0$, contradiction. See Appendix C. $\square$

3.4. Separation Impossibility for the Hopf Link

Theorem 3.7 (Link separation impossibility). *Let $X, Y \subset \mathbb{R}^3$ be disjoint simple closed curves with $\text{lk}(X, Y) \neq 0$. Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be any width-3 feedforward network with affine transformations and ReLU activations. Then*

Figure 2. Rank-deficient transformations force intersection.



$F(X)$ and $F(Y)$ are not linearly separable, and perfect classification is impossible (Appendix A.2).

Proof sketch. Assume for contradiction that linear separability is achieved, implying $\text{lk}(F(X), F(Y)) = 0$ by Lemma 3.3 and disjoint images throughout all layers. We analyze each layer:

Invertible affine layers: Homeomorphisms of $\mathbb{R}^3$ preserve linking numbers (up to sign).

ReLU activations: We apply homotopies *one coordinate at a time*. For coordinate i , define $H_t^{(i)}$ that interpolates $x_i \rightarrow \text{ReLU}(x_i)$ while other coordinates stay fixed. Suppose paths collide: $H_t^{(i)}(x)_i = H_t^{(i)}(y)_i$ for some t . If $x_i \geq 0$, the path is constant at x_i ; if $x_i < 0$, it traverses $[x_i, 0]$. In all cases, collision implies $\text{ReLU}(x_i) = \text{ReLU}(y_i)$. Since $\sigma(X) \cap \sigma(Y) = \emptyset$ by assumption, no collision occurs. Concatenating $H^{(1)}, H^{(2)}, H^{(3)}$ gives $\text{lk}(\sigma(X), \sigma(Y)) = \text{lk}(X, Y)$.

Rank-deficient layers: By Lemma 3.6, these create intersections when $\text{lk} \neq 0$.

Since invertible affines and ReLU preserve lk (up to sign for affines), changing it from non-zero to zero requires rank-deficient layers. But these force intersections. Complete proof in Appendix C. $\square$

This example demonstrates the core mechanism: linking numbers create barriers that narrow monotonic networks cannot overcome, regardless of depth. The next section generalizes to higher-dimensional linked manifolds and arbitrary coordinate-wise monotonic activations.

4. Higher Dimension, More Activations

The Hopf link example demonstrates the core mechanism in a visually intuitive setting. We now generalize in two directions: (1) from curves in $\mathbb{R}^3$ to higher-dimensional linked manifolds, and (2) from ReLU to arbitrary coordinate-wise monotonic activations.

4.1. Higher-Dimensional Linking Numbers

The concept of linking generalizes from curves to higher-dimensional manifolds. For two closed, oriented, disjoint

manifolds M^m and N^n in ambient space $\mathbb{R}^d$ where $d = m + n + 1$, the linking number is well-defined via degree theory.

Definition 4.1 (Higher-dimensional linking number). For two disjoint, closed, oriented manifolds M^m and N^n in $\mathbb{R}^d$ where $d = m + n + 1$, the linking number $\text{lk}(M, N)$ is defined via the Gauss map $G : M \times N \rightarrow S^{d-1}$ given by $G(x, y) = \frac{x-y}{|x-y|}$:

$$\text{lk}(M, N) = \deg(G)$$

where $\deg(G)$ is the topological degree (see Appendix C for the formal definition).

Example 4.2 (Higher Hopf links: $S^n \sqcup S^n$ in $\mathbb{R}^{2n+1}$). Two n -spheres can be linked in $\mathbb{R}^{2n+1}$ with $\text{lk} = \pm 1$. For $n = 1$, this recovers the classical Hopf link ($S^1 \sqcup S^1 \subset \mathbb{R}^3$). For $n = 2$, we obtain linked 2-spheres $S^2 \sqcup S^2 \subset \mathbb{R}^5$, which we use in experiments (Section 7.5). The construction embeds each sphere into complementary coordinate subspaces via stereographic projection; see Appendix J.5 for explicit parametrizations.

The key lemmas from Section 3 extend to higher dimensions:

Lemma 4.3 (Higher-dimensional homotopy invariance). Let $M^m, N^n \subset \mathbb{R}^d$ (with $d = m + n + 1$) be disjoint manifolds. If M' is homotopic to M via a homotopy avoiding N , then $\text{lk}(M', N) = \text{lk}(M, N)$.

Lemma 4.4 (Higher-dimensional linear separability). If M^m and N^n in $\mathbb{R}^d$ (with $d = m + n + 1$) are linearly separable, then $\text{lk}(M, N) = 0$.

Theorem 4.5 (Higher-dimensional separation impossibility). Let M^m and N^n be disjoint, closed, oriented manifolds in $\mathbb{R}^d$ (with $d = m + n + 1$) such that $\text{lk}(M, N) \neq 0$. Then no width- d feedforward network with affine transformations and coordinate-wise monotonic activations can linearly separate M and N , and perfect classification is impossible.

Proof sketch. The argument follows Theorem 3.7: invertible affines preserve lk up to sign as homeomorphisms; monotonic activations preserve lk via the same straight-line homotopy argument (which works in any dimension); rank-deficient transformations force intersections when $\text{lk} \neq 0$. See Appendix C. $\square$

4.2. General Monotonic Activations

The Hopf link analysis used ReLU, but the impossibility extends to any *coordinate-wise monotonic* activation $\sigma(x) = (\sigma_1(x_1), \dots, \sigma_n(x_n))$ where each σ_i is continuous and monotonic (e.g., sigmoid, tanh, Leaky ReLU, ELU). This shows the obstruction is fundamental to monotonicity, not an artifact of ReLU's piecewise-linear form.

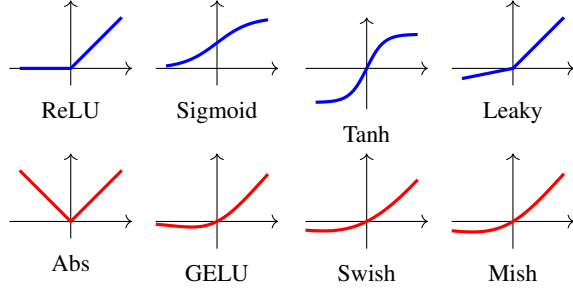


Figure 3. Monotonic (top) vs non-monotonic (bottom) activations.

The key to generalizing beyond ReLU is the concept of *link homotopy*: a simultaneous continuous deformation of multiple components that keeps them disjoint throughout. Unlike single-component homotopy (where one set moves while others stay fixed), link homotopy moves all components together via the same formula $H_t(x) = (1-t)x + t\sigma(x)$. This simultaneity is essential—if only one component moves, it can collide with stationary components even for nondecreasing activations.

Lemma 4.6 (Monotonic activation preservation). *Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be coordinate-wise monotonic with m nonincreasing coordinates. If $X, Y \subset \mathbb{R}^n$ are disjoint compact manifolds such that $\sigma(X)$ and $\sigma(Y)$ remain disjoint, then:*

$$\text{lk}(\sigma(X), \sigma(Y)) = (-1)^m \text{lk}(X, Y)$$

Proof sketch. The straight-line homotopy $H_t(x) = (1-t)x + t\sigma(x)$ applied simultaneously to both X and Y defines a link homotopy when σ is nondecreasing: if $x_i < y_i$, then monotonicity ensures $H_t(x)_i < H_t(y)_i$ for all t , so the moving components never collide. For nonincreasing coordinates, we decompose $\sigma = R \circ \sigma'$ where R is a coordinate reflection and σ' is nondecreasing; reflections contribute the sign factor $(-1)^m$. See Appendix D for the complete proof. $\square$

Theorem 4.7 (General impossibility theorem). *Let $X, Y \subset \mathbb{R}^n$ be disjoint compact manifolds with $\text{lk}(X, Y) \neq 0$ (assuming appropriate dimensionality conditions). Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be any width- n feedforward network with affine transformations and coordinate-wise monotonic activations. Then $F(X)$ and $F(Y)$ are not linearly separable, and perfect classification is impossible.*

Proof sketch. By Lemma 4.6, monotonic activations preserve lk up to sign via link homotopy. The proof structure of Theorem 3.7 carries through: invertible affines preserve lk up to sign, monotonic activations preserve lk up to sign, and since $\text{lk} \neq 0$ implies $\pm \text{lk} \neq 0$, rank-deficient layers are required to achieve $\text{lk} = 0$. But rank-deficient layers force intersections. $\square$

Remark 4.8 (Smooth vs piecewise-linear). The result holds for smooth activations (sigmoid, tanh) as well as piecewise-linear (ReLU). The crucial property is monotonicity, not differentiability. This shows the topological barrier is intrinsic to the coordinate-wise monotonic structure.

5. Extensions

Beyond two-component linking, topological obstructions arise from multi-component entanglement and single-object complexity. For *multi-component links*, the Borromean rings demonstrate that three curves can be collectively inseparable despite pairwise linking numbers all being zero; Milnor’s μ -invariant (Milnor, 1954) detects this higher-order entanglement and is preserved by monotonic activations. For *single objects*, knot type (e.g., trefoil vs unknot) is preserved by width-3 networks with monotonic activations whenever the network remains injective. Both results generalize to higher dimensions; full statements and proofs appear in Appendices E and F.

6. Breaking Topological Constraints: Architectural Mechanisms

The impossibility results in Sections 3–5 demonstrate fundamental limitations of narrow networks with coordinate-wise monotonic activations. However, these constraints can be overcome through several architectural approaches while preserving narrow width. We explore three complementary mechanisms.

6.1. Non-Monotonic Activation Functions

The first approach breaks the monotonicity constraint underlying our impossibility theorems. While ReLU and other monotonic activations preserve topological rigidity, non-monotonic functions create geometric flexibility needed to overcome linking and knotting obstructions.

Important non-monotonic activations (Figure 3, bottom) include GELU (Hendrycks & Gimpel, 2016) ($x \cdot \Phi(x)$, standard in Transformers), Swish/SiLU (Ramachandran et al., 2017; Elfving et al., 2018) ($x \cdot \sigma(x)$), and Mish (Misra, 2020) ($x \cdot \tanh(\text{softplus}(x))$). The impossibility results critically depend on Lemma 3.5: monotonic activations preserve linking numbers via straight-line homotopies that avoid intersections. This fails for non-monotonic functions.

Example 6.1 (Non-monotonic unlinking). Coordinate-wise $|x|$ can unlink the Hopf link and unknot the trefoil through “folding” operations that collapse signed coordinates into the positive octant (Appendix I). By translating data appropriately, each folding layer affects only one coordinate at a time: $(x_1, x_2, x_3) \mapsto (|x_1|, x_2, x_3) \mapsto (|x_1|, |x_2|, x_3) \mapsto \dots$ Figure 4 illustrates.

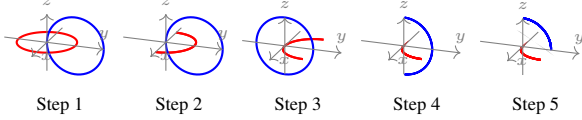


Figure 4. Hopf link unlinking via $|x|$ activations (full resolution: Figure 13; mathematical details: Appendix I.2).

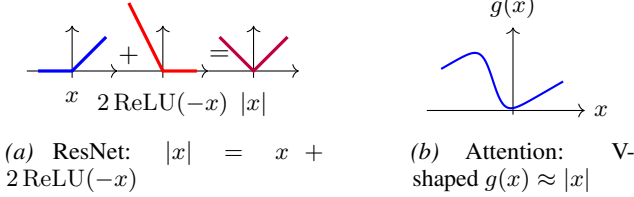


Figure 5. Mechanisms for expressing $|x|$ with monotonic primitives.

This extends to any activation with local extrema. For compact data and an activation σ with a local maximum or minimum (e.g., GELU near $x \approx -0.5$, Swish near $x \approx -1.3$), affine layers can scale and translate the data into the non-monotonic region, enabling the same topological transformations.

6.2. Skip Connections and Residual Networks

Residual networks (He et al., 2016) maintain monotonic activations but break the purely feedforward constraint through skip connections. The key insight is that ResNet can express non-monotonic functions:

Theorem 6.2 (ResNet topological expressivity). *Width- n ResNet architectures using only ReLU activations can perform the same topological transformations as networks with non-monotonic activations, including unlinking and unknotting.*

Proof sketch. ResNet blocks $\mathcal{F}(x) = x + \mathcal{G}(x)$ can express absolute value via the identity:

$$|x| = x + 2 \text{ReLU}(-x) \quad (3)$$

Setting $\mathcal{G}(x) = 2 \text{ReLU}(-x)$ yields $\mathcal{F}(x) = |x|$. This enables the same coordinate-wise folding operations as non-monotonic activations. Complete construction in Appendix I. $\square$

Remark 6.3 (ResNet vs Neural ODEs). While ResNets can be viewed as discretizations of Neural ODEs (Chen et al., 2018), the discrete formulation is strictly more expressive for topological transformations. Neural ODEs generate continuous flows (diffeomorphisms) that must preserve topological invariants. Discrete ResNet blocks can implement “jumps” that break these invariants—the $|x| = x + 2 \text{ReLU}(-x)$ identity relies crucially on discrete, non-infinitesimal residuals.

6.3. Attention Mechanisms

The Transformer architecture (Vaswani et al., 2017) introduces self-attention mechanisms enabling global information routing. We study the topological expressivity of attention by considering *pure transformers* (Appendix A.1).

Theorem 6.4 (Transformer topological expressivity). *Width- n pure transformers can express coordinate-wise non-monotonic functions resembling absolute value, thereby performing the same topological transformations as non-monotonic activations.*

Proof sketch. We use two-token sequences (x, x) as a surrogate for autoregressive generation: given input x , the Transformer first generates a copy token to form (x, x) , then processes this sequence. With distinct positional encodings, attention weights computed via softmax create non-monotonic behavior—the resulting function $g(x)$ exhibits a V-shaped profile locally approximating $|x|$ (Figure 5b). Since each coordinate can be processed independently, Transformers inherit topological capabilities from non-monotonic constructions. Details in Appendix I. $\square$

6.4. Width Threshold and Design Implications

Width $\geq n + 1$ eliminates all constraints. Our impossibility results are tight: width- $(n + 1)$ networks with any monotonic activation can achieve universal approximation on compact subsets of $\mathbb{R}^n$ (Hanin & Sellke, 2017). This enables mapping any link configuration to an unlinked one. The topological obstructions are purely a consequence of the dimension constraint $\text{width} \leq n$. See Appendix G for the complete construction.

Design guidance. For topologically complex data: (1) expand MLP width slightly beyond input dimension in early layers before narrowing; (2) prefer non-monotonic activations (GELU, SiLU, Mish) in bottleneck layers; (3) use skip connections; or (4) leverage attention mechanisms.

7. Experiments

We validate our theoretical results with experiments on the Hopf link classification task. These experiments verify both the impossibility results for monotonic activations and the escape mechanisms described in the previous section.

7.1. Experimental Setup

Thickened Hopf link. We construct volumetric data by sampling points from two thickened interlinked tori. Each curve is thickened by adding small perturbations normal to the curve surface. Dataset: 6000 points (3000 per class), split 80/20 train/validation.

Table 2. Mean test accuracy (%) on Hopf link classification across depths, ReLU vs GELU.

Depth	3	5	8	12	16	20
ReLU	87.0	77.8	82.1	86.2	88.0	87.5
GELU	91.2	95.1	97.5	98.4	98.7	98.9

Table 3. Mean test accuracy (%) on Hopf link classification across depths, plain ReLU vs ResNet.

Depth	3	4	5	6	7	8
Plain ReLU	83.1	90.3	50.7*	50.7*	50.7*	91.0
ResNet	92.8	98.2	100.0	99.3	99.7	99.0

*Training collapsed for some seeds.

Training protocol. Width-3 networks throughout all experiments. Adam optimizer with learning rate 10^{-3} , up to 800 epochs with early stopping (patience 100–200). Binary cross-entropy loss.

7.2. ReLU vs GELU on Hopf Link

We compare monotonic (ReLU) versus non-monotonic (GELU) activations across 12 depths (3–20 layers) with 30 random seeds each (720 total experiments).

Table 2 confirms the theory: ReLU plateaus at 77–88% regardless of depth (Theorem 3.7), while GELU breaks the barrier, reaching $\sim 99\%$. Detailed observations appear in Appendix J.4.

7.3. ResNet vs Plain ReLU

We compare plain ReLU feedforward networks against ReLU ResNets, both at width 3.

ResNet achieves **100% accuracy at depth 5**, demonstrating that skip connections provide a mechanism to overcome topological barriers even with monotonic activations, as predicted by Theorem 6.2.

7.4. Mechanistic Interpretability: How ResNet Unlinks S^0 - S^1 in $\mathbb{R}^2$

The disk-annulus separation task is the $n = 2$ case of our higher-dimensional linking framework (Theorem 4.5): the disk center (S^0) links with the annulus boundary (S^1) with linking number ± 1 . By our theory, width-2 monotonic feedforward networks cannot separate them, but width-2 ResNet can.

Figure 6 shows the layer-by-layer transformation. ResNet implements “folding” operations via $|x| = x + 2 \text{ReLU}(-x)$ (Equation 3), enabling non-monotonic transformations that separate the topologically linked components.

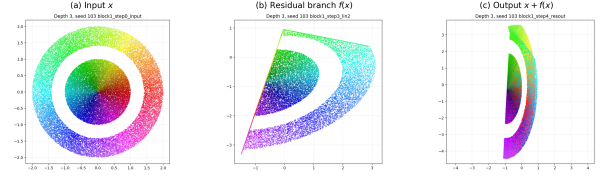


Figure 6. ResNet skip connection implementing $|x| = x + 2 \text{ReLU}(-x)$ on linked disk-annulus (S^0 - S^1 link). (a) Input x . (b) Residual branch $f(x) \approx 2 \text{ReLU}(-x)$. (c) Output $x + f(x)$: folding separates components. Explanation in Appendix J.10.

Table 4. Best test accuracy (%) across 5 seeds for total linking number k (i.e., k disjoint copies of linked $S^2 \sqcup S^2$) in $\mathbb{R}^5$. Width-5, depth-5 networks.

Model	$k=1$	$k=2$	$k=5$	$k=10$	$k=20$	$k=50$
ReLU	99.0	96.3	90.6	88.8	90.4	90.3
ReLU+Skip	100	99.7	98.7	91.6	98.8	98.0
GELU	99.6	99.8	93.3	91.2	92.8	91.3
Swish	99.4	94.4	98.0	88.6	91.3	99.1
Gap	+1.0	+3.4	+8.1	+2.8	+8.3	+7.7

7.5. Higher-Dimensional Linking: $S^n \sqcup S^n$ in $\mathbb{R}^{2n+1}$

We extend experiments to higher dimensions using the standard linked spheres construction (Appendix J.5): two n -spheres embedded in $\mathbb{R}^{2n+1}$ with linking number ± 1 . For $n = 2$, this gives $S^2 \sqcup S^2 \subset \mathbb{R}^5$ classified by width-5 networks. We quantify topological difficulty via linking number: a single link has only one local entanglement region, so networks can achieve high accuracy by being wrong only in that small patch; placing k disjoint copies forces the network to handle k independent entanglement regions (Table 4).

At $k = 1$, all architectures achieve $> 97\%$ accuracy, but as k increases, plain ReLU drops to $\sim 90\%$ while ReLU+skip maintains $\sim 98\%$ —an 8% gap confirming topological barriers persist in higher dimensions. The effect at $k = 1$ is weaker than for $n = 1$ (Table 2) due to the linking region’s smaller volume fraction in higher dimensions; extended 100-seed experiments across 17 model variants (22,100 experiments total; Appendix J.7) confirm these patterns.

7.6. Linking in Real Data: CIFAR-10

We investigate whether topological linking exists in real image data and affects classification. Using the linking detection algorithm (Appendix K), we analyze CIFAR-10 class manifolds projected to 3D via PCA.

Link detection. With $20\times$ data augmentation (1.05M samples) and k -NN graph construction ($k = 15$, mutual edges), we detect linked cycles between bird and deer classes at $\varepsilon = 0.034$ (0.22% of the 3D bounding box diago-

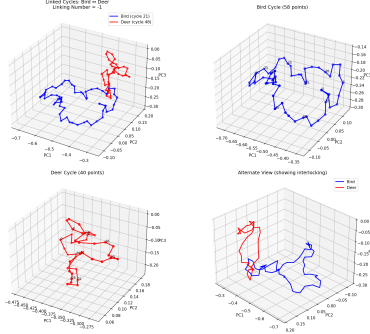


Figure 7. Linked cycles in CIFAR-10: bird (blue, 58 pts) and deer (red, 40 pts) with $lk = -1$ at $\varepsilon = 0.034$. Full resolution in Appendix L.

Table 5. Binary classification accuracy at L8 no-skip: linked pair (deer-dog) vs unlinked control (frog-ship). All 3 non-monotonic activations outperform all 4 monotonic activations on the linked pair; no such pattern on the unlinked pair.

Activation	Linked (deer-dog)		Unlinked (frog-ship)	
	Type	Acc (%)	Type	Acc (%)
GELU	NONM	90.8	NONM	97.9
Mish	NONM	90.8	NONM	97.7
Swish	NONM	90.1	NONM	97.8
ReLU	MONO	89.6	MONO	97.8
LeakyReLU	MONO	89.6	MONO	97.7
ELU	MONO	89.1	MONO	97.2
SELU	MONO	87.4	MONO	97.6
Gap		+1.2%		+0.1%

nal). The witness cycles have linking number $lk = -1$ with Gauss integral -1.004 (Figure 7). This demonstrates that topological linking—previously studied only in synthetic data—occurs naturally in real image manifolds.

10-class linking consistency. Running detection 11 times across all 45 class pairs yields a linking consistency matrix (Appendix L). Of 45 pairs: 27 are strongly linked ($>70\%$ consistency), 8 are unlinked ($<30\%$), and 10 are ambiguous. Semantically similar classes (deer-dog 91%, bird-cat 91%, cat-dog 82%) show high linking, while dissimilar pairs (frog-ship 18%, airplane-horse 18%) are unlinked.

Classification experiments. We train width-bounded CNNs (all intermediate layers $\leq 3072D$) on both binary and 10-class classification tasks, comparing monotonic (ReLU, ELU, SELU, LeakyReLU) versus non-monotonic (GELU, Swish, Mish) activations (Appendix L).

At L8 no-skip (Table 5), all non-monotonic activations outperform all monotonic activations on the linked deer-dog pair (+1.2% gap), while no such pattern exists on the unlinked frog-ship pair (control). With skip connec-

tions, the effect disappears—monotonic networks match non-monotonic performance, as predicted by Theorem 6.2. See Appendix L for full details; Appendix L.8 shows that linking consistency can be used to analyze dataset difficulty, reliably predicting class-pair confusion within CIFAR-10 ($r \approx 0.48$, $p < 0.001$) and outperforming standard distance metrics.

Expressivity vs. optimization. Our theoretical results concern *expressivity*—what functions network architectures can represent—not *optimization*—what functions gradient descent actually finds. Even when two architectures have identical expressivity ceilings, they may exhibit different training dynamics: convergence speed, sensitivity to learning rate, propensity to find particular local minima, etc. The observed CIFAR-10 gaps thus reflect a combination of (i) the topological expressivity barrier predicted by our theory, and (ii) optimization dynamics that our theory does not address.

8. Conclusion

This work introduces a paradigm shift in neural network expressivity analysis: from approximation rates to geometric transformation requirements. Rather than asking whether a network can approximate a target function, we ask what geometric operations it can perform on embedded data. This perspective goes beyond expressivity and extends to mechanistic understanding: we identify *why* modern architectures succeed—ResNet skip connections, attention mechanisms, and non-monotonic activations share a unifying principle of coordinate-wise non-monotonicity (“folding”).

Our framework focuses on *ambient-sensitive* (extrinsic) topological invariants, fundamentally distinguishing it from traditional topological data analysis which studies intrinsic properties. The linking number of two manifolds depends on *how* they are embedded in $\mathbb{R}^n$, not merely on the manifolds themselves. This extrinsic perspective captures the algorithmic reality: neural networks must transform data within the ambient space, making embedding geometry as important as intrinsic data structure.

We expect this work to open broader applications of low-dimensional topology and geometric topology to machine learning. The invariants we study—linking numbers, Milnor’s μ -invariants, knot types—represent initial examples; richer invariants from knot theory and link theory likely yield further correspondences. Similarly, the task correspondences we establish (unlinking $\leftrightarrow$ classification, unknotting $\leftrightarrow$ injective embedding) likely represent initial examples of a broader pattern yet to be fully explored. Understanding this interplay between data geometry and algorithmic capability offers new directions for both theoretical foundations and principled architecture design.

Impact Statement

This paper presents theoretical and empirical work on neural network expressivity. We identify fundamental limitations of certain architectures, which may inform more principled architecture design. We also propose a link detection algorithm that characterizes topological complexity in datasets, which may inform dataset difficulty analysis.

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A. Network Architecture

This appendix provides precise mathematical definitions for neural network architectures (Section A.1) and establishes the connection between classifier heads and linear separability (Section A.2).

A.1. Network Architectures

Definition A.1 (Width- n Feedforward Network). A width- n feedforward network with activation σ is a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of the form:

$$F = A_L \circ \sigma \circ A_{L-1} \circ \sigma \circ \cdots \circ \sigma \circ A_1$$

where:

1. Each A_i is an affine transformation: $A_i(x) = W_i x + b_i$ where $W_i \in \mathbb{R}^{n \times n}$ and $b_i \in \mathbb{R}^n$
2. σ is an activation function: $\sigma(x) = (\sigma(x_1), \sigma(x_2), \dots, \sigma(x_n))$ applied coordinate-wise

The network has uniform width n , meaning all intermediate representations remain in $\mathbb{R}^n$.

Definition A.2 (ResNet architecture). A width- n ResNet with L blocks is a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of the form $F = B_L \circ B_{L-1} \circ \cdots \circ B_1$ where each residual block $B_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as:

$$B_i(x) = x + R_i(x)$$

and each residual mapping $R_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ has the form $R_i = A_{i,2} \circ \sigma \circ A_{i,1}$ where each $A_{i,j}$ is affine with $W_{i,j} \in \mathbb{R}^{n \times n}$ and σ is coordinate-wise monotonic (typically ReLU).

Definition A.3 (Pure transformer architecture). A *pure transformer* is a transformer network with both residual connections and layer normalization removed. Each block consists of: (1) self-attention $\text{Attention}(X) = \text{softmax}\left(\frac{XW_Q(XW_K)^T}{\sqrt{d_k}}\right)XW_V$, (2) an affine layer, (3) a feedforward sublayer with coordinate-wise activation. The width constraint requires all intermediate dimensions $\leq n$.

Definition A.4 (Autoencoder architecture). An autoencoder with bottleneck dimension d is a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of the form $F = D \circ E$ where:

- The encoder $E : \mathbb{R}^n \rightarrow \mathbb{R}^d$ compresses via $E = B_{L_E} \circ \sigma \circ \cdots \circ \sigma \circ A_1$ where $A_1 : \mathbb{R}^n \rightarrow \mathbb{R}^d$ is the initial projection and subsequent layers maintain width d
- The decoder $D : \mathbb{R}^d \rightarrow \mathbb{R}^n$ reconstructs via $D = A_L \circ \sigma \circ \cdots \circ \sigma \circ B_1$ where $A_L : \mathbb{R}^d \rightarrow \mathbb{R}^n$ is the final expansion
- All intermediate layers maintain width d with coordinate-wise monotonic activations

Corollary A.5 (Autoencoder topological impossibility). *Let $X, Y \subset \mathbb{R}^n$ be disjoint compact manifolds contained in a d -dimensional affine subspace $H \subset \mathbb{R}^n$ with $\text{lk}(X, Y) \neq 0$ (for appropriate dimension). Then no autoencoder with bottleneck dimension d and coordinate-wise monotonic activations can transform X and Y into linearly separable images while preserving disjointness.*

Proof. Consider an autoencoder with input/output dimension n and bottleneck width d . Without loss of generality, assume $H = \{(x_1, \dots, x_d, 0, \dots, 0) : x_i \in \mathbb{R}\}$ (by applying an appropriate rotation).

The first affine transformation $A_1 : \mathbb{R}^n \rightarrow \mathbb{R}^d$ restricted to H can be written as $A_1(x_1, \dots, x_d, 0, \dots, 0) = A'_1(x_1, \dots, x_d) + c'_1$ where $A'_1 : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is affine. Similarly, the final layer $A_L : \mathbb{R}^d \rightarrow \mathbb{R}^n$ can be decomposed as $A_L = (A'_L \oplus 0_{n-d}) + c_L$ where $A'_L : \mathbb{R}^d \rightarrow \mathbb{R}^d$.

All intermediate layers operate in $\mathbb{R}^d$ with width d . Therefore, the restriction of the autoencoder to H is equivalent to a width- d feedforward network $F' = A'_L \circ \sigma \circ \cdots \circ \sigma \circ A'_1 : \mathbb{R}^d \rightarrow \mathbb{R}^d$. By Theorem 4.7, no such width- d network can transform $X, Y \subset \mathbb{R}^d$ with $\text{lk}(X, Y) \neq 0$ into linearly separable images. $\square$

Remark A.6 (Extension to other link homotopy invariants). The same proof technique applies to any link homotopy invariant preserved by narrow feedforward networks, including Milnor's $\bar{\mu}$ -invariants for multi-component links (Appendix E). The key insight is that the autoencoder's bottleneck reduces the problem to a width- d feedforward network, inheriting all topological obstructions of that width class.

Remark A.7 (Implications for dimensionality reduction). This reveals a fundamental limitation: when compressing from $\mathbb{R}^n$ to a d -dimensional bottleneck, the bottleneck dimension d determines topological constraints, not the input dimension n . If linked structures exist in any d -dimensional subspace of the input, compression to that dimension forces topological obstructions. A potential workaround is to employ several full-width (n -dimensional) layers before compression, allowing non-monotonic activations or skip connections to resolve topological complexity before dimensional reduction.

A.2. Classifier Heads and Linear Separability

Our theoretical framework analyzes when network outputs become *linearly separable*. We now establish why this criterion directly captures practical classification performance.

Definition A.8 (Logit-based classification). A standard classifier head maps feature representations $z \in \mathbb{R}^d$ to class logits: for K classes, the network outputs $\ell = Wz + b \in \mathbb{R}^K$ where $W \in \mathbb{R}^{K \times d}$ and $b \in \mathbb{R}^K$, and the predicted class is $\hat{y} = \arg \max_i \ell_i$.

Definition A.9 (Decision regions). For K -class logit-based classification, the **decision region** for class i is:

$$D_i = \{z \in \mathbb{R}^K : z_i > z_j \text{ for all } j \neq i\}$$

Each D_i is the intersection of $K - 1$ open half-spaces $\{z_i > z_j\}$, hence *convex*. The decision regions $\{D_1, \dots, D_K\}$ are pairwise disjoint.

Proposition A.10 (Binary classification equivalence). *For binary classification ($K = 2$), the decision rule $\hat{y} = \arg \max\{\ell_0, \ell_1\}$ is equivalent to thresholding a linear function: $\hat{y} = \mathbf{1}[\ell_0 - \ell_1 > 0] = \mathbf{1}[(w_0 - w_1)^\top z + (b_0 - b_1) > 0]$. Thus, two classes are perfectly classified if and only if their feature representations are linearly separable by a hyperplane.*

Theorem A.11 (Multiclass classification and link homotopy). *Let $X_1, \dots, X_K \subset \mathbb{R}^n$ be K disjoint compact subsets with a non-trivial multi-component link homotopy invariant (e.g., pairwise linking number $\text{lk}(X_i, X_j) \neq 0$ for some pair, or Milnor's $\bar{\mu}(X_1, X_2, X_3) \neq 0$). If $n \geq K$, then no width- n feedforward network with coordinate-wise monotonic activations can achieve perfect K -class classification on $(X_1, \dots, X_K)$.*

Proof. Suppose for contradiction that a width- n network $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ achieves perfect K -class classification. This means the final K coordinates of $F(x)$ serve as logits, with $F(X_i) \subset \tilde{D}_i$ for each i , where

$$\tilde{D}_i = \{z \in \mathbb{R}^n : z_i > z_j \text{ for all } j \in \{1, \dots, K\}, j \neq i\}$$

is the decision region for class i extended to $\mathbb{R}^n$ (the remaining $n - K$ coordinates are unconstrained).

Each $\tilde{D}_i$ is convex (intersection of half-spaces) and the $\tilde{D}_i$ are pairwise disjoint. Since convex sets are contractible, each $F(X_i) \subset \tilde{D}_i$ can be continuously contracted to a point $p_i \in \tilde{D}_i$ while remaining within $\tilde{D}_i$. Because the $\tilde{D}_i$ are disjoint, these contractions can be performed simultaneously without any two components intersecting.

This means the images $(F(X_1), \dots, F(X_K))$ are *link homotopically trivial*: they can be simultaneously shrunk to K distinct points while maintaining pairwise disjointness. Therefore, all multi-component link homotopy invariants vanish: $\text{lk}(F(X_i), F(X_j)) = 0$ for all pairs, and $\bar{\mu}(F(X_{i_1}), \dots, F(X_{i_m})) = 0$ for all higher-order Milnor invariants.

However, by our main preservation theorems, width- n networks with monotonic activations preserve link homotopy invariants (up to sign). If $(X_1, \dots, X_K)$ has a non-trivial invariant, so does $(F(X_1), \dots, F(X_K))$. Contradiction. $\square$

Remark A.12 (Width requirement). The condition $n \geq K$ ensures the network can represent K logits. When $n < K$, the network cannot directly output K class logits, and a separate projection layer would be needed, which is outside our width- n constraint. In typical applications where input dimension n far exceeds the number of classes K , this condition is easily satisfied.

This correspondence justifies our theoretical focus on linear separability and linking: when we prove that a network cannot separate linked components, we are equivalently proving that no classifier head can achieve perfect accuracy on those components.

B. Invertible Architectures and Ambient Homeomorphisms

This section establishes why ambient homeomorphisms—including normalizing flows, Neural ODEs, and other invertible architectures—face the strongest topological constraints.

Definition B.1 (Ambient homeomorphism). An *ambient homeomorphism* of $\mathbb{R}^n$ is a continuous bijection $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ whose inverse h^{-1} is also continuous. Equivalently, h is a homeomorphism from $\mathbb{R}^n$ to itself. An ambient homeomorphism acts on subsets $X \subset \mathbb{R}^n$ by $X \mapsto h(X)$, preserving all topological properties determined by the embedding of X in the ambient space $\mathbb{R}^n$.

B.1. Fundamental Properties

Two classical results establish that continuous injective neural networks act as homeomorphisms on their domains:

Lemma B.2 (Compact case). *A continuous injective map from a compact space to a Hausdorff space is a homeomorphism onto its image.*

Theorem B.3 (Invariance of Domain, Brouwer 1912 (Munkres, 2000, Theorem 36.5)). *Let $U \subseteq \mathbb{R}^n$ be open. A continuous injective map $f : U \rightarrow \mathbb{R}^n$ is an open map (maps open sets to open sets), hence a homeomorphism onto its image $f(U)$.*

Together, these results establish that for width- n neural networks $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that are continuous and injective, the restriction to any open or compact domain is a homeomorphism onto its image. Thus, topological constraints apply broadly across different types of embedded datasets.

B.2. Topological Preservation by Ambient Homeomorphisms

We now establish why ambient homeomorphisms receive all failure marks in Table 1.

Lemma B.4 (Ambient homeomorphisms preserve component count). *Let $X_1, \dots, X_k \subset \mathbb{R}^n$ be disjoint compact connected sets. If $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an ambient homeomorphism, then $h(X_1), \dots, h(X_k)$ are also k disjoint compact connected sets.*

Proof. Continuity preserves connectedness: each $h(X_i)$ is connected. Bijectivity and disjointness of X_i ensure $h(X_i) \cap h(X_j) = \emptyset$ for $i \neq j$. Thus component count is preserved. $\square$

Lemma B.5 (Ambient homeomorphisms preserve linking numbers). *For disjoint simple closed curves $X, Y \subset \mathbb{R}^3$ and any ambient homeomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$:*

$$\text{lk}(h(X), h(Y)) = \pm \text{lk}(X, Y)$$

where the sign is $+1$ for orientation-preserving homeomorphisms and -1 for orientation-reversing ones.

Proof. The linking number is defined via the degree of the Gauss map $\Gamma : X \times Y \rightarrow S^2$ (Definition C.1). Ambient homeomorphisms induce maps on the Gauss map that preserve degree up to sign, depending on orientation. $\square$

Lemma B.6 (Ambient homeomorphisms preserve knot type). *If $K \subset \mathbb{R}^3$ is a knot and $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an ambient homeomorphism, then $h(K)$ has the same knot type as K (or its mirror image if h is orientation-reversing).*

Proof. Ambient homeomorphisms induce ambient isotopies, and isotopy invariants (knot group, Alexander polynomial, etc.) are preserved. For orientation-preserving h , this follows from the standard fact that ambient homeomorphisms are isotopic to the identity on any compact subset. $\square$

These lemmas generalize to all higher invariants: Milnor μ -invariants, Alexander polynomials, ambient cobordism classes, and any other ambient-sensitive topological invariant are preserved by ambient homeomorphisms.

B.3. Invertible Architectures in Machine Learning

Several important classes of neural network architectures implement or approximate ambient homeomorphisms, and therefore face the topological constraints we have established.

Reversible residual networks. Reversible architectures such as RevNet (Gomez et al., 2017) and i-RevNet (Jacobsen et al., 2018) maintain invertibility via explicit construction for memory-efficient training. As homeomorphisms, they cannot change topological invariants.

Normalizing flows. Normalizing flows (Dinh et al., 2017; Kingma & Dhariwal, 2018) learn invertible transformations $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ for density estimation. These architectures use coupling layers, affine transformations, and invertible 1×1 convolutions to maintain bijectivity by construction. As ambient homeomorphisms, they face all topological constraints in Table 1. They cannot unlink curves, unknot knots, or change any ambient-sensitive invariant.

ODE-based flows. Neural ODEs (Chen et al., 2018) and continuous normalizing flows (Grathwohl et al., 2019) learn vector fields $v_\theta(z, t) : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}^n$ that generate flows via ODE integration: $\frac{dz}{dt} = v_\theta(z, t)$, $z(0) = z_0$. A fundamental theorem of ODEs establishes that when v_θ is Lipschitz continuous, the flow map $\phi_T : z_0 \mapsto z(T)$ is a *diffeomorphism*—a smooth ambient homeomorphism of $\mathbb{R}^n$. This implies these models face all topological constraints, including inability to change linking numbers or unknot knots.

This theoretical prediction is validated empirically by Dupont et al. (2019), who demonstrate that Neural ODEs fail on classification tasks involving concentric circles: since ODE trajectories cannot cross, points in an inner circle cannot be mapped through an outer annulus. Their proposed solution—*augmented* Neural ODEs that embed data in higher-dimensional space—is precisely the width-expansion mechanism, confirming that topological constraints are the fundamental barrier.

Flow matching. Recent developments including Flow Matching (Lipman et al., 2023) similarly learn ODE-based flows, and thus produce ambient homeomorphisms subject to the same topological constraints.

One-step direct evaluation: MeanFlow. A fundamentally different approach is exemplified by MeanFlow (Geng et al., 2025), which performs *one-step direct evaluation* rather than ODE integration: $z_0 = z_1 - u_\theta(z_1, 0, 1)$ where u_θ is a neural network (typically a Transformer). Critically, this direct evaluation is *not* constructed via ODE integration, and thus need not be a homeomorphism. MeanFlow escapes topological constraints via two mechanisms: (1) architectural features that break coordinate-wise monotonicity (attention, non-monotonic activations, residual structure), and (2) an algebraic structure that avoids homeomorphism requirements by allowing many-to-one mappings. This contrasts with Flow Matching models that use Transformers but still perform ODE integration—such models remain homeomorphic regardless of architectural expressivity.

Remark B.7 (ResNets vs Neural ODEs). Neural ODEs are sometimes interpreted as continuous-time limits of ResNets, but this analogy is fundamentally limited. ResNets implement *discrete* layer-by-layer transformations $x_{i+1} = x_i + f_i(x_i)$ whose composition need not be a homeomorphism. In contrast, Neural ODEs produce *continuous* flows that are necessarily homeomorphic. The discrete structure of ResNets enables topological expressivity precisely because it avoids the homeomorphism constraint inherent to continuous flows.

Remark B.8 (Practical success despite constraints). Two factors explain the practical success of ODE-based generative models despite these theoretical constraints: (1) *Numerical integration:* In practice, ODEs are solved via numerical methods (Euler, Runge-Kutta), not analytically. Discretization errors break the exact uniqueness of continuous flows, creating approximate homeomorphisms where trajectories may effectively “cross.” (2) *Task structure:* Generative modeling typically maps from simple connected distributions (Gaussian noise) to complex but still connected data manifolds—neither requiring component separation nor topological unlinking.

C. Proofs for Section 3: ReLU Networks in $\mathbb{R}^3$

This appendix provides complete proofs for the results in Section 3, which concern width-3 ReLU networks acting on linked curves in $\mathbb{R}^3$. The proofs use elementary geometric arguments specific to this setting.

C.1. Linking Number in $\mathbb{R}^3$

Definition C.1 (Link and linking number in $\mathbb{R}^3$). A *link* is a finite collection of disjoint simple closed curves in $\mathbb{R}^3$. For disjoint oriented simple closed curves $X, Y \subset \mathbb{R}^3$, the *linking number* $\text{lk}(X, Y) \in \mathbb{Z}$ can be computed by the *combinatorial formula*: for any regular projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$,

$$\text{lk}(X, Y) = \frac{1}{2} \sum_{p \in \pi(X) \cap \pi(Y)} \epsilon_p$$

where $\epsilon_p = \pm 1$ is the sign of each crossing (determined by the right-hand rule applied to the tangent vectors and crossing direction). Equivalently, via the *Gauss integral*:

$$\text{lk}(X, Y) = \frac{1}{4\pi} \oint_X \oint_Y \frac{(x - y) \cdot (dx \times dy)}{|x - y|^3}$$

The linking number is always an integer and is independent of the choice of projection.

C.2. ReLU Activation Preserves Linking Number

For ReLU activations specifically, we can prove linking number preservation using a simple sequential argument that processes one curve at a time, one coordinate at a time. This elementary approach avoids the need for the more sophisticated “link homotopy” framework required for general monotonic activations (Appendix D).

Lemma C.2 (Linking number homotopy invariance). *Let $X, Y \subset \mathbb{R}^3$ be disjoint simple closed curves. If $H : X \times [0, 1] \rightarrow \mathbb{R}^3$ is a continuous homotopy such that $H(X, t) \cap Y = \emptyset$ for all $t \in [0, 1]$, then $\text{lk}(H(X, 1), Y) = \text{lk}(X, Y)$.*

Proof. The Gauss map $G_t : X \times Y \rightarrow S^2$ given by $G_t(x, y) = (H(x, t) - y)/|H(x, t) - y|$ is well-defined for all t since $H(X, t) \cap Y = \emptyset$. As degree is a homotopy invariant, $\deg(G_1) = \deg(G_0)$. $\square$

Lemma C.3 (ReLU preserves linking number). *Let $X, Y \subset \mathbb{R}^3$ be disjoint simple closed curves with $\sigma(X) \cap \sigma(Y) = \emptyset$, where σ is coordinate-wise ReLU. Then $\text{lk}(\sigma(X), \sigma(Y)) = \text{lk}(X, Y)$.*

Proof. We decompose ReLU as $\sigma = G_3 \circ G_2 \circ G_1$ where G_j applies ReLU only to coordinate j :

$$G_1(x_1, x_2, x_3) = (\max(0, x_1), x_2, x_3)$$

and similarly for G_2, G_3 . We show each G_j preserves the linking number by processing curves sequentially.

Step 1: Homotope X while Y is fixed.

Consider the straight-line homotopy $H_t^X(x) = (1 - t)x + tG_1(x)$ for $x \in X$. We claim $H_t^X(X) \cap Y = \emptyset$ for all $t \in [0, 1]$.

Suppose for contradiction that $H_t^X(x) = y$ for some $x \in X, y \in Y$, and $t \in (0, 1]$. Then:

- Coordinates 2 and 3: $x_2 = y_2$ and $x_3 = y_3$ (these coordinates are unchanged by G_1)
- Coordinate 1: $(1 - t)x_1 + t \max(0, x_1) = y_1$

If $x_1 \geq 0$, then $H_t^X(x)_1 = x_1$ for all t , so $x_1 = y_1$. Combined with coordinates 2 and 3, this gives $x = y$, contradicting $X \cap Y = \emptyset$.

If $x_1 < 0$, then $H_t^X(x)_1 = (1 - t)x_1 \in [x_1, 0)$ for $t \in (0, 1]$. For collision with y , we need $y_1 \in [x_1, 0)$. But then $G_1(x) = (0, x_2, x_3)$ and $G_1(y) = (0, y_2, y_3) = (0, x_2, x_3)$, so $G_1(x) = G_1(y)$, contradicting $\sigma(X) \cap \sigma(Y) = \emptyset$.

By Lemma C.2, $\text{lk}(G_1(X), Y) = \text{lk}(X, Y)$.

Step 2: Homotope Y while $G_1(X)$ is fixed.

By an identical argument, the homotopy $H_t^Y(y) = (1 - t)y + tG_1(y)$ satisfies $G_1(X) \cap H_t^Y(Y) = \emptyset$. Therefore $\text{lk}(G_1(X), G_1(Y)) = \text{lk}(G_1(X), Y) = \text{lk}(X, Y)$.

Steps 3–6: Repeat for coordinates 2 and 3.

Applying the same argument to G_2 and G_3 , we obtain:

$$\text{lk}(\sigma(X), \sigma(Y)) = \text{lk}(G_3 \circ G_2 \circ G_1(X), G_3 \circ G_2 \circ G_1(Y)) = \text{lk}(X, Y)$$

$\square$

Figure 8 illustrates this “disjoint tubes” argument: as each coordinate is processed, the homotopy paths of X and Y remain in disjoint regions.

C.3. Proof of Lemma 3.6: Rank-Deficient Intersection

Complete Proof. We decompose the rank-deficient affine transformation $f(x) = Ax + b$ into elementary operations. Since A is rank-deficient, $\text{rank}(A) \leq 2$. We can write $A = U\Sigma V^T$ where U, V are orthogonal matrices and Σ is diagonal with at least one zero entry.

Consider the sequence of transformations: $x \mapsto V^T x \mapsto \Sigma V^T x \mapsto U\Sigma V^T x + b$. Since rotations and translations are ambient homeomorphisms that preserve both intersections and linking numbers, we may reduce to the case where $f(x) = \Sigma x$.

Note that any rank-deficient transformation can be decomposed as $f = g \circ h$ where h has rank exactly 2 and g is a further rank-deficient map. Specifically, if $\text{rank}(A) \leq 1$, we can write $\Sigma = \Sigma_2 \Sigma_1$ where $\Sigma_1 = \text{diag}(1, 1, 0)$ (rank 2) and Σ_2 reduces the rank further.

First consider the rank-2 case: $h(x) = \Sigma_1 x$ where $\Sigma_1 = \text{diag}(1, 1, 0)$. This is equivalent to orthogonal projection $P : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $P(x_1, x_2, x_3) = (x_1, x_2)$.

Suppose for contradiction that $P(X) \cap P(Y) = \emptyset$. We can view P as a regular projection for computing the linking number using the combinatorial formula:

$$\text{lk}(X, Y) = \frac{1}{2} \sum_{p \in P(X) \cap P(Y)} \epsilon_p$$

Since $P(X) \cap P(Y) = \emptyset$, the sum is empty, giving $\text{lk}(X, Y) = 0$. This contradicts our assumption that $\text{lk}(X, Y) \neq 0$.

Therefore $P(X) \cap P(Y) \neq \emptyset$. If $\text{rank}(A) \leq 1$, then $f(X) = g(P(X))$ and $f(Y) = g(P(Y))$ where g maps into an even lower-dimensional space. Since $P(X)$ and $P(Y)$ already intersect, their images under any map g will also intersect.

Therefore, in all cases, $f(X) \cap f(Y) \neq \emptyset$. □

Figure 2 illustrates this projection argument: linked curves in $\mathbb{R}^3$ must intersect when projected to $\mathbb{R}^2$.

C.4. Proof of Theorem 3.7: Link Separation Impossibility

Complete Proof. Assume for contradiction that a width-3 ReLU network $F = F_L \circ \dots \circ F_1$ can linearly separate the curves X and Y .

Linear separability requires $F(X)$ and $F(Y)$ to be disjoint (separated by a hyperplane), hence no intersection can occur throughout the layers. In particular, for every intermediate representation $(X^{(j)}, Y^{(j)})$ after layer j , we must have $X^{(j)} \cap Y^{(j)} = \emptyset$.

Since $\text{lk}(X, Y) \neq 0$, and by Lemma 3.3, linear separability of $F(X)$ and $F(Y)$ implies $\text{lk}(F(X), F(Y)) = 0$. Therefore, the network must change the linking number from a non-zero value to zero.

We analyze how each type of layer affects the linking number:

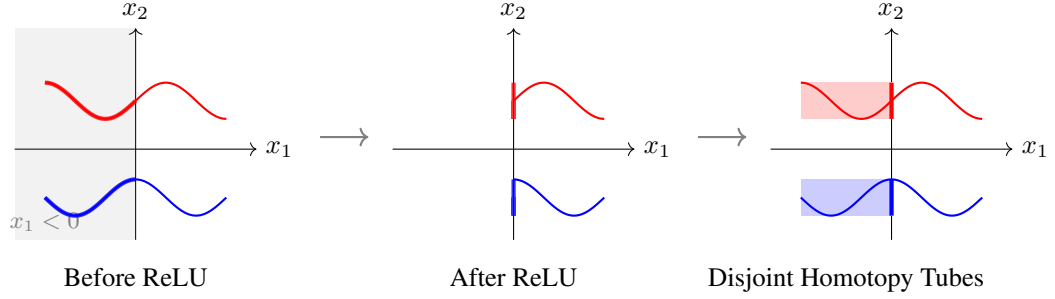
Invertible affine layers. Consider an invertible affine transformation $A(v) = Wv + b$ where W is invertible. Since A is a homeomorphism of $\mathbb{R}^3$, it preserves topological properties. Thus $\text{lk}(A(X), A(Y)) = \pm \text{lk}(X, Y)$ (with sign depending on orientation).

ReLU activation layers. For a ReLU activation layer $\sigma(v) = (\max(0, v_1), \max(0, v_2), \max(0, v_3))$, we apply the sequential homotopy argument from Lemma C.3. Since the curves remain disjoint throughout (by our assumption that linear separability is achieved), the lemma guarantees $\text{lk}(\sigma(X), \sigma(Y)) = \text{lk}(X, Y)$.

Rank-deficient affine layers. By Lemma 3.6, applying a rank-deficient transformation to curves with non-zero linking number creates an intersection: $f(X) \cap f(Y) \neq \emptyset$.

Conclusion. Since invertible affine layers preserve the linking number up to sign and ReLU layers preserve it exactly, and since $\text{lk} \neq 0$ implies $\pm \text{lk} \neq 0$, the only way to change the linking number from non-zero to zero is through a rank-deficient affine transformation. But rank-deficient layers force intersections, contradicting the disjointness required for linear separability. □

Figure 8. ReLU Homotopy and Disjoint Tubes



The ReLU homotopy folds negative x_1 values to zero while maintaining separation between the red and blue curves. The shaded regions show the “tubes” swept by each curve during the homotopy—they remain disjoint, preserving linking number.

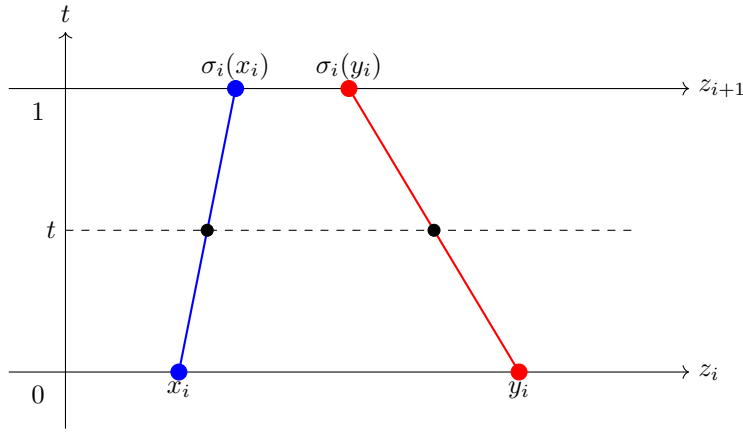


Figure 9. Monotonic activation homotopy in the x_i - t plane. For $x_i < y_i$ and monotonic σ_i , we have $\sigma_i(x_i) < \sigma_i(y_i)$. The linear interpolations $(1-t)x_i + t\sigma_i(x_i)$ and $(1-t)y_i + t\sigma_i(y_i)$ maintain their separation for all $t \in [0, 1]$, ensuring the paths never intersect.

D. Proofs for Section 4: General Monotonic Activations

This appendix provides proofs for the generalized results in Section 4, extending from ReLU in $\mathbb{R}^3$ to arbitrary coordinate-wise monotonic activations in $\mathbb{R}^n$. The key tool is *link homotopy*: a simultaneous deformation of multiple components that maintains pairwise disjointness throughout.

D.1. Link Homotopy Framework

Definition D.1 (Link homotopy). A *link homotopy* of disjoint compact subsets $X_1, \dots, X_k \subset \mathbb{R}^n$ is a continuous map $H : (\bigsqcup_{i=1}^k X_i) \times [0, 1] \rightarrow \mathbb{R}^n$ such that:

1. $H(\cdot, 0)$ is the inclusion map
2. For all $t \in [0, 1]$, the sets $H(X_i, t)$ remain pairwise disjoint

Two configurations are *link homotopic* if connected by such a homotopy.

The linking number $\text{lk}(X, Y)$ is a link homotopy invariant: it depends only on the link homotopy class of the configuration (X, Y) .

D.2. Higher-Dimensional Linking Number

Definition D.2 (Degree of a map). The *degree* $\deg(G)$ of a continuous map $G : X \rightarrow S^k$ from a closed oriented k -dimensional manifold X to the k -sphere can be computed as follows: for any regular value $y \in S^k$,

$$\deg(G) = \sum_{x \in G^{-1}(y)} \text{sign}(\det(DG_x))$$

The degree is independent of the choice of regular value, measures how many times G wraps X around S^k (counted with orientation), and is always an integer. Constant maps have degree zero.

Definition D.3 (Higher-dimensional linking number). For two disjoint, closed, oriented manifolds M^m and N^n in $\mathbb{R}^d$ where $d = m + n + 1$, the *linking number* $\text{lk}(M, N)$ is defined as the degree of the Gauss map $G : M \times N \rightarrow S^{d-1}$:

$$G(x, y) = \frac{x - y}{|x - y|}$$

Since $\dim(M \times N) = m + n = d - 1 = \dim(S^{d-1})$, the degree is well-defined.

Lemma D.4 (Link homotopy invariance). *Let $M^m, N^n \subset \mathbb{R}^d$ (with $d = m + n + 1$) be disjoint manifolds. If (M', N') is link homotopic to (M, N) —i.e., there exist continuous $H_M : M \times [0, 1] \rightarrow \mathbb{R}^d$ and $H_N : N \times [0, 1] \rightarrow \mathbb{R}^d$ with $H_M(M, t) \cap H_N(N, t) = \emptyset$ for all t —then $\text{lk}(M', N') = \text{lk}(M, N)$.*

Proof. The linking number $t \mapsto \text{lk}(H_M(M, t), H_N(N, t))$ is a continuous function from $[0, 1]$ to $\mathbb{Z}$. Since the Gauss map $G_t : H_M(M, t) \times H_N(N, t) \rightarrow S^{d-1}$ varies continuously with t (as long as disjointness is maintained), and degree is a homotopy invariant, the linking number remains constant throughout the link homotopy. $\square$

Lemma D.5 (Higher-dimensional linear separability). *Let M^m and N^n be two disjoint, closed, oriented manifolds in $\mathbb{R}^d$ (with $d = m + n + 1$). If M and N are linearly separable, then $\text{lk}(M, N) = 0$.*

Proof. Linear separability implies there exists a hyperplane such that M and N lie in disjoint open half-spaces H^+ and H^- . We construct a link homotopy contracting M to a point $p \in H^+$ and N to a point $q \in H^-$ simultaneously: since the half-spaces are disjoint and convex, the straight-line contractions within each half-space never intersect, maintaining $H_M(M, t) \cap H_N(N, t) = \emptyset$ for all t .

By link homotopy invariance (Lemma D.4), $\text{lk}(M, N) = \text{lk}(\{p\}, \{q\})$. The Gauss map $G : \{p\} \times \{q\} \rightarrow S^{d-1}$ is the constant map $G(p, q) = \frac{p-q}{|p-q|}$. Since constant maps have degree zero, $\text{lk}(M, N) = 0$. $\square$

D.3. Monotonic Activation Preservation via Link Homotopy

Lemma D.6 (General monotonic activation preservation). *Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be coordinate-wise monotonic with m nonincreasing coordinates. If $X, Y \subset \mathbb{R}^n$ are disjoint compact manifolds such that $\sigma(X)$ and $\sigma(Y)$ remain disjoint, then:*

$$\text{lk}(\sigma(X), \sigma(Y)) = (-1)^m \text{lk}(X, Y)$$

Proof. We decompose $\sigma = R \circ \sigma'$ where:

- $R(x_1, \dots, x_n) = (r_1 x_1, \dots, r_n x_n)$ with $r_i = -1$ if σ_i is nonincreasing and $r_i = 1$ otherwise
- $\sigma'_i(x_i) = \sigma_i(x_i)$ if σ_i is nondecreasing, and $\sigma'_i(x_i) = -\sigma_i(-x_i)$ otherwise

Then σ' is coordinate-wise nondecreasing.

Stage 1: Apply σ' via link homotopy. The straight-line homotopy $H_t(x) = (1 - t)x + t\sigma'(x)$ applied simultaneously to both X and Y defines a link homotopy when σ' is nondecreasing.

To verify disjointness: suppose $H_t(x) = H_t(y)$ for $x \in X, y \in Y$, and $t \in [0, 1]$. Since $X \cap Y = \emptyset$, there exists coordinate i with $x_i \neq y_i$. Assume $x_i < y_i$. Since σ'_i is nondecreasing, $\sigma'_i(x_i) \leq \sigma'_i(y_i)$. Then:

$$H_t(x)_i = (1 - t)x_i + t\sigma'_i(x_i) < (1 - t)y_i + t\sigma'_i(y_i) = H_t(y)_i$$

for all $t \in [0, 1)$. Equality at $t = 1$ requires $\sigma'_i(x_i) = \sigma'_i(y_i)$, but then $\sigma'(X) \cap \sigma'(Y) \neq \emptyset$, contradicting our assumption.

Since H is a link homotopy, $\text{lk}(\sigma'(X), \sigma'(Y)) = \text{lk}(X, Y)$.

Stage 2: Apply reflection R . The reflection R is a homeomorphism. Reflections in m coordinates multiply the linking number by $(-1)^m$ (each reflection reverses orientation in one coordinate). Thus:

$$\text{lk}(R(\sigma'(X)), R(\sigma'(Y))) = (-1)^m \text{lk}(\sigma'(X), \sigma'(Y)) = (-1)^m \text{lk}(X, Y)$$

□

D.4. General Rank-Deficient Intersection

The following lemma generalizes Lemma 3.6 from $\mathbb{R}^3$ to arbitrary dimensions using a kernel translation argument.

Lemma D.7 (General rank-deficient intersection). *Let $X, Y \subset \mathbb{R}^n$ be disjoint compact manifolds with $\text{lk}(X, Y) \neq 0$ (assuming appropriate dimensionality). For any rank-deficient affine $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, we have $f(X) \cap f(Y) \neq \emptyset$.*

Proof. Suppose for contradiction that $f(X) \cap f(Y) = \emptyset$.

Let $f(x) = Ax + b$ where A has rank $r < n$. Choose a unit vector $v \in \ker(A)$. Since X and Y are compact, let $R = \max\{\sup_{x \in X} |x \cdot v|, \sup_{y \in Y} |y \cdot v|\}$.

Define translated sets:

$$X' = X + L_1 v, \quad Y' = Y + L_2 v$$

where $L_1 = 0$ and $L_2 = 3R + 1$. This ensures X' and Y' have disjoint projections onto the v -direction: for any $x \in X'$ and $y \in Y'$:

$$x \cdot v \leq R < 2R + 1 < L_2 - R \leq y \cdot v$$

Since $v \in \ker(A)$:

$$f(X') = f(X + L_1 v) = AX + b = f(X)$$

$$f(Y') = f(Y + L_2 v) = AY + b = f(Y)$$

so the images are unchanged by translation along v .

The translated sets X' and Y' are linearly separable (by the hyperplane $\{z : z \cdot v = (R + L_2 - R)/2\}$). By Lemma D.5, $\text{lk}(X', Y') = 0$.

Translation along v is a link homotopy: if $(X + t_1 v)$ and $(Y + t_2 v)$ were to collide at some point z during the homotopy, then $x + t_1 v = z = y + t_2 v$ for some $x \in X, y \in Y$. Since $v \in \ker(A)$:

$$f(x) = Ax + b = A(x + t_1 v) + b = f(z) = A(y + t_2 v) + b = Ay + b = f(y)$$

This implies $f(X) \cap f(Y) \neq \emptyset$, contradicting our assumption. Therefore disjointness is preserved throughout, and by link homotopy invariance (Lemma D.4), $\text{lk}(X', Y') = \text{lk}(X, Y) \neq 0$. Contradiction. □

D.5. Proof of Theorem 4.5: Higher-Dimensional Separation Impossibility

Complete Proof. Let M^m and N^n be disjoint, closed, oriented manifolds in $\mathbb{R}^d$ (with $d = m + n + 1$) such that $\text{lk}(M, N) \neq 0$. Suppose for contradiction that a width- d feedforward network F with coordinate-wise monotonic activations achieves linear separability.

By Lemma D.5, linear separability requires $\text{lk}(F(M), F(N)) = 0$. We show this is impossible:

- **Invertible affine layers** preserve lk up to sign (homeomorphisms preserve degree).
- **Monotonic activations** preserve lk up to sign via link homotopy (Lemma D.6).
- **Rank-deficient layers** force intersections when $\text{lk} \neq 0$ (Lemma D.7).

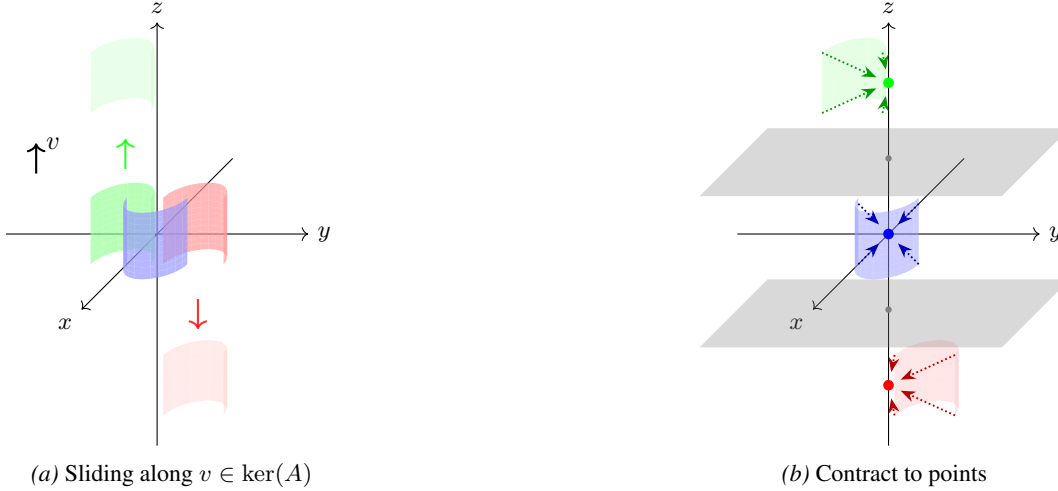


Figure 10. Kernel translation argument. If a rank-deficient affine map with v in its kernel does not create intersections, we can slide components along v (unchanged by f) to achieve linear separation, then contract each to a point, forcing $\text{lk} = 0$.

Since $\text{lk}(M, N) \neq 0$ and each layer preserves lk up to sign (so $\pm \text{lk} \neq 0$), no sequence of such layers can achieve $\text{lk} = 0$ without a rank-deficient layer. But rank-deficient layers create intersections, contradicting disjointness required for linear separability. $\square$

D.6. Proof of Theorem 4.7: General Impossibility Theorem

Complete Proof. Suppose for contradiction that a width- n feedforward network $F = A_L \circ \sigma_L \circ \dots \circ A_1 \circ \sigma_1$ with affine transformations A_i and coordinate-wise monotonic activations σ_i achieves linear separability of X and Y .

Let $(X^{(j)}, Y^{(j)})$ denote the images after the first j layers, with $(X^{(0)}, Y^{(0)}) = (X, Y)$.

We prove by induction that for all j :

1. $X^{(j)} \cap Y^{(j)} = \emptyset$
2. $\text{lk}(X^{(j)}, Y^{(j)}) = \pm \text{lk}(X, Y) \neq 0$

Base case: By hypothesis, $X \cap Y = \emptyset$ and $\text{lk}(X, Y) \neq 0$.

Inductive step: Assume the claim holds for $j - 1$.

If layer j is an invertible affine transformation, it preserves lk up to sign and preserves disjointness.

If layer j is rank-deficient, Lemma D.7 implies $A_j(X^{(j-1)}) \cap A_j(Y^{(j-1)}) \neq \emptyset$, contradicting disjointness.

If layer j is a monotonic activation σ_j , Lemma D.6 gives $\text{lk}(\sigma_j(X^{(j-1)}), \sigma_j(Y^{(j-1)})) = \pm \text{lk}(X^{(j-1)}, Y^{(j-1)}) \neq 0$.

By induction, $\text{lk}(F(X), F(Y)) = \pm \text{lk}(X, Y) \neq 0$. But linear separability requires $\text{lk} = 0$ (Lemma D.5). Contradiction. $\square$

E. Multi-Component Linking in $\mathbb{R}^3$: Milnor's $\bar{\mu}$ -Invariants

Appendix D established that pairwise linking numbers obstruct separation of two-component links in any dimension. For multi-component links in $\mathbb{R}^3$, a subtler phenomenon arises: *all pairwise linking numbers can vanish, yet the link remains inseparable*. Detecting such higher-order entanglement requires Milnor's $\bar{\mu}$ -invariants (Milnor, 1954), which capture collective relationships among three or more components invisible to pairwise invariants.

This extension is significant for multi-class classification. While two-component linking corresponds to binary classification, real-world tasks such as ImageNet involve $k = 1000$ classes. Even when no pair of class manifolds is linked, higher-order

$\bar{\mu}$ -invariants can detect collective entanglement among three or more classes that obstructs simultaneous separation—a phenomenon with no analogue in the two-class setting.

The key technical distinction from the two-component case is that higher $\bar{\mu}$ -invariants require *simultaneous* link homotopy—all components must be deformed together while maintaining pairwise disjointness throughout—rather than the sequential approach (deform one component, then the other) that suffices for pairwise linking numbers. Fortunately, the straight-line homotopy induced by monotonic activations naturally provides such simultaneous deformations, so the proof structure from Appendix D extends directly.

E.1. Milnor’s $\bar{\mu}$ -Invariants

Definition E.1 (Milnor’s $\bar{\mu}$ -invariants). Let $L = X_1 \cup \dots \cup X_k$ be a k -component link of disjoint oriented simple closed curves in $\mathbb{R}^3$. For each sequence $I = (i_1, \dots, i_m)$ of distinct indices from $\{1, \dots, k\}$ (i.e., no repetitions), Milnor (Milnor, 1954) defined an integer-valued invariant $\bar{\mu}_L(I) = \bar{\mu}_L(i_1, \dots, i_m)$, constructed from the lower central series of the link group $\pi_1(\mathbb{R}^3 \setminus L)$.

The invariants satisfy:

- $\bar{\mu}(i, j) = \text{lk}(X_i, X_j)$, the pairwise linking number
- For $m \geq 3$, $\bar{\mu}(i_1, \dots, i_m)$ detects higher-order entanglement not captured by lower-order invariants
- The invariant $\bar{\mu}(i_1, \dots, i_m)$ is well-defined modulo the greatest common divisor of all lower-order $\bar{\mu}$ -invariants involving subsets of the indices; when all lower-order invariants vanish, it is a well-defined integer

Example E.2 (Three-component invariant). For a 3-component link $L = X \cup Y \cup Z$ with $\text{lk}(X, Y) = \text{lk}(Y, Z) = \text{lk}(X, Z) = 0$, the triple invariant $\bar{\mu}(1, 2, 3)$ is a well-defined integer that can be computed geometrically as follows (Mellor & Melvin, 2003): let S_Y and S_Z be oriented Seifert surfaces with $\partial S_Y = Y$ and $\partial S_Z = Z$. Then

$$\bar{\mu}(1, 2, 3) = \text{lk}(X, S_Y \cap S_Z)$$

where $S_Y \cap S_Z$ is the (generically 1-dimensional) intersection of the surfaces, oriented appropriately.

The fundamental property of Milnor’s invariants is their invariance under link homotopy:

Theorem E.3 (Link homotopy invariance (Milnor, 1954)). All $\bar{\mu}$ -invariants with non-repeating indices are link homotopy invariants. That is, if $(X'_1, \dots, X'_k)$ is link homotopic to $(X_1, \dots, X_k)$, then $\bar{\mu}_{L'}(I) = \bar{\mu}_L(I)$ for all sequences I of distinct indices.

Lemma E.4 (Invertible affine maps and $\bar{\mu}$ -invariants). Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be an invertible affine transformation with $A(x) = Mx + b$, and let $L \subset \mathbb{R}^3$ be a k -component link. For any sequence $I = (i_1, \dots, i_m)$ of distinct indices:

1. If $\det(M) > 0$ (orientation-preserving), then $\bar{\mu}_{A(L)}(I) = \bar{\mu}_L(I)$.
2. If $\det(M) < 0$ (orientation-reversing), then $\bar{\mu}_{A(L)}(I) = (-1)^m \bar{\mu}_L(I)$, where $m = |I|$ is the length of the sequence.

In particular, $\bar{\mu}_L(I) \neq 0$ if and only if $\bar{\mu}_{A(L)}(I) \neq 0$ for any invertible A .

Proof. (1) An orientation-preserving invertible affine map $A(x) = Mx + b$ with $\det(M) > 0$ is ambient isotopic to the identity via the path $A_t(x) = ((1-t)I + tM)x + tb$ for $t \in [0, 1]$. The determinant $\det((1-t)I + tM)$ varies continuously from 1 to $\det(M) > 0$, never passing through 0. This ambient isotopy induces a link homotopy from L to $A(L)$, so by Theorem E.3, $\bar{\mu}_{A(L)}(I) = \bar{\mu}_L(I)$.

(2) An orientation-reversing map can be written as the composition of an orientation-preserving map with a reflection. By part (1), it suffices to consider reflections. A reflection provides an isomorphism of link groups but reverses the handedness of the meridian-longitude peripheral system. Since the $\bar{\mu}$ -invariants of length m are extracted from the m -th term of the lower central series via the Magnus expansion of the longitude, this reversal introduces a factor of $(-1)^m$ (Milnor, 1957). $\square$

Milnor further established that these invariants completely detect link homotopy equivalence:

Theorem E.5 (Milnor’s completeness theorem (Milnor, 1954; Habegger & Lin, 1990)). *A link of circles in $\mathbb{R}^3$ is link-homotopically trivial (i.e., link homotopic to an unlink of k disjoint circles in disjoint balls) if and only if all $\bar{\mu}$ -invariants with non-repeating indices vanish. Milnor (Milnor, 1954) proved this for $k \leq 3$ components; Habegger and Lin (Habegger & Lin, 1990) established the complete classification for arbitrary k .*

Corollary E.6. *If a k -component link $L \subset \mathbb{R}^3$ cannot be separated into k disjoint convex regions, then at least one $\bar{\mu}$ -invariant of L is non-zero.*

E.2. Separation Impossibility for Multi-Component Links

We now establish the key lemmas, paralleling the structure of Appendix D.

Lemma E.7 (Linear separability implies all $\bar{\mu}$ -invariants vanish). *Let $X_1, \dots, X_k \subset \mathbb{R}^3$ be pairwise disjoint simple closed curves. If the components can be simultaneously placed in k pairwise disjoint convex regions, then $\bar{\mu}(i_1, \dots, i_m) = 0$ for all sequences of distinct indices.*

Proof. Since each X_j lies in a disjoint convex region C_j , we can simultaneously contract each X_j to a point $p_j \in C_j$ via the straight-line homotopy $H_t^{(j)}(x) = (1-t)x + tp_j$ for $x \in X_j$. The convexity of each C_j ensures that $H_t^{(j)}(X_j) \subset C_j$ for all $t \in [0, 1]$. Since the C_j are pairwise disjoint, this defines a link homotopy: all components remain pairwise disjoint throughout.

By Theorem E.3, $\bar{\mu}_L(I) = \bar{\mu}_{\{p_1, \dots, p_k\}}(I)$ for all I . For k distinct points in general position, all $\bar{\mu}$ -invariants vanish (the link group of the complement of k points is free, with trivial lower central series quotients). $\square$

Lemma E.8 (Monotonic activations preserve $\bar{\mu}$ -invariants up to sign). *Let $\sigma : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a coordinate-wise monotonic function with m nonincreasing coordinates. If $X_1, \dots, X_k \subset \mathbb{R}^3$ are pairwise disjoint simple closed curves such that $\sigma(X_1), \dots, \sigma(X_k)$ remain pairwise disjoint, then*

$$\bar{\mu}_{\sigma(L)}(I) = (-1)^{m|I|} \bar{\mu}_L(I)$$

for all sequences I of distinct indices with $|I|$ denoting the length of I (where we identify $\sigma(L) = \sigma(X_1) \cup \dots \cup \sigma(X_k)$). In particular, $\bar{\mu}_L(I) \neq 0$ if and only if $\bar{\mu}_{\sigma(L)}(I) \neq 0$.

Proof. Following the approach of Lemma D.6, we decompose $\sigma = R \circ \sigma'$ where:

- $R(x_1, x_2, x_3) = (r_1x_1, r_2x_2, r_3x_3)$ with $r_i = -1$ if σ_i is nonincreasing and $r_i = 1$ otherwise
- $\sigma'_i(x_i) = \sigma_i(x_i)$ if σ_i is nondecreasing, and $\sigma'_i(x_i) = -\sigma_i(-x_i)$ otherwise

Then σ' is coordinate-wise nondecreasing.

Stage 1: Apply σ' via link homotopy. The straight-line homotopy $H_t(x) = (1-t)x + t\sigma'(x)$, applied simultaneously to all components, defines a link homotopy from L to $\sigma'(L)$ when σ' is nondecreasing.

The disjointness verification is identical to Lemma D.6: suppose $H_t(x) = H_t(y)$ for some $x \in X_i, y \in X_j$ with $i \neq j$. Since $X_i \cap X_j = \emptyset$, there exists a coordinate ℓ with $x_\ell \neq y_\ell$. Assume $x_\ell < y_\ell$. Since σ'_ℓ is nondecreasing, $\sigma'_\ell(x_\ell) \leq \sigma'_\ell(y_\ell)$, and

$$H_t(x)_\ell = (1-t)x_\ell + t\sigma'_\ell(x_\ell) < (1-t)y_\ell + t\sigma'_\ell(y_\ell) = H_t(y)_\ell$$

for all $t \in [0, 1)$. Equality at $t = 1$ requires $\sigma'_\ell(x_\ell) = \sigma'_\ell(y_\ell)$, but then $\sigma'(X_i) \cap \sigma'(X_j) \neq \emptyset$, contradicting our assumption.

Since H_t is a link homotopy, Theorem E.3 implies $\bar{\mu}_{\sigma'(L)}(I) = \bar{\mu}_L(I)$.

Stage 2: Apply reflection R . The reflection R is an invertible affine map with $\det(R) = (-1)^m$. By Lemma E.4: if m is even then $\bar{\mu}_{R(\sigma'(L))}(I) = \bar{\mu}_{\sigma'(L)}(I)$; if m is odd then $\bar{\mu}_{R(\sigma'(L))}(I) = (-1)^{|I|} \bar{\mu}_{\sigma'(L)}(I)$. In either case, $\bar{\mu}_{\sigma(L)}(I) = (-1)^{m|I|} \bar{\mu}_L(I)$. $\square$

Lemma E.9 (Rank-deficient transformations force intersection). *Let $X_1, \dots, X_k \subset \mathbb{R}^3$ be pairwise disjoint simple closed curves with some $\bar{\mu}$ -invariant non-zero. For any rank-deficient affine transformation $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, at least one pair $f(X_i) \cap f(X_j)$ is non-empty.*

Proof. The argument is identical to Lemma D.7, extended to k components. Suppose for contradiction that all $f(X_i)$ are pairwise disjoint.

Let $f(x) = Ax + b$ where A has rank $r < 3$, and let $v \in \ker(A)$ be a unit vector. Since each X_j is compact, let $R = \max_j \sup_{x \in X_j} |x \cdot v|$. Define translated components

$$X'_j = X_j + (3jR)v$$

for $j = 1, \dots, k$. These translations separate the components along the v -direction: for any $x \in X'_i$ and $y \in X'_j$ with $i < j$,

$$y \cdot v - x \cdot v \geq 3jR - R - (3iR + R) = 3(j - i)R - 2R \geq R > 0$$

Thus $(X'_1, \dots, X'_k)$ are linearly separable by hyperplanes orthogonal to v .

Since $v \in \ker(A)$, we have $f(X'_j) = f(X_j + (3jR)v) = AX_j + b = f(X_j)$ for all j . By assumption, the images $f(X_j)$ are pairwise disjoint, so the images $f(X'_j)$ are also pairwise disjoint.

Translation along v defines a link homotopy: $H_t^{(j)}(x) = x + t(3jR)v$ moves each component along parallel lines, so disjointness is preserved throughout. By Theorem E.3, all $\bar{\mu}$ -invariants are preserved: $\bar{\mu}_{L'}(I) = \bar{\mu}_L(I)$ for all I .

But $L' = (X'_1, \dots, X'_k)$ is linearly separable, so by Lemma E.7, all $\bar{\mu}_{L'}(I) = 0$. This contradicts the assumption that some $\bar{\mu}_L(I) \neq 0$. $\square$

Theorem E.10 (Multi-component separation impossibility). *Let $L = X_1 \cup \dots \cup X_k \subset \mathbb{R}^3$ be a k -component link of pairwise disjoint simple closed curves. If any Milnor $\bar{\mu}$ -invariant of L is non-zero, then no width-3 feedforward network with coordinate-wise monotonic activations can simultaneously place all components in pairwise disjoint convex regions.*

In particular, perfect k -class classification of $(X_1, \dots, X_k)$ is impossible for such networks.

Proof. Let $F = A_L \circ \sigma_{L-1} \circ \dots \circ \sigma_1 \circ A_1$ be a width-3 network with affine transformations A_j and coordinate-wise monotonic activations σ_j . Suppose for contradiction that $F(X_1), \dots, F(X_k)$ lie in pairwise disjoint convex regions.

We prove by induction that the link $L^{(j)}$ after j layers has some non-zero $\bar{\mu}$ -invariant, while maintaining pairwise disjointness.

Base case: The input link L has some $\bar{\mu}_L(I) \neq 0$ by assumption.

Inductive step: Let $L^{(j)} = (X_1^{(j)}, \dots, X_k^{(j)})$ denote the images after j layers, with $L^{(0)} = L$.

Case 1: Layer $j + 1$ is an invertible affine transformation A . By Lemma E.4, $L^{(j)}$ has some non-zero $\bar{\mu}$ -invariant if and only if $A(L^{(j)})$ does. Pairwise disjointness is preserved since A is a bijection.

Case 2: Layer $j + 1$ is a coordinate-wise monotonic activation σ . If $\sigma(X_1^{(j)}), \dots, \sigma(X_k^{(j)})$ are pairwise disjoint, then by Lemma E.8, all $\bar{\mu}$ -invariants are preserved up to sign, so some $\bar{\mu}$ -invariant remains non-zero.

Case 3: Layer $j + 1$ is a rank-deficient affine transformation A . By Lemma E.9, since some $\bar{\mu}_{L^{(j)}}(I) \neq 0$, at least one pair $A(X_i^{(j)}) \cap A(X_\ell^{(j)}) \neq \emptyset$. This intersection propagates through all subsequent layers, contradicting the assumption that $F(X_1), \dots, F(X_k)$ are pairwise disjoint.

Thus all affine layers must be invertible, and by induction, some $\bar{\mu}_{F(L)}(I) \neq 0$. But disjoint convex regions imply $\bar{\mu}_{F(L)}(I) = 0$ for all I by Lemma E.7. Contradiction.

The classification impossibility follows from Theorem A.11: perfect k -class classification requires decision regions that are pairwise disjoint and convex. $\square$

Corollary E.11 (Complete obstruction for links in $\mathbb{R}^3$). *By Milnor's completeness theorem (Theorem E.5), if a link $L \subset \mathbb{R}^3$ is not link-homotopically trivial, then at least one $\bar{\mu}$ -invariant is non-zero. Hence: every non-trivial link of circles in $\mathbb{R}^3$ cannot be separated by any width-3 feedforward network with coordinate-wise monotonic activations.*

E.3. Example: The Borromean Rings

Example E.12 (Borromean rings). The Borromean rings (Figure 11) consist of three circles $X_1, X_2, X_3 \subset \mathbb{R}^3$ arranged so that:

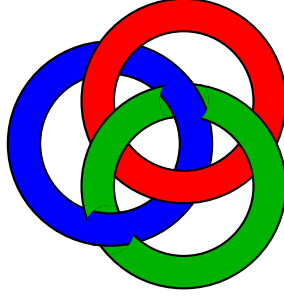


Figure 11. The Borromean rings: three circles with all pairwise linking numbers zero ($\text{lk}(X_i, X_j) = 0$ for all $i \neq j$), yet collectively inseparable due to the non-vanishing triple Milnor invariant $\bar{\mu}(1, 2, 3) = 1$.

- All pairwise linking numbers vanish: $\bar{\mu}(i, j) = \text{lk}(X_i, X_j) = 0$ for all $i \neq j$
- The triple Milnor invariant is non-zero: $\bar{\mu}(1, 2, 3) = 1$

This is the simplest example where pairwise invariants fail to detect linking. Removing any single ring causes the other two to become unlinked, yet all three together cannot be separated.

By Theorem E.10, no width-3 feedforward network with monotonic activations can separate the Borromean rings into three disjoint convex regions, despite the fact that every pair of rings has linking number zero.

Remark E.13 (Open problems in higher dimensions). While Milnor’s $\bar{\mu}$ -invariants are specific to links of circles in $\mathbb{R}^3$, higher-dimensional multi-component linking remains an open problem with potentially abundant invariants yet to be discovered (Koschorke, 1988). Most techniques developed in this appendix—preservation under monotonic maps, obstruction from non-trivial invariants, architectural mechanisms for breaking constraints—generalize naturally to new invariants as they are discovered. This opens a productive interaction between empirical search for linked structures in high-dimensional datasets and theoretical proposals of mathematically interesting invariants: new linking phenomena found in data can motivate theoretical investigation, while new invariants from topology can guide the search for dataset structures that create learning barriers.

F. Knot Preservation: Full Details

This section provides a complete treatment of knot preservation by width-constrained networks. Unlike the inter-object constraints from linking (where multiple curves are entangled), knot preservation concerns *intra-object* constraints arising from internal topological complexity of a single curve.

F.1. Knot Theory Preliminaries

Definition F.1 (Knot and unknot). A *knot* is a simple closed curve (i.e., a homeomorphic image of S^1) embedded in $\mathbb{R}^3$. The *unknot* (or *trivial knot*) is the standard circle $\{(x, y, 0) : x^2 + y^2 = 1\}$ in $\mathbb{R}^3$.

Definition F.2 (Trefoil knot). The *trefoil knot* is the simplest non-trivial knot, given parametrically by:

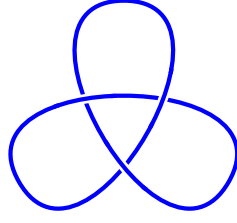
$$T(t) = (\sin(t) + 2 \sin(2t), \cos(t) - 2 \cos(2t), -\sin(3t))$$

for $t \in [0, 2\pi]$.

Definition F.3 (Isotopy and ambient isotopy). Let $X \subset \mathbb{R}^n$ be a compact subset.

- An *isotopy of X* is a continuous family of embeddings $f_t : X \rightarrow \mathbb{R}^n$ for $t \in [0, 1]$ with f_0 the inclusion map and $(x, t) \mapsto f_t(x)$ continuous.
- An *ambient isotopy* is a continuous family of homeomorphisms $\phi_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$ for $t \in [0, 1]$ such that $\phi_0 = \text{id}_{\mathbb{R}^n}$.
- Two compact subsets $X, Y \subset \mathbb{R}^n$ are *ambient isotopic* if there exists an ambient isotopy ϕ_t with $\phi_1(X) = Y$.

Figure 12. The Trefoil Knot



The trefoil knot with three-fold rotational symmetry. This is the simplest non-trivial knot and cannot be unknotted by any width-3 network with monotonic activations while maintaining injectivity.

Definition F.4 (Knot equivalence). Two knots K_1 and K_2 in $\mathbb{R}^3$ are *equivalent* if they are ambient isotopic. In this work, we consider a knot and its mirror image to be of the same knot type, as our focus is on whether knots can be unknotted rather than on chirality.

Theorem F.5 (Ambient isotopy extension theorem). *Let $X \subset \mathbb{R}^n$ be a compact smooth (or piecewise-linear) submanifold and $f_t : X \rightarrow \mathbb{R}^n$ be a smooth (or PL) isotopy with f_0 the inclusion map. Then there exists an ambient isotopy $\phi_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $\phi_t|_X = f_t$.*

This classical result (Palais, 1960; Hudson & Zeeman, 1964) allows us to prove knot equivalence by constructing isotopies of the knots themselves rather than of the entire ambient space.

F.2. Characterization of the Unknot

Lemma F.6 (Disk-bounding characterization). *A knot $K \subset \mathbb{R}^3$ is isotopic to the unknot if and only if it bounds an embedded disk in $\mathbb{R}^3$.*

Proof. The forward direction is immediate: the standard circle bounds a disk in the xy -plane.

For the reverse direction, if K bounds an embedded disk D , then by the Alexander trick, one can construct an isotopy of $\mathbb{R}^3$ that carries the disk to the standard position, thereby unknotting K . $\square$

Lemma F.7 (Planar knots are unknotted). *Any knot $K \subset \mathbb{R}^3$ that lies entirely in a plane is equivalent to the unknot.*

Proof. Without loss of generality, assume K lies in the xy -plane. By the Jordan curve theorem, K divides this plane into exactly two regions: one bounded and one unbounded. By the Schönflies theorem in dimension 2, the bounded region is homeomorphic to a closed disk. Thus K bounds an embedded disk in $\mathbb{R}^3$. By Lemma F.6, K is isotopic to the unknot. $\square$

This lemma reveals the key tension: to unknot a non-trivial knot, a network must somehow “flatten” it into a plane, but doing so while maintaining injectivity is impossible.

F.3. Layer-by-Layer Analysis

Lemma F.8 (Homotopy-isotopy dichotomy). *Let $K \subset \mathbb{R}^n$ be a compact subset and $H : K \times [0, 1] \rightarrow \mathbb{R}^n$ be a continuous homotopy with $H(\cdot, 0) = \text{id}_K$. Then either:*

1. *H creates self-intersections: $\exists t \in (0, 1]$ and distinct $x, y \in K$ with $H(x, t) = H(y, t)$, or*
2. *H is an isotopy from K to $H(K, 1)$.*

Proof. If $H(\cdot, t)$ is injective on K for all $t \in [0, 1]$, then each $H(\cdot, t) : K \rightarrow \mathbb{R}^n$ is an embedding (as K is compact and $\mathbb{R}^n$ is Hausdorff). The homotopy H is then an isotopy. $\square$

Lemma F.9 (Invertible affine transformations preserve knot type). *Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be an invertible affine transformation and K be any knot. Then $A(K)$ is equivalent to K (or its mirror if A is orientation-reversing).*

Proof. Let $A(x) = Mx + b$ with $M \in GL(3, \mathbb{R})$. If $\det(M) > 0$, then $M \in GL_+(3, \mathbb{R})$, which is path-connected. Thus there exists a continuous path $\gamma : [0, 1] \rightarrow GL_+(3, \mathbb{R})$ with $\gamma(0) = I$ and $\gamma(1) = M$. The family $\phi_t(x) = \gamma(t)x + tb$ is an ambient isotopy from the identity to A , so $A(K)$ is ambient isotopic to K .

If $\det(M) < 0$, decompose $M = M'R$ where R is a reflection and $\det(M') > 0$. Then $A(K)$ is the mirror image of a knot equivalent to K . $\square$

Lemma F.10 (Rank-deficient transformations destroy injectivity on non-trivial knots). *Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a rank-deficient affine transformation and K be a non-trivial knot. Then $A|_K$ is not injective.*

Proof. Let v be a unit vector in $\ker(A)$. Consider the homotopy $H : K \times [0, 1] \rightarrow \mathbb{R}^3$ defined by:

$$H(x, t) = x - t(x \cdot v)v$$

This projects each point along v toward the hyperplane $v^\perp$.

We claim H is an isotopy. Suppose $H(x, t) = H(y, t)$ for $x, y \in K$ and some $t \in [0, 1]$. Then:

$$x - y = t((x - y) \cdot v)v$$

So $x - y$ is parallel to v . If $A|_K$ were injective, and since $A(x) = A(y)$ when $x - y \in \ker(A)$, we would need $x = y$.

Thus if $A|_K$ is injective, H is an isotopy from K to $H(K, 1)$, which lies entirely in the hyperplane $v^\perp$. By Lemma F.7, $H(K, 1)$ is unknotted. By the ambient isotopy extension theorem, K would be equivalent to the unknot, contradicting the assumption that K is non-trivial.

Therefore $A|_K$ cannot be injective. $\square$

Lemma F.11 (ReLU preserves knot type when injective). *Let $\sigma : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the coordinate-wise ReLU function and K be a knot. If $\sigma|_K$ is injective, then $\sigma(K)$ is equivalent to K .*

Proof. Consider the straight-line homotopy $H(p, t) = (1 - t)p + t\sigma(p)$ for $p \in K$. For each coordinate:

$$H_i(p, t) = \begin{cases} p_i & \text{if } p_i \geq 0 \\ (1 - t)p_i & \text{if } p_i < 0 \end{cases}$$

We verify $H(\cdot, t)$ is injective for all $t \in [0, 1]$. Suppose $H(p, t) = H(q, t)$ for $p, q \in K$:

- If $p_i, q_i \geq 0$: $H_i(p, t) = p_i$ and $H_i(q, t) = q_i$, so $p_i = q_i$.
- If $p_i, q_i < 0$: $H_i(p, t) = (1 - t)p_i$ and $H_i(q, t) = (1 - t)q_i$, so $p_i = q_i$.
- If $p_i < 0$ and $q_i \geq 0$: $(1 - t)p_i = q_i$ requires $t > 1$ (impossible since $p_i < 0, q_i \geq 0$).

Thus H is an isotopy, and by Theorem F.5, $\sigma(K)$ is ambient isotopic to K . $\square$

Theorem F.12 (Knot type preservation). *Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a width-3 feedforward network with monotonic activations and $K \subset \mathbb{R}^3$ a non-trivial knot. If $F|_K$ is injective, then $F(K)$ is equivalent to K (or its mirror).*

Proof. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a width-3 ReLU network and K a non-trivial knot with $f|_K$ injective.

Write $f = F_L \circ \dots \circ F_1$ where each F_i is either an affine transformation or ReLU activation. Since $f|_K$ is injective, each intermediate restriction must be injective: letting $K^{(i)} = (F_i \circ \dots \circ F_1)(K)$, we have $F_{i+1}|_{K^{(i)}}$ injective.

By our layer analysis:

- **Invertible affine layers** preserve knot type (Lemma F.9).
- **Rank-deficient affine layers** would destroy injectivity on non-trivial knots (Lemma F.10).
- **ReLU activations** preserve knot type when injective (Lemma F.11).

Since injectivity is maintained throughout, no rank-deficient layer can be applied to a non-trivial knot. Thus all layers either preserve knot type or are ReLU layers that also preserve knot type. Therefore $f(K)$ is equivalent to K .

In particular, since the unknot is not equivalent to any non-trivial knot, f cannot transform K into an unknot while maintaining injectivity. $\square$

Corollary F.13 (Trefoil unknotting impossibility). *No width-3 feedforward network with monotonic activations can transform the trefoil knot into an unknot while maintaining an injective restriction to the trefoil.*

Remark F.14 (Implications for representation learning). If a neural network learns an embedding that is injective on a knotted subset of its input data, the network cannot simplify the topology of that subset without destroying embedding quality. This creates a fundamental tension between topological simplification and representation fidelity.

Remark F.15 (Generalization to higher dimensions). The techniques in this section generalize naturally to $\mathbb{R}^n$ for any $n \geq 3$. The core arguments—straight-line homotopy for monotonic activations, rank-deficient projection to hyperplanes, and hyperplanar triviality—are dimension-independent. Any *nonplanar isotopy invariant* (isotopy-invariant, trivial on hyperplanes) creates analogous obstructions for width- n networks. Examples include knotted 2-spheres in $\mathbb{R}^4$, surface genus for embedded surfaces, and higher-dimensional linking invariants. This opens connections between high-dimensional topological data analysis and neural network expressivity.

G. Width Upper Bound for Topological Unlinking

This section establishes that removing width restrictions entirely eliminates the impossibility results, demonstrating that topological obstructions are purely a consequence of dimension constraints. Specifically, we show that width- $(n + 1)$ networks suffice to unlink any configuration.

We begin with two foundational results that enable our main construction. The first is a separation result for compact sets:

Lemma G.1 (Urysohn lemma). *Let $X_1, \dots, X_k \subset \mathbb{R}^n$ be disjoint compact subsets. Then for each i , there exists an open neighborhood U_i of X_i such that $\overline{U_i} \cap \overline{U_j} = \emptyset$ for $i \neq j$, and a continuous function $\psi_i : \mathbb{R}^n \rightarrow [0, 1]$ such that:*

- $\psi_i(x) = 1$ for all $x \in X_i$
- $\psi_i(x) = 0$ for all $x \notin U_i$

Proof. Since compact subsets of a metric space are closed, and disjoint closed sets can be separated by open neighborhoods, we can find disjoint open neighborhoods U_i of each X_i . The existence of the continuous functions ψ_i follows from the standard Urysohn lemma for normal spaces (and $\mathbb{R}^n$ is normal). $\square$

The following universal approximation result enables width-unrestricted constructions:

Lemma G.2 (Hanin-Sellke). *For any compact set $K \subset \mathbb{R}^n$ and continuous function $f : K \rightarrow \mathbb{R}$, and any $\varepsilon > 0$, there exists a feedforward neural network $F : \mathbb{R}^n \rightarrow \mathbb{R}$ with ReLU activations and width $n + 1$ such that $|F(x) - f(x)| < \varepsilon$ for all $x \in K$.*

Proof. This is the main result of (Hanin & Sellke, 2017). The construction uses “max-min strings” to build continuous functions by repeatedly cutting off corners of convex hulls, controlled by the modulus of continuity. $\square$

Theorem G.3 (Unlinking via width-unrestricted networks). *Let $X_1, \dots, X_k \subset \mathbb{R}^n$ be any collection of disjoint compact subsets, regardless of their linking configuration. For any $\varepsilon > 0$ with $\varepsilon < 1/2$, there exists a feedforward neural network $F : \mathbb{R}^n \rightarrow \mathbb{R}$ with ReLU activations and intermediate width unconstrained such that:*

1. F achieves linear separation: $i - 1/2 < F(x) < i + 1/2$ for all $x \in X_i$

2. The images $F(X_i)$ lie in disjoint convex (hence contractible) intervals, implying the link $X_1 \cup \dots \cup X_k$ is effectively unlinked after transformation
3. The network can be realized with any standard monotonic activation function (ReLU, sigmoid, tanh, etc.)

Complete Proof. We construct a scalar classification function that separates the components into disjoint intervals.

Step 1: Separating disjoint compact sets. By Lemma G.1, for each i there exist disjoint open neighborhoods U_i of X_i and continuous functions $\psi_i : \mathbb{R}^n \rightarrow [0, 1]$ with $\psi_i(x) = 1$ on X_i and $\psi_i(x) = 0$ outside U_i .

Step 2: Constructing the target function. Define the continuous target function $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$ by:

$$\tilde{f}(x) = \sum_{i=1}^k i \cdot \psi_i(x)$$

Note that $\tilde{f}(x) = i$ for $x \in X_i$ and $\tilde{f}(x) = 0$ for x outside all neighborhoods U_i .

Step 3: Neural network approximation. By Lemma G.2, there exists a feedforward network $F : \mathbb{R}^n \rightarrow \mathbb{R}$ with ReLU activations and width $n + 1$ such that $|F(x) - \tilde{f}(x)| < \varepsilon$ uniformly on the compact set $K = \bigcup_{i=1}^k \overline{U_i}$.

Step 4: Verifying linear separation. For any $x \in X_i$, we have:

$$|F(x) - i| = |F(x) - \tilde{f}(x)| < \varepsilon < 1/2$$

Therefore, $i - 1/2 < F(x) < i + 1/2$.

Step 5: Topological interpretation and dimension augmentation. After applying F , each component X_i is mapped into the interval $(i - 1/2, i + 1/2)$. These intervals are disjoint and convex, hence contractible. This means the configuration $F(X_1) \cup \dots \cup F(X_k) \subset \mathbb{R}$ is unlinked, regardless of the original linking structure. To maintain input-output dimension equality, we can define $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $G(x) = (F(x), 0, \dots, 0)$, which maps each X_i into a hyperplane slice achieving geometric separation. $\square$

This construction shows that width- $(n + 1)$ networks can overcome all the linking obstructions from our main results. The key insight is that with width $n + 1$, neural networks can implement arbitrary continuous functions (by Hanin-Sellke), including those that map linked configurations to unlinked ones. This establishes an **upper bound**: width $n + 1$ suffices for topological unlinking.

H. Width Lower Bound for Universal Approximation

Our topological framework yields a theoretical application to classical approximation theory: we establish that width n is *insufficient* for universal approximation, regardless of network depth. This complements the upper bound of the previous section (width $n + 1$ suffices) and improves upon existing results by removing Lipschitz assumptions on the activation function.

Theorem H.1 (Width lower bound for universal approximation). *Consider any continuous coordinate-wise monotonic activation function (ReLU, leaky-ReLU, sigmoid, tanh, etc.). The minimum width required for a feedforward network $F : \mathbb{R}^n \rightarrow \mathbb{R}$ to be a uniform universal approximator on compact sets is:*

$$w_{\min} \geq n + 1$$

where n is the input dimension.

Proof. Suppose for contradiction that width- n feedforward networks with some monotonic activation function σ are dense in $C(K)$ for all compact $K \subset \mathbb{R}^n$.

Consider the Hopf link $X \cup Y \subset \mathbb{R}^3$ (or its higher-dimensional analogue in $\mathbb{R}^n$) with $\text{lk}(X, Y) = 1$. The continuous function $f : X \cup Y \rightarrow \mathbb{R}$ defined by $f|_X = 0$ and $f|_Y = 1$ can be extended to a continuous function on $\mathbb{R}^n$ (by Tietze extension).

If width- n networks are universal approximators, they can approximate this function arbitrarily well, achieving linear separation of X and Y . But our main results (Theorem 3.7 and its generalizations) establish that width- n networks with monotonic activations preserve linking numbers, making linear separation of linked components impossible.

This contradiction shows that width- n networks with monotonic activations cannot be universal approximators on $\mathbb{R}^n$, establishing $w_{\min} \geq n + 1$. $\square$

Remark H.2 (Comparison with existing lower bound techniques). Our approach improves upon existing results by removing technical conditions:

- **(Hanin & Sellke, 2017)**: Uses bounded level set components and shows width- n networks have structural restrictions on level sets, exhibiting a continuous function with a compact level set that disconnects a bounded region.
- **(Johnson, 2019)**: Analyzes topology of level sets for activations uniformly approximable by injections, showing all level sets must be unbounded for width- $\leq n$ networks.
- **(Rochau et al., 2024)**: Requires monotone *Lipschitz* activations and establishes impossibility results through approximation theory on homeomorphisms.
- **Our approach (Theorem H.1)**: Uses ambient topological invariants to establish exact impossibility results without approximation considerations and without requiring Lipschitz conditions on activations. Our framework applies to *any* continuous coordinate-wise monotonic activation, including non-Lipschitz functions, bounded or unbounded in range.

The key advantage of our method is that it works through exact topological preservation rather than approximation-theoretic arguments, yielding a cleaner and more general result.

I. Detailed Constructions Breaking Topological Constraints

I.1. Why Non-Monotonic Activations Break Homotopy Preservation

The impossibility results established in earlier sections critically depend on lemmas showing that coordinate-wise monotonic activations preserve linking numbers and knot types. Both preservation properties fail when the activation function is not monotonic.

Consider the absolute value function $\sigma(x) = |x|$ applied coordinate-wise. The absolute value function is not monotonic—it decreases on $(-\infty, 0)$ and increases on $(0, \infty)$. This non-monotonicity breaks the homotopy constructions that underlie our impossibility theorems.

Specifically, the intermediate homotopy $h_t(x) = (1 - t)x + t|x|$ connecting the identity to the absolute value function can create self-intersections within individual curve components. When $x < 0$, the path from x to $|x| = -x$ crosses through $x = 0$, potentially creating intersections that violate the disjointness preservation required for linking number preservation (multi-component) and knot type preservation (single-object).

For example, consider a curve that passes through points with both positive and negative coordinates. Under the homotopy h_t , different parts of the curve move toward the “fold line” at $x = 0$ and can collide before reaching the final folded configuration. This failure of the straight-line homotopy to preserve disjointness means that both linking numbers $\text{lk}(\sigma(X), \sigma(Y))$ and single-object invariants like knot type are not necessarily preserved, breaking the topological rigidity that underlies our impossibility results.

This breakdown of topological preservation opens the possibility for constructive approaches: we can now explicitly construct transformations that unlink linked curves and unknot knotted curves using non-monotonic activations.

I.2. Hopf Link Unlinking via Absolute Value Activations

Complete Construction. We provide an explicit construction that systematically unlinks the Hopf link using absolute value activations. The key insight is that absolute value “folding” can be applied coordinate-wise to reduce crossing numbers while preserving disjointness.

Consider the Hopf link with components:

$$X(t) = (\cos t, \sin t, 0) \text{ for } t \in [0, 2\pi] \quad (4)$$

$$Y(s) = (0, 1 + \cos s, \sin s) \text{ for } s \in [0, 2\pi] \quad (5)$$

Step 1: Initial Configuration

$$X(t) = (\cos t, \sin t, 0) \quad (6)$$

$$Y(s) = (0, 1 + \cos s, \sin s) \quad (7)$$

Step 2: Apply ABS to y -coordinate

$$X_1(t) = (\cos t, |\sin t|, 0) \quad (8)$$

$$Y_1(s) = (0, 1 + \cos s, \sin s) \quad (9)$$

Step 3: Combined affine transformation $(x, y, z) \mapsto (x, 1 - y, z)$

$$X_2(t) = (\cos t, 1 - |\sin t|, 0) \quad (10)$$

$$Y_2(s) = (0, 1 - (1 + \cos s), \sin s) = (0, -\cos s, \sin s) \quad (11)$$

Step 4: Apply ABS to both x and y coordinates

$$X_3(t) = (|\cos t|, 1 - |\sin t|, 0) \quad (12)$$

$$Y_3(s) = (0, |\cos s|, \sin s) \quad (13)$$

Step 5: Apply ABS to z -coordinate

$$X_4(t) = (|\cos t|, 1 - |\sin t|, 0) \quad (14)$$

$$Y_4(s) = (0, |\cos s|, |\sin s|) \quad (15)$$

The final curves X_4 and Y_4 are both contained in the positive octant and are clearly unlinked with linking number zero. They are linearly separable by a hyperplane, demonstrating that the absolute value activation successfully breaks the topological obstruction. $\square$

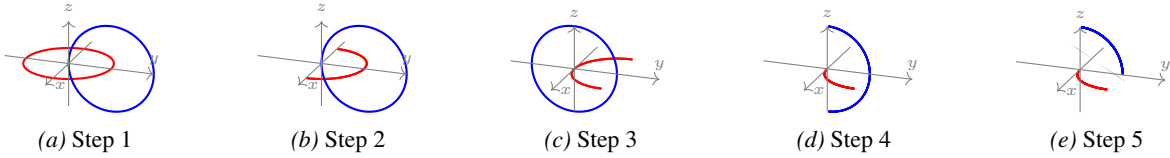


Figure 13. Hopf link unlinking via absolute value activations. The shaded plane in Step 5 shows linear separability.

I.3. Trefoil Unknotting via Absolute Value Activations

The same principle applies to unknotting individual knots. Consider the trefoil parametrization:

$$x(t) = \sin(t) + 2 \sin(2t) \quad (16)$$

$$y(t) = \cos(t) - 2 \cos(2t) \quad (17)$$

$$z(t) = -\sin(3t), \quad t \in [0, 2\pi] \quad (18)$$

The ABS folding $y \mapsto c + |y - c|$ with $c = -1.8$ eliminates topological constraints by folding the crossings.

Remark I.1 (General non-monotonic activations). The constructions above extend to any activation function that exhibits non-monotonic behavior in some neighborhood. Given compact subsets $X, Y \subset \mathbb{R}^n$ and an activation σ with a non-monotonic region, we can always apply affine transformations to bring the relevant portions of X and Y into the non-monotonic neighborhood of σ . The local folding behavior then enables the same topological transformations as demonstrated with absolute value. This includes activations such as sine, cosine, or any function with local maxima or minima—including modern activations like GELU, Swish/SiLU, and Mish that exhibit non-monotonic behavior near the origin.

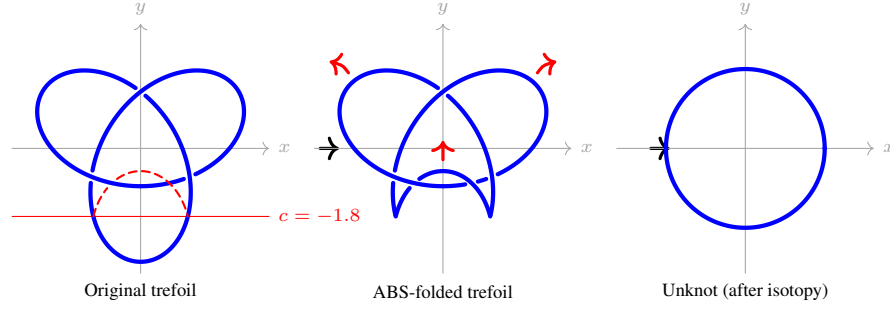


Figure 14. Trefoil unknotting via ABS activation. Left: Original trefoil (blue) with crossing structure. Middle: ABS-folded trefoil (red dashed shows reflected portion). Right: Final unknotted configuration.

I.4. ResNet Absolute Value Synthesis

Complete Proof of Theorem 6.2 (ResNet Topological Expressivity). We establish that ResNet architectures can express coordinate-wise absolute value functions through the fundamental identity

$$|x| = x + 2 \operatorname{ReLU}(-x). \quad (19)$$

For any coordinate $i \in \{1, 2, \dots, n\}$, this identity depends only on x_i itself, enabling coordinate-wise transformations without requiring information from other coordinates.

Consider a ResNet block of the form $\mathcal{F}(x) = x + \mathcal{G}(x)$ where $\mathcal{G} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is implemented by a feedforward network with ReLU activations. To express the absolute value function $|x_i|$ for each coordinate, we set

$$\mathcal{G}_i(x_i) = 2 \operatorname{ReLU}(-x_i). \quad (20)$$

This yields

$$\mathcal{F}_i(x) = x_i + 2 \operatorname{ReLU}(-x_i) = |x_i|. \quad (21)$$

For a width- n implementation, the residual branch $\mathcal{G}$ takes the form

$$\mathcal{G}(x) = W \cdot \operatorname{ReLU}(-x) \quad (22)$$

where $W = 2I_n$ with I_n denoting the $n \times n$ identity matrix. Since affine transformations are available, the general folding operation $x_i \mapsto c_i + |x_i - c_i|$ can be achieved by composing translations with the absolute value function.

Since ResNet can express coordinate-wise absolute value transformations, it inherits the topological transformation capabilities demonstrated above—specifically, unlinking linked curves and unknotting knots. $\square$

Remark I.2 (Residual networks vs neural ODEs). While ResNets can be viewed as discretizations of neural ODEs (Chen et al., 2018) in the limit of many layers with small residuals, the discrete ResNet formulation is strictly more expressive for topological transformations. Neural ODEs are constrained to generate continuous flows (diffeomorphisms), which must preserve topological invariants. In contrast, discrete ResNet blocks can implement “jumps” in function space that break these invariants. The ability to express absolute value through $|x| = x + 2 \operatorname{ReLU}(-x)$ relies crucially on the discrete, non-infinitesimal nature of the residual connection—this transformation cannot be expressed as the solution to any ODE with Lipschitz continuous dynamics.

Reversible variants such as RevNet (Gomez et al., 2017) and i-RevNet (Jacobsen et al., 2018) maintain invertibility while providing computational benefits. The residual structure $y = x + f(x)$ appears in modern generative models, demonstrating how residual-like transformations enable topological expressivity while avoiding the homeomorphism constraints of continuous flows.

I.5. Transformer Absolute Value Synthesis

Complete Proof of Theorem 6.4 (Transformer Topological Expressivity). We construct an input sequence with two tokens to properly utilize the attention mechanism. For coordinate x_i , we create the minimal sequence (x_i, x_i) , where the same value appears in both positions. Since we employ no causal masking, both positions attend to each other, and by symmetry, both output positions yield identical values.

Consider a single-head, scalar-valued attention layer augmented with positional encodings. Let (p_1, p_2) be distinct positional encodings, producing layer inputs $(z_1, z_2) = (x + p_1, x + p_2)$. The attention computation proceeds with scalar parameters w_q, w_k, w_v, w_o (projection weights) and b_q, b_k, b_v, b_o (projection biases).

For the second output position:

$$q_2 = (x + p_2)w_q + b_q \quad (23)$$

$$k_j = (x + p_j)w_k + b_k \quad \text{for } j \in \{1, 2\} \quad (24)$$

$$v_j = (x + p_j)w_v + b_v \quad \text{for } j \in \{1, 2\} \quad (25)$$

The attention weights via softmax become:

$$\alpha_{21} = \frac{\exp(q_2 k_1)}{\exp(q_2 k_1) + \exp(q_2 k_2)} = \text{sigmoid}(q_2(k_1 - k_2)) \quad (26)$$

where $\text{sigmoid}(z) = 1/(1 + e^{-z})$ is the sigmoid activation.

The output is then:

$$g(x) = w_o \cdot (\alpha_{21} v_1 + (1 - \alpha_{21}) v_2) + b_o \quad (27)$$

With the specific parameter values:

$$w_q = -5.0, \quad w_k = 5.0, \quad w_v = 1.0, \quad w_o = 1.0 \quad (28)$$

$$p_1 = 0.0, \quad p_2 = 1.0, \quad b_q = 4.3, \quad b_k = b_v = b_o = 0 \quad (29)$$

the resulting function $g(x)$ exhibits non-monotonic behavior with a local minimum near $x = 0$. Specifically, $g(x)$ decreases for $x < 0$ and increases for $x > 0$ in a neighborhood of the origin, creating the characteristic V-shape of the absolute value function.

This demonstrates that pure attention mechanisms can express non-monotonic functions through the interaction of multiple tokens. The key insight is that attention weights, computed via softmax over multiple positions, introduce the essential non-linearity that escapes the constraints of coordinate-wise monotonic functions.

Why coordinate-wise non-monotonicity suffices. The unlinking and unknotting constructions in Appendix I apply absolute value *coordinate-wise*—each coordinate is folded independently via $x_i \mapsto |x_i|$, without requiring interactions between coordinates. This coordinate-wise structure is essential: it means that any mechanism capable of expressing a non-monotonic function in each coordinate independently can perform the same topological transformations.

Since the attention construction above processes each coordinate independently (the same two-token mechanism applies separately to each input dimension), and the resulting function $g(x)$ topologically resembles $|x|$ (monotonically decreasing then increasing with a minimum at the origin), the pure Transformer inherits the topological transformation capabilities of absolute value activations. Specifically, by applying this attention construction coordinate-wise, a width- n Transformer can express the coordinate-wise absolute value map $(x_1, \dots, x_n) \mapsto (|x_1|, \dots, |x_n|)$, enabling the unlinking and unknotting operations that overcome the topological obstructions identified in our impossibility theorems. $\square$

Remark I.3 (Geometric interpretation of attention non-monotonicity). The attention mechanism can be geometrically interpreted as computing a convex combination of values v_1 and v_2 at the two positions:

$$o_2 = \alpha_{21} v_1 + \alpha_{22} v_2 \quad (30)$$

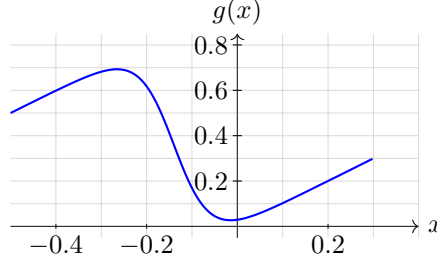


Figure 15. The non-monotonic function $g(x)$ expressed by a pure attention layer. The function exhibits a V-shaped profile near the origin, locally approximating $|x|$.

where $\alpha_{21}, \alpha_{22} \geq 0$ and $\alpha_{21} + \alpha_{22} = 1$ due to the softmax normalization. This convex combination property enables the attention mechanism to break topological constraints that restrict purely feedforward transformations. The crucial distinction is that the combination coefficients α_{ij} depend on the input coordinate x , creating an adaptive, input-dependent mixing that varies across the domain. This coordinate-dependent weighting allows the attention mechanism to implement different local behaviors—linear growth in some regions, rapid transitions in others—thereby achieving the non-monotonic profile necessary for topological transformations.

J. Experimental Details

J.1. Hopf Link Parametrization and Thickening

Hopf link parametrization.

$$\begin{aligned} X(t) &= (\cos t, \sin t, 0) \\ Y(s) &= (1 + \cos s, 0, \sin s) \end{aligned}$$

Thickening procedure. Points sampled as $\gamma(t) + \epsilon \cdot \mathbf{n}(t)$ where $\mathbf{n}(t)$ is a unit vector normal to the curve, $\epsilon \sim \mathcal{U}(0, r)$ with $r = 0.15$, and high-frequency oscillations $0.3 \sin(100t)$ added to preserve topology.

J.2. Network Architectures

- **FFN:** Width-3, depths 3–20, fully connected layers
- **ResNet:** Width-3, residual blocks $x \mapsto x + g(x)$ where g is a 2-layer subnetwork

J.3. Training Protocol

Adam optimizer, learning rate 10^{-3} , batch size 128, up to 800 epochs, early stopping with patience 100–200, binary cross-entropy loss. Dataset: 6000 points (3000 per class), 80/20 train/validation split.

J.4. ReLU vs GELU: Detailed Observations

Table 2 reveals three key patterns:

1. **ReLU plateau:** Accuracy remains 77–88% regardless of depth, consistent with Theorem 3.7. The network cannot achieve 100% due to topological constraints.
2. **GELU breaks barrier:** Accuracy increases with depth toward $\sim 99\%$, confirming that non-monotonic activations escape the constraint (Section 6.1).
3. **Depth cannot compensate:** Unlike universal approximation where depth substitutes for width, depth provides no escape from topological barriers.

J.5. Higher-Dimensional Linked Spheres Construction

For the higher-dimensional experiments (Section 7.5), we construct two n -spheres linked in $\mathbb{R}^{2n+1}$ with linking number ± 1 . The construction uses explicit parametrizations $\tilde{A}, \tilde{B} : S^n \rightarrow \mathbb{R}^{2n+1}$ derived from stereographic projection.

Parametrization. Let $u = (u_0, u_1, \dots, u_n) \in S^n \subset \mathbb{R}^{n+1}$ with $\|u\| = 1$. Write $u = (u_0, u')$ where $u' = (u_1, \dots, u_n) \in \mathbb{R}^n$. The embedding $\tilde{A} : S^n \rightarrow \mathbb{R}^{2n+1}$ is defined as:

$$a = 1 - \frac{u_0}{\sqrt{2}} \quad (31)$$

$$\tilde{A}(u) = \left(\frac{u'}{a}, -\frac{u_0}{\sqrt{2} \cdot a}, \mathbf{0}_n \right) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n = \mathbb{R}^{2n+1} \quad (32)$$

where $\mathbf{0}_n$ denotes the zero vector in $\mathbb{R}^n$. Similarly, for $v = (v_0, v') \in S^n$:

$$b = 1 - \frac{v_0}{\sqrt{2}} \quad (33)$$

$$\tilde{B}(v) = \left(\mathbf{0}_n, \frac{v_0}{\sqrt{2} \cdot b}, \frac{v'}{b} \right) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n = \mathbb{R}^{2n+1} \quad (34)$$

The images $\tilde{A}(S^n)$ and $\tilde{B}(S^n)$ are two disjoint n -spheres in $\mathbb{R}^{2n+1}$ with linking number $\text{lk}(\tilde{A}(S^n), \tilde{B}(S^n)) = 1$, verified numerically via the higher-dimensional Gauss linking integral.

Geometric structure. The embedding places $\tilde{A}(S^n)$ in the X - Z subspace (first n coordinates plus middle coordinate, with $Y = \mathbf{0}$) and $\tilde{B}(S^n)$ in the Z - Y subspace (middle coordinate plus last n coordinates, with $X = \mathbf{0}$). The spheres intersect only at the shared middle coordinate axis, and the linking arises from their interlocked configuration around this axis.

Minimum separation. The minimum distance between points on the two spheres is $d_{\min} = 2(\sqrt{2} - 1) \approx 0.828$, independent of n . This ensures the spheres remain well-separated.

Targeted thickening. To create training data with non-trivial volume, we use *targeted thickening*: each sphere is thickened only within its complementary (normal) subspace to preserve the linking structure. Recall the coordinate structure is (X, Z, Y) where $X, Y \in \mathbb{R}^n$ and $Z \in \mathbb{R}$. Since $\tilde{A}$ lies in the X - Z subspace (with $Y = 0$), we thicken it in the Y -direction. Conversely, $\tilde{B}$ lies in the Z - Y subspace (with $X = 0$), so we thicken it in the X -direction:

$$\tilde{A}_\rho(u, \eta) = \tilde{A}(u) + (0, 0, \eta), \quad \eta \sim \text{Uniform}(B_n(\rho)) \quad (35)$$

$$\tilde{B}_\rho(v, \zeta) = \tilde{B}(v) + (\zeta, 0, 0), \quad \zeta \sim \text{Uniform}(B_n(\rho)) \quad (36)$$

where $B_n(\rho)$ is the n -dimensional ball of radius ρ . This targeted approach thickens each sphere within its normal bundle while maintaining the essential linking geometry. Our experiments use $\rho = 0.5$.

Multi-copy placement. To amplify the topological barrier (Section 7.5), we place k disjoint copies of the linked pair using *L^1 -ordered grid placement*: copy centers are integer lattice points in $\mathbb{Z}^{2n+1}$ enumerated in nondecreasing L^1 norm order, scaled by spacing $s = 10$. Specifically, we enumerate shells $\{v \in \mathbb{Z}^{2n+1} : \|v\|_1 = m\}$ for $m = 0, 1, 2, \dots$ and take the first k vectors. This spreads copies isotropically rather than along a single axis, preventing networks from exploiting directional biases. Each copy contributes an independent entanglement region, so networks must overcome k local obstructions simultaneously.

J.6. Linking Scaling Experiments: Detailed Variance Analysis

Figure 16 provides detailed variance analysis for the higher-dimensional linking experiments (Table 4). The plot shows accuracy versus total linking number k (i.e., k disjoint copies of a unit link) on a log-log scale, with error bars indicating the interquartile range (IQR) and whiskers showing the full min-max range across seeds. Lines connect the best (maximum) accuracy at each k , highlighting the expressivity ceiling for each architecture.

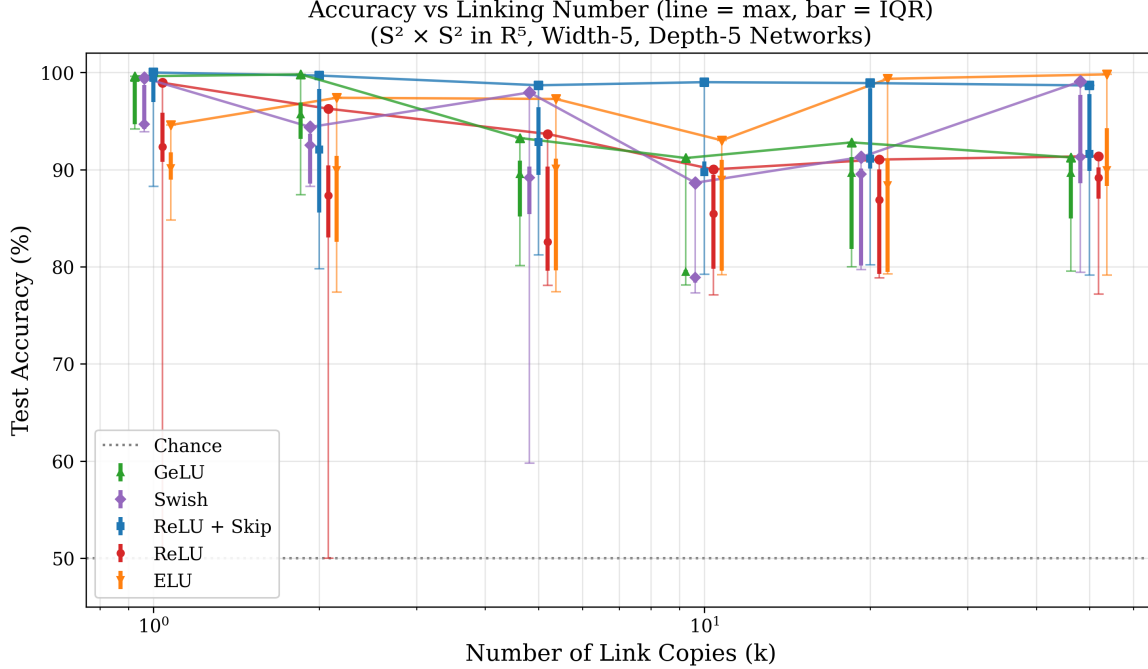


Figure 16. Accuracy vs linking number with detailed variance. Markers show median; thick bars show IQR (25th–75th percentile); thin whiskers show min/max across seeds. Lines connect the maximum accuracy at each k , showing the best-case expressivity trend. ReLU+Skip (blue) maintains consistently high accuracy with low variance, while plain ReLU (red) shows both lower ceiling and higher variance at large k .

Key observations from variance analysis.

- **ReLU+Skip has tight variance:** IQR spans only 2–3% across seeds, indicating robust performance regardless of initialization.
- **Plain ReLU has high variance:** Some seeds achieve $>95\%$ while others collapse to $\sim 50\%$ (chance level), suggesting optimization sensitivity when approaching topological barriers.
- **Non-monotonic activations are inconsistent:** GELU and Swish occasionally match ReLU+Skip but show large seed-to-seed variance, making them less reliable for topologically constrained tasks.
- **Gap increases with k :** The best-case gap between ReLU+Skip and plain ReLU grows from $\sim 1\%$ at $k = 1$ to $\sim 8\%$ at $k \geq 5$, confirming that multiple copies amplify the topological barrier.

Why higher dimensions require larger k to reveal topological barriers. An important phenomenon is that for the classical Hopf link ($n = 1$, $S^1 \times S^1$ in $\mathbb{R}^3$), the topological advantage of non-monotonic activations and skip connections is already prominent at $k = 1$ (Table 2), whereas for $n = 2$ ($S^2 \times S^2$ in $\mathbb{R}^5$), this requires $k > 5$ copies. This dimensional scaling has a natural geometric explanation.

Networks with monotonic activations can achieve relatively high overall accuracy by *sacrificing accuracy locally*—i.e., allowing decision regions to overlap in a small patch near the linking region. The characteristic size ε of this overlap region is determined by several factors: architecture capacity, learning rate’s ability to discover subtle solutions, numerical precision, thickening diameter ρ , and stochastic factors from initialization and noise realization.

Crucially, the *volume fraction* of this sacrificed region scales as $\Theta(\varepsilon^{2n+1})$ since the ambient space is $\mathbb{R}^{2n+1}$. For $n = 1$ (ambient $\mathbb{R}^3$), a single link’s error region might constitute $\sim 10\%$ of the data volume. For $n = 2$ (ambient $\mathbb{R}^5$), the same ε yields only $\sim \varepsilon^5 / \varepsilon^3 \approx \varepsilon^2 \approx 1\%$ of the volume. Indeed, the accuracy drop from $n = 1$ to $n = 2$ at $k = 1$ (from $\sim 90\%$ to $\sim 99\%$) suggests $\varepsilon \approx 0.3$, since $0.3^2 \approx 0.1$ matches the observed ratio.

However, in natural high-dimensional datasets, we expect *more linking opportunities*—more locations where topological entanglement can occur—with volume growing as $\Theta(\text{poly}(n))$. This growth compensates the shrinking per-link error fraction, potentially making the net topological effect stable across dimensions. Our multi-copy experimental design (k disjoint copies) synthetically replicates this phenomenon: by placing k independent links, we restore the cumulative topological obstruction to a level comparable to lower dimensions, allowing controlled study of how architectures handle topological barriers.

J.7. Full 100-Seed Results: Higher-Dimensional Linking

We ran comprehensive experiments with 100 random seeds across 17 model configurations and 13 values of k (total linking number), yielding 22,100 experiments. The large number of repetitions is necessary because our synthetic dataset requires narrow networks (width 5) to test the theoretical predictions—wide networks could trivially separate any finite dataset regardless of topology. However, optimizing narrow networks is notoriously difficult: the loss landscape is highly non-convex and results are stochastic, depending strongly on weight initialization and optimization trajectory. Running 100 seeds serves two purposes: (1) the *maximum* accuracy across seeds allows models to approach their theoretical expressivity upper bound, revealing what architectures *can* achieve; (2) the *mean* accuracy provides a reliable estimate of typical performance with sufficient sample size to reduce variance.

Tables 6, 7, and 8 report the maximum, mean, and median test accuracy, respectively. Bold entries indicate the best performance in each column.

Table 6. Maximum test accuracy (%) across 100 seeds for each model and linking number k . Width-5, depth-5 networks on $S^2 \sqcup S^2 \subset \mathbb{R}^5$.

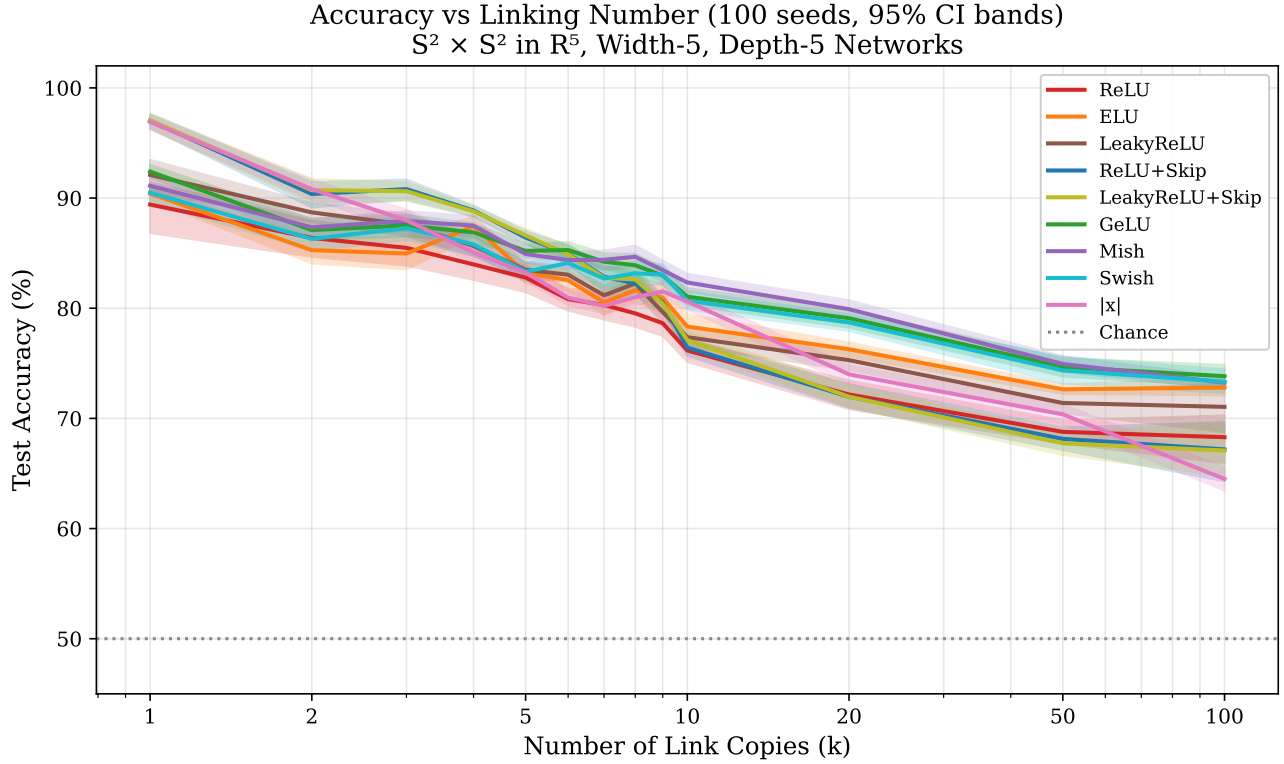
Model	1	2	3	4	5	6	7	8	9	10	20	50	100
<i>Monotonic, no skip</i>													
relu	98.6	98.0	98.3	95.4	93.5	91.5	93.5	93.2	89.6	88.0	84.5	80.2	80.6
elu	96.6	98.4	96.6	92.6	93.0	93.0	92.8	94.3	91.0	91.2	86.0	83.2	78.1
leaky_relu	98.8	98.5	99.0	92.3	91.3	91.1	93.5	93.6	90.7	87.5	87.9	85.3	80.8
<i>Monotonic, skip</i>													
relu+skip	100	98.7	98.1	95.1	94.1	92.2	93.2	92.5	87.6	85.4	83.6	80.9	77.7
elu+skip	100	98.4	98.7	94.6	94.2	93.2	92.9	93.0	89.8	86.8	87.5	81.6	77.7
leaky+skip	100	98.8	98.2	93.6	95.3	91.9	92.8	90.4	87.4	86.7	89.8	78.5	80.7
<i>Non-monotonic, no skip</i>													
gelu	98.8	98.4	96.4	93.8	92.5	91.9	93.2	94.6	91.9	91.1	88.6	84.8	82.3
swish	97.0	98.3	98.5	93.2	92.0	93.7	93.6	94.5	93.3	91.6	88.9	84.3	81.1
mish	97.4	97.7	98.4	95.0	93.8	92.7	93.3	95.2	91.8	90.2	90.2	86.0	80.9
<i>Non-monotonic, skip</i>													
gelu+skip	100	99.2	97.3	95.2	93.4	94.7	92.8	92.9	88.9	85.8	85.4	80.5	78.2
swish+skip	100	99.0	96.4	94.4	92.4	93.0	91.0	91.2	90.5	85.3	86.5	79.1	77.7
mish+skip	100	98.3	95.8	96.5	96.0	93.6	92.0	92.1	90.8	91.9	86.3	80.0	78.4
<i>Special activations (no skip)</i>													
$ x $ (abs)	100	98.7	97.4	93.2	90.8	88.4	90.2	89.0	91.5	90.1	83.4	82.7	83.5
negslope _{-0.01}	98.9	98.6	97.7	92.4	92.0	91.9	90.8	92.9	89.4	87.8	86.4	84.6	79.4
negslope _{-0.1}	99.3	98.0	98.4	94.6	91.0	91.0	92.1	93.7	90.0	88.8	87.8	85.1	81.7
negslope _{-0.5}	100	99.0	97.2	96.1	90.6	88.4	86.7	90.6	87.9	88.5	87.6	82.5	78.6
negslope _{-1.0}	100	98.8	97.4	93.2	90.8	88.4	89.7	89.0	91.5	90.1	83.5	86.8	83.4

Table 7. Mean test accuracy (%) across 100 seeds for each model and linking number k . Width-5, depth-5 networks on $S^2 \sqcup S^2 \subset \mathbb{R}^5$.

Model	1	2	3	4	5	6	7	8	9	10	20	50	100
<i>Monotonic, no skip</i>													
relu	89.4	86.4	85.5	84.0	82.8	80.8	80.3	79.5	78.6	76.2	72.0	68.8	67.9
elu	90.4	85.3	85.0	87.4	83.2	82.6	80.5	81.6	80.9	78.3	76.4	72.7	73.1
leaky_relu	92.1	88.7	87.6	85.7	83.5	83.0	81.2	82.3	79.7	77.4	75.1	71.5	69.5
<i>Monotonic, skip</i>													
relu+skip	97.0	90.4	90.8	88.9	86.4	84.9	82.9	82.3	80.5	76.4	71.9	68.2	67.1
elu+skip	94.4	89.0	88.3	88.9	84.9	83.1	81.1	80.2	79.2	76.9	72.9	69.1	68.2
leaky+skip	97.1	90.7	90.6	88.8	86.6	84.9	82.8	82.6	80.4	77.1	71.8	67.7	67.0
<i>Non-monotonic, no skip</i>													
gelu	92.4	87.1	87.5	86.9	85.2	85.3	84.2	83.9	83.0	81.0	79.0	74.7	73.9
swish	90.5	86.3	87.3	85.8	83.3	84.1	82.7	83.2	83.1	80.7	78.7	74.2	73.6
mish	91.1	87.4	87.9	87.5	84.9	84.4	84.4	84.7	83.5	82.3	79.9	74.9	73.2
<i>Non-monotonic, skip</i>													
gelu+skip	97.1	91.4	91.3	89.5	86.9	86.1	84.5	84.1	81.3	78.1	73.4	68.1	67.3
swish+skip	93.7	90.0	90.2	89.4	86.6	85.6	83.5	83.1	81.3	77.5	75.3	69.2	68.2
mish+skip	95.7	91.1	90.8	89.8	86.8	85.8	84.1	83.8	80.9	77.8	75.2	68.6	68.0
<i>Special activations (no skip)</i>													
$ x $ (abs)	96.9	90.8	88.0	85.1	83.2	81.0	80.2	81.0	81.5	80.6	74.0	70.4	64.5
negslope _{-0.01}	90.2	88.1	87.2	86.4	83.4	82.7	81.7	81.7	79.0	77.3	73.5	70.6	69.2
negslope _{-0.1}	92.1	88.2	88.2	86.0	83.9	82.4	81.0	80.7	79.7	78.1	72.9	69.1	68.6
negslope _{-0.5}	95.6	90.0	87.0	85.0	82.9	81.1	80.1	79.3	78.7	77.9	72.1	66.8	63.7
negslope _{-1.0}	96.9	90.8	88.0	85.2	83.1	81.0	80.2	81.0	81.7	80.7	74.1	70.5	65.4

Table 8. Median test accuracy (%) across 100 seeds for each model and linking number k . Width-5, depth-5 networks on $S^2 \sqcup S^2 \subset \mathbb{R}^5$.

Model	1	2	3	4	5	6	7	8	9	10	20	50	100
<i>Monotonic, no skip</i>													
relu	91.8	88.2	86.5	84.2	83.6	81.2	80.2	79.6	78.1	76.5	72.7	69.4	70.1
elu	90.3	86.0	88.9	90.0	84.6	83.1	81.1	81.1	82.0	77.8	75.0	72.3	72.7
leaky_relu	92.8	89.4	88.7	87.1	83.2	82.9	81.0	81.6	78.7	77.3	75.0	72.2	71.3
<i>Monotonic, skip</i>													
relu+skip	98.3	91.4	90.9	89.4	86.6	85.2	83.1	82.9	81.2	77.1	72.2	68.8	68.0
elu+skip	94.4	90.2	90.4	89.9	85.2	84.4	83.1	83.8	79.7	77.5	73.5	70.1	69.2
leaky+skip	98.5	91.3	90.7	89.4	86.7	85.1	83.0	83.6	80.5	77.2	72.0	68.1	67.8
<i>Non-monotonic, no skip</i>													
gelu	92.9	88.1	88.9	87.3	85.5	85.6	84.3	84.2	83.2	80.7	79.3	74.5	73.5
swish	90.7	88.3	89.4	86.7	84.5	84.9	83.5	83.6	82.7	80.7	78.3	72.6	73.1
mish	91.2	89.0	89.9	89.8	85.4	84.9	84.6	85.2	83.9	82.7	79.7	73.6	72.3
<i>Non-monotonic, skip</i>													
gelu+skip	98.3	91.3	91.0	90.2	86.8	86.1	84.1	85.0	82.0	78.4	73.4	67.9	68.2
swish+skip	93.3	90.1	90.4	89.8	86.9	85.6	84.2	84.4	81.6	78.3	75.0	69.1	68.4
mish+skip	95.8	90.8	90.2	86.7	85.6	84.3	84.8	81.2	78.2	75.0	67.9	68.9	
<i>Special activations (no skip)</i>													
$ x $ (abs)	98.1	91.3	88.5	85.9	83.6	80.9	79.6	80.9	82.3	81.0	74.3	70.4	63.8
negslope _{-0.01}	91.7	88.8	88.2	87.3	83.5	83.2	81.1	81.9	78.5	76.7	73.7	71.1	71.2
negslope _{-0.1}	91.6	88.7	88.7	87.4	84.5	82.9	81.4	80.6	79.6	77.8	73.2	69.3	71.0
negslope _{-0.5}	97.2	90.2	88.9	85.9	83.3	81.5	80.1	79.7	78.5	78.0	72.1	65.9	62.0
negslope _{-1.0}	98.1	91.1	88.6	85.9	83.5	81.0	79.7	81.0	82.5	81.2	74.2	70.6	65.0


 Figure 17. Accuracy vs linking number k with 95% confidence interval bands (100 seeds). Models grouped by type: monotonic no-skip (ReLU, ELU, LeakyReLU), monotonic with skip (ReLU+Skip, LeakyReLU+Skip), non-monotonic (GeLU, Mish, Swish), and direct $|x|$ activation. Non-monotonic activations maintain higher accuracy at large k .

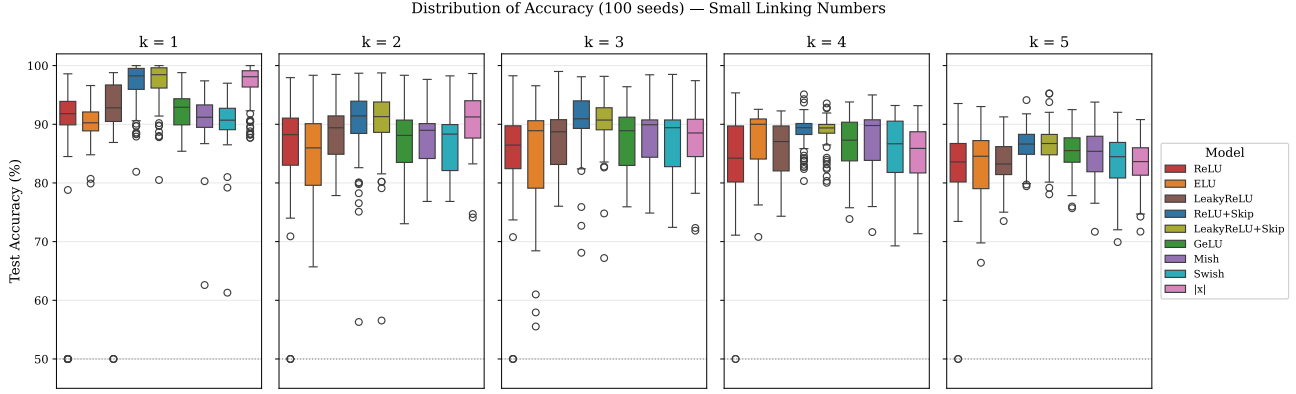


Figure 18. Distribution of test accuracy across 100 seeds for small linking numbers ($k = 1, 2, 3, 4, 5$). Box plots show median, IQR, and outliers. At $k = 1$, skip connections and non-monotonic activations achieve near-perfect accuracy, while monotonic no-skip models show a ceiling below 100%.

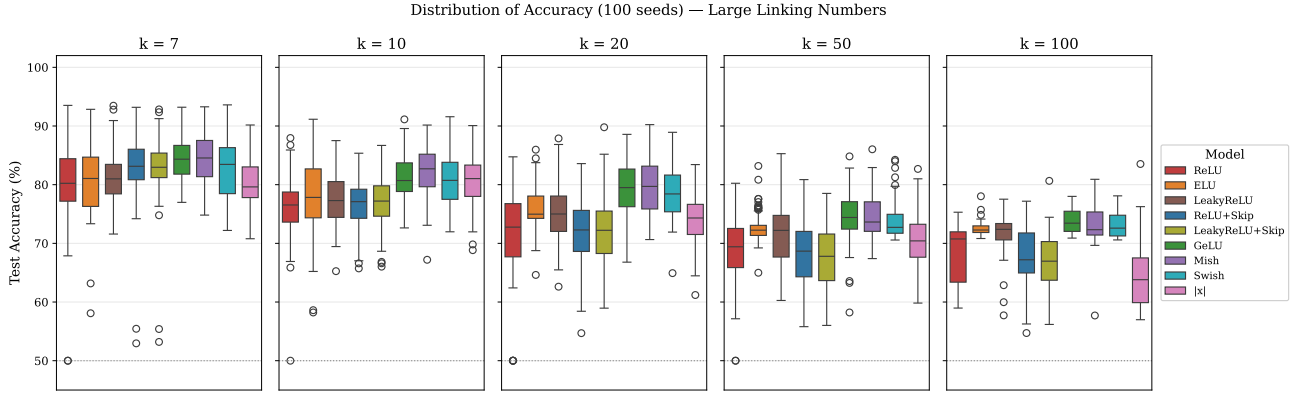


Figure 19. Distribution of test accuracy across 100 seeds for large linking numbers ($k = 7, 10, 20, 50, 100$). Non-monotonic activations (GeLU, Mish, Swish) consistently outperform monotonic activations at high k . The $|x|$ activation shows high variance despite theoretical optimality.

Key findings from 100-seed experiments.

- Skip connections enable 100% at $k = 1$:** All 6 skip models achieve 100% maximum accuracy at $k = 1$, confirming that skip connections synthesize $|x|$ via $|x| = x + 2 \cdot \text{ReLU}(-x)$.
- Monotonic no-skip models hit ceiling:** `relu` (98.6%), `elu` (96.6%), `leaky_relu` (98.8%) never reach 100% at $k = 1$, confirming the topological obstruction for coordinate-wise monotonic activations.
- Non-monotonic no-skip also reach 100%:** `abs`, `negslope-0.5`, `negslope-1.0` achieve 100% at $k = 1$, confirming that non-monotonic activations overcome the barrier without skip connections.
- `negslope-1.0` $\approx |x|$:** The activation $f(x) = x$ for $x \geq 0$ and $f(x) = -x$ for $x < 0$ is exactly $|x|$; results track `abs` closely.
- Non-monotonic no-skip dominates at high k :** For $k \geq 8$, `mish` (no-skip) achieves the best mean accuracy, followed by `gelu` and `swish`. At $k = 100$, `gelu` (73.9%) beats all skip models.
- Skip hurts at high k :** For non-monotonic activations, skip variants have *lower* mean accuracy than no-skip at $k \geq 50$ (e.g., `gelu+skip` 67.3% vs. `gelu` 73.9% at $k = 100$).
- `abs` paradox:** Achieves 100% max at $k = 1$ and best max at $k = 100$ (83.5%), but mean drops sharply (64.5%)—indicating optimization difficulty despite theoretical optimality.

J.8. Sample Efficiency: Training Dynamics Across Epochs

Beyond final accuracy, we analyze how validation accuracy evolves during training. Figure 20 shows mean validation accuracy at epochs 50, 100, and 200 for selected k values, revealing striking training dynamics.

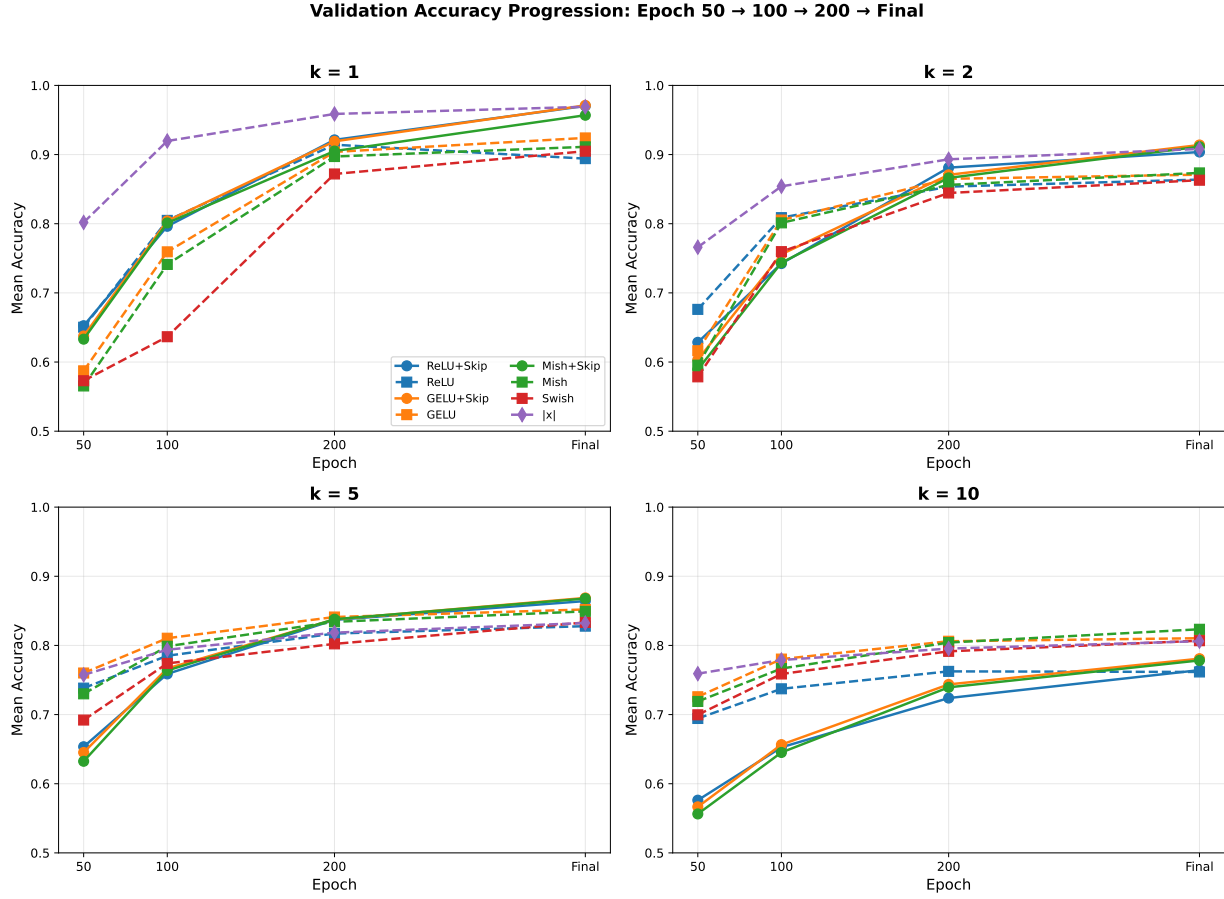


Figure 20. Validation accuracy progression across training epochs for $k \in \{1, 2, 5, 10\}$. Skip-connection models (dashed) start slower but overtake no-skip variants at low k , while non-monotonic activations show consistent advantages at high k . The crossover patterns at $k = 1$ suggest skip connections require longer to synthesize $|x|$.

Key observations. (1) **Skip models start slow, finish strong at low k :** At $k = 1$, skip models achieve only 65% at epoch 50 vs 80% for abs, but reach 97% final accuracy vs 89% for plain ReLU. (2) **Ranking reversals diminish with k :** At high k , model rankings stabilize by epoch 100; at low k , rankings remain fluid until late training. (3) **Non-monotonic activations converge faster:** GELU, Swish, and Mish achieve high accuracy earlier than skip-based ReLU at $k \geq 5$. (4) **abs leads early, struggles late at high k :** abs achieves highest epoch-50 accuracy but shows sharpest degradation at $k = 100$ (80% $\rightarrow$ 64.5%).

Tables 9–11 provide complete validation accuracy data at epochs 50, 100, and 200 for all 17 models and 13 k values.

J.9. Pure Transformer Results

We also evaluated pure transformers (width-5, depth-5, 2 tokens, learnable positional encoding, *no* LayerNorm, *no* residual connections) on the same higher-dimensional linking task. Table 12 reports the results across 100 seeds.

Table 12. Pure transformer test accuracy (%) across 100 seeds. Width-5, depth-5, no LayerNorm, no residuals.

Metric	1	2	3	4	5	6	7	8	9	10	20	50	100
Max	97.4	95.6	98.6	91.7	89.5	84.1	86.3	88.4	83.9	81.8	81.1	80.2	76.2
Mean	71.1	71.0	72.6	71.8	68.5	68.1	67.4	67.9	67.0	67.5	67.3	66.2	67.3
Median	77.0	77.4	78.7	76.7	73.0	73.9	73.4	74.0	72.6	73.7	73.4	71.0	71.4
Std	16.8	15.5	15.5	15.3	13.1	11.8	12.2	12.1	11.0	10.9	10.6	9.3	9.4
Fail%	35	29	28	27	29	27	28	26	25	25	24	22	21

Transformer vs FFN comparison. Table 13 compares the pure transformer against feedforward networks.

Table 13. Maximum test accuracy comparison: Transformer vs FFN (100 seeds each).

k	Transformer	ReLU	GELU	TFM-ReLU	TFM-GELU
1	97.4	98.6	98.8	-1.2	-1.4
2	95.6	98.0	98.4	-2.4	-2.8
3	98.6	98.3	96.4	+0.3	+2.2
5	89.5	93.5	92.5	-4.0	-3.0
10	81.8	88.0	91.1	-6.2	-9.4
50	80.2	80.2	84.8	0.0	-4.7
100	76.2	80.6	82.3	-4.3	-6.0

Key transformer findings.

1. **Transformer underperforms FFN:** Both ReLU and GELU FFNs consistently outperform the pure transformer at most k values.
2. **High failure rate:** 21–35% of runs get stuck at random chance ($\sim 50\%$), indicating severe optimization instability without LayerNorm/residuals.

Conclusion: Pure transformers without LayerNorm and residual connections are not a practical solution for overcoming topological constraints—the optimization landscape is too difficult.

J.10. ResNet Skip Connection Visualization

Figure 21 shows the full resolution visualization of the ResNet skip connection mechanism on the disk-annulus (S^0 - S^1) separation task. This experiment demonstrates that a depth-3 width-2 ResNet with ReLU activations learns to implement the folding operation $|x| = x + 2 \text{ReLU}(-x)$ predicted by Theorem 6.2.

Training details. The ResNet architecture uses 3 residual blocks with width-2 hidden layers. Training uses Adam optimizer with learning rate 0.001 for 5000 epochs on 50,000 points per class. The visualization shows the block 1 transformation (input $\rightarrow$ first residual block output). Multiple random seeds (including seed 103 shown) consistently learn similar folding transformations, demonstrating that the theoretical mechanism is reliably discovered by gradient descent.

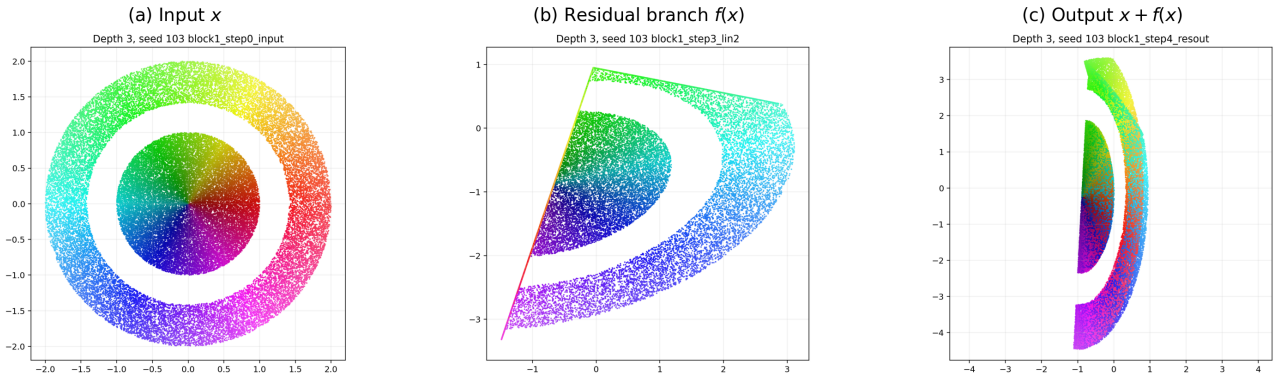


Figure 21. Full resolution ResNet skip connection visualization. (a) **Input x :** The nested disk (inner) and annulus (outer) form an S^0 - S^1 link with $\text{lk} = \pm 1$. Each point is assigned a color based on its angular position in the input space; this color is preserved across all three panels, allowing one to track where each input point is mapped by the network. Since colors vary continuously with position, a point’s trajectory can be identified by locating the dot (or small patch of similar color) in each panel. (b) **Residual branch output $f(x)$:** The learned transformation approximates $2 \text{ReLU}(-x)$, “folding” negative coordinate values toward zero. Note the characteristic triangular shape created by the ReLU. (c) **Skip connection output $x + f(x)$:** Adding the residual to the input implements $|x|$, folding the configuration and separating the two components into vertically disjoint regions. The disk and annulus are now linearly separable. This experiment validates that ResNet can overcome topological barriers through the skip connection mechanism.

K. Linking Detection Algorithm for Point Cloud Data

This section presents a complete algorithm for detecting topological linking between point cloud datasets. Given two finite point sets $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^d$ representing different classes, the algorithm determines whether their geometric configurations exhibit non-trivial linking by constructing spatial graphs, extracting cycle bases, and computing linking numbers.

K.1. Algorithmic Overview

The detection pipeline consists of four stages:

1. **Dimensionality Reduction:** Project high-dimensional data to $\mathbb{R}^3$ via PCA or optimized affine subspace search
2. **Spatial Graph Construction:** Build k -nearest neighbor graphs with distance thresholding
3. **Cycle Basis Extraction:** Compute fundamental cycle bases using spanning forests
4. **Linking Number Computation:** Evaluate Gauss linking integral between cycle pairs

The key theoretical insight enabling efficient computation is that the fundamental cycle basis generates the first homology group $H_1(G; \mathbb{Z})$, and linking number extends bilinearly to homology classes. This reduces an infinite search over all cycles to a finite computation over basis cycles.

K.2. Graph Construction with Distance Thresholding

Given a point cloud $\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathbb{R}^3$, we construct an undirected graph $G = (V, E)$ with vertex set $V = \{1, \dots, n\}$ and edge set determined by proximity.

Definition K.1 (ϵ -filtered k -NN graph). For parameters $k \in \mathbb{N}$ (number of neighbors) and $\epsilon > 0$ (maximum edge length), the ϵ -filtered k -NN graph has edge set $E = \{(i, j) : j \in \text{kNN}(i) \text{ and } \|x_i - x_j\| \leq \epsilon\}$, where $\text{kNN}(i)$ denotes the k nearest neighbors of point x_i .

The distance threshold ϵ prevents spurious long edges that could create topologically meaningless cycles. In practice, ϵ is set based on the empirical edge length distribution—typically at a percentile (e.g., median or 70th percentile) of nearest-neighbor distances.

Definition K.2 (Mutual k -NN graph). For more robust graph construction, the *mutual* k -NN graph includes edge (i, j) only if both $j \in \text{kNN}(i)$ and $i \in \text{kNN}(j)$: $E_{\text{mutual}} = \{(i, j) : j \in \text{kNN}(i), i \in \text{kNN}(j), \|x_i - x_j\| \leq \epsilon\}$.

Mutual k -NN graphs are sparser but more symmetric, reducing sensitivity to outliers and density variations.

K.3. Fundamental Cycle Basis via Spanning Forest

To enumerate cycles in the spatial graph, we compute the *fundamental cycle basis*—a minimal generating set for the cycle space.

Definition K.3 (Fundamental cycle basis). Let $G = (V, E)$ be a connected graph with $|V| = n$ vertices and $|E| = m$ edges. A spanning tree $T \subset E$ contains $n - 1$ edges. Each non-tree edge $e \in E \setminus T$ creates exactly one cycle when added to T . The collection of these $m - (n - 1) = m - n + 1$ cycles forms the **fundamental cycle basis**.

For disconnected graphs, we compute a spanning *forest* (one spanning tree per connected component) and the analysis proceeds component-wise.

Algorithm 1 Fundamental Cycle Basis Extraction

Require: Graph $G = (V, E)$
Ensure: List of cycles $\mathcal{C}$

- 1: Compute spanning forest T via BFS, recording parent pointers and depths
 - 2: $\mathcal{C} \leftarrow \emptyset$
 - 3: **for** each non-tree edge $(u, v) \in E \setminus T$ **do**
 - 4: Find path $P_{u \rightarrow v}$ in T using parent pointers
 - 5: $C \leftarrow P_{u \rightarrow v} \cup \{(v, u)\}$ {Close the cycle}
 - 6: $\mathcal{C} \leftarrow \mathcal{C} \cup \{C\}$
 - 7: **end for**
 - 8: **return** $\mathcal{C}$
-

Proposition K.4 (Homological completeness). *The fundamental cycle basis $\mathcal{C} = \{C_1, \dots, C_\beta\}$ generates the first homology group: $H_1(G; \mathbb{Z}) = \bigoplus_{i=1}^{\beta} \mathbb{Z} \cdot [C_i]$, where $\beta = m - n + c$ is the first Betti number and c is the number of connected components. Every cycle in G can be written as an integer linear combination of basis cycles.*

K.4. Gauss Linking Number Computation

For two disjoint closed curves $\gamma_1, \gamma_2 : S^1 \rightarrow \mathbb{R}^3$, the linking number is $\text{lk}(\gamma_1, \gamma_2) = \frac{1}{4\pi} \oint_{\gamma_1} \oint_{\gamma_2} \frac{(\gamma_1 - \gamma_2) \cdot (d\gamma_1 \times d\gamma_2)}{|\gamma_1 - \gamma_2|^3}$. For piecewise-linear curves represented as vertex sequences, we discretize using midpoint quadrature:

Algorithm 2 Gauss Linking Number (Numerical)

Require: Cycles $C_1 = (p_0, p_1, \dots, p_{L_1} = p_0)$, $C_2 = (q_0, q_1, \dots, q_{L_2} = q_0)$ as 3D point sequences

Require: Subdivision parameter N_{sub}
Ensure: Linking number $\text{lk} \in \mathbb{Z}$

- 1: $S \leftarrow 0$
 - 2: **for** each edge (p_i, p_{i+1}) in C_1 **do**
 - 3: $\Delta p \leftarrow (p_{i+1} - p_i) / N_{\text{sub}}$
 - 4: **for** $k = 0$ to $N_{\text{sub}} - 1$ **do**
 - 5: $x \leftarrow p_i + (k + 0.5) \cdot \Delta p$ {Midpoint of sub-segment}
 - 6: **for** each edge (q_j, q_{j+1}) in C_2 **do**
 - 7: $\Delta q \leftarrow (q_{j+1} - q_j) / N_{\text{sub}}$
 - 8: **for** $\ell = 0$ to $N_{\text{sub}} - 1$ **do**
 - 9: $y \leftarrow q_j + (\ell + 0.5) \cdot \Delta q$
 - 10: $r \leftarrow x - y$
 - 11: **if** $\|r\| > \epsilon_{\text{tol}}$ **then**
 - 12: $S \leftarrow S + \frac{r \cdot (\Delta p \times \Delta q)}{\|r\|^3}$
 - 13: **end if**
 - 14: **end for**
 - 15: **end for**
 - 16: **end for**
 - 17: **end for**
 - 18: **return** $\text{round}(S/4\pi)$
-

K.5. Homological Definition of Linking Number

We now present the rigorous homological foundation for linking numbers, which reveals their bilinearity as an immediate structural consequence. This perspective, based on Alexander duality, provides the theoretical justification for our detection algorithm.

K.5.1. ALEXANDER DUALITY

Alexander duality is a fundamental theorem relating the homology of a subspace to the cohomology of its complement. We state the version relevant to our setting.

Theorem K.5 (Alexander Duality). *Let $K \subset S^n$ be a compact, locally contractible subspace of the n -sphere. Then $\tilde{H}_k(K; \mathbb{Z}) \cong \tilde{H}^{n-k-1}(S^n \setminus K; \mathbb{Z})$, where $\tilde{H}_k$ denotes reduced homology and $\tilde{H}^{n-k-1}$ denotes reduced cohomology.*

For our purposes, we work in $\mathbb{R}^3 \cong S^3 \setminus \{\infty\}$, and the relevant case is when $K = \gamma$ is a simple closed curve (homeomorphic to S^1).

Corollary K.6. *Let $\gamma \subset \mathbb{R}^3$ be a simple closed curve. Then $H_1(\mathbb{R}^3 \setminus \gamma; \mathbb{Z}) \cong \mathbb{Z}$, with generator represented by any small meridian circle linking γ once.*

Proof. We compactify $\mathbb{R}^3$ to S^3 by adding a point at infinity. Since γ is compact, $S^3 \setminus \gamma = (\mathbb{R}^3 \setminus \gamma) \cup \{\infty\}$, and this inclusion induces an isomorphism on H_1 (the point at infinity does not affect first homology).

Applying Alexander duality with $n = 3$ and $k = 0$: $\tilde{H}_0(\gamma; \mathbb{Z}) \cong \tilde{H}^2(S^3 \setminus \gamma; \mathbb{Z})$. Since $\gamma \cong S^1$ is connected, $\tilde{H}_0(\gamma) = 0$, so $\tilde{H}^2(S^3 \setminus \gamma) = 0$.

For $k = 1$: $\tilde{H}_1(\gamma; \mathbb{Z}) \cong \tilde{H}^1(S^3 \setminus \gamma; \mathbb{Z})$. Since $H_1(S^1; \mathbb{Z}) \cong \mathbb{Z}$, we have $H^1(S^3 \setminus \gamma; \mathbb{Z}) \cong \mathbb{Z}$. By the Universal Coefficient Theorem (Hatcher, 2002, Theorem 3.2), since $H_0(S^3 \setminus \gamma) \cong \mathbb{Z}$ is free, we have $H^1 \cong \text{Hom}(H_1, \mathbb{Z})$. Thus H_1 has free rank 1, and since knot complements have torsion-free first homology, $H_1(S^3 \setminus \gamma; \mathbb{Z}) \cong \mathbb{Z}$. $\square$

K.5.2. HOMOLOGICAL DEFINITION OF LINKING NUMBER

The Alexander duality isomorphism provides a canonical way to define linking numbers purely in terms of homology.

Definition K.7 (Homological linking number). Let $\gamma_1, \gamma_2 \subset \mathbb{R}^3$ be disjoint simple closed curves. The **linking number** $\text{lk}(\gamma_1, \gamma_2)$ is defined as follows:

Since $\gamma_1 \subset \mathbb{R}^3 \setminus \gamma_2$, the curve γ_1 represents a homology class $[\gamma_1] \in H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z})$. By Corollary K.6, $H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z}) \cong \mathbb{Z}$, with generator $[\mu]$ represented by a meridian of γ_2 . Define $\text{lk}(\gamma_1, \gamma_2) := n$ where $[\gamma_1] = n \cdot [\mu] \in H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z})$.

Proposition K.8 (Equivalence of definitions). *The homological linking number (Definition K.7) coincides with the Gauss integral formula:*

$$\text{lk}(\gamma_1, \gamma_2) = \frac{1}{4\pi} \oint_{\gamma_1} \oint_{\gamma_2} \frac{(\gamma_1 - \gamma_2) \cdot (d\gamma_1 \times d\gamma_2)}{|\gamma_1 - \gamma_2|^3}$$

Proof sketch. The Gauss integral computes the degree of the Gauss map $G : \gamma_1 \times \gamma_2 \rightarrow S^2$ defined by $G(x, y) = (x - y)/|x - y|$ (Hatcher, 2002, Chapter 2). This degree equals the algebraic intersection number of γ_1 with any Seifert surface Σ bounded by γ_2 . By Alexander duality (Hatcher, 2002, Theorem 3.44), this intersection number measures how many times $[\gamma_1]$ wraps around the generator of $H_1(\mathbb{R}^3 \setminus \gamma_2)$. $\square$

K.5.3. BILINEARITY FROM THE HOMOLOGICAL PERSPECTIVE

The homological definition immediately yields bilinearity as a structural consequence.

Theorem K.9 (Bilinearity of linking number). *The linking number extends to a bilinear pairing on homology groups. Specifically, let $\gamma_2 \subset \mathbb{R}^3$ be a fixed simple closed curve. Then linking with γ_2 defines a homomorphism $\text{lk}(\cdot, \gamma_2) : H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z}) \rightarrow \mathbb{Z}$. Consequently, for any 1-cycles α, α' in $\mathbb{R}^3 \setminus \gamma_2$ and integers $a, a' \in \mathbb{Z}$: $\text{lk}(a[\alpha] + a'[\alpha'], \gamma_2) = a \cdot \text{lk}(\alpha, \gamma_2) + a' \cdot \text{lk}(\alpha', \gamma_2)$. By symmetry of the linking number, bilinearity holds in the second argument as well.*

Proof. By Definition K.7, the linking number $\text{lk}(\gamma_1, \gamma_2)$ is defined as the unique integer n such that $[\gamma_1] = n \cdot [\mu]$ in $H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z}) \cong \mathbb{Z}$. This is precisely the image of $[\gamma_1]$ under the isomorphism $H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z}) \xrightarrow{\sim} \mathbb{Z}$ that sends the meridian class $[\mu]$ to 1.

Since this isomorphism is a group homomorphism, we have:

$$\text{lk}(a[\alpha] + a'[\alpha'], \gamma_2) = \phi(a[\alpha] + a'[\alpha']) \quad (37)$$

$$= a \cdot \phi([\alpha]) + a' \cdot \phi([\alpha']) \quad (38)$$

$$= a \cdot \text{lk}(\alpha, \gamma_2) + a' \cdot \text{lk}(\alpha', \gamma_2) \quad (39)$$

where $\phi : H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z}) \rightarrow \mathbb{Z}$ is the isomorphism.

For bilinearity in the second argument, we use the symmetry $\text{lk}(\gamma_1, \gamma_2) = \text{lk}(\gamma_2, \gamma_1)$. This follows from the Gauss integral: under interchange $\gamma_1 \leftrightarrow \gamma_2$, we have $(\gamma_1 - \gamma_2) \mapsto -(\gamma_1 - \gamma_2)$ and $(d\gamma_1 \times d\gamma_2) \mapsto -(d\gamma_1 \times d\gamma_2)$, so the two sign changes cancel. $\square$

Corollary K.10 (Homological invariance). *If γ_1 and γ'_1 are homologous in $\mathbb{R}^3 \setminus \gamma_2$ (i.e., $[\gamma_1] = [\gamma'_1]$ in $H_1(\mathbb{R}^3 \setminus \gamma_2; \mathbb{Z})$), then $\text{lk}(\gamma_1, \gamma_2) = \text{lk}(\gamma'_1, \gamma_2)$.*

Proof. Immediate from the fact that linking number is defined on homology classes. $\square$

K.5.4. THE LINKING PAIRING

The bilinearity can be understood more symmetrically through the *linking pairing*, which is the algebraic structure underlying all linking computations.

Definition K.11 (Linking pairing). For disjoint compact subsets $K_1, K_2 \subset \mathbb{R}^3$, the **linking pairing** is the bilinear map $\text{lk} : H_1(K_1; \mathbb{Z}) \times H_1(K_2; \mathbb{Z}) \rightarrow \mathbb{Z}$, defined by viewing cycles in K_1 as cycles in $\mathbb{R}^3 \setminus K_2$ and applying the homological linking number.

Theorem K.12 (Full bilinearity). *For 1-chains $\alpha = \sum_i a_i C_i$ and $\beta = \sum_j b_j D_j$ representing homology classes in disjoint compact sets K_1 and K_2 respectively: $\text{lk}(\sum_i a_i [C_i], \sum_j b_j [D_j]) = \sum_{i,j} a_i b_j \cdot \text{lk}(C_i, D_j)$.*

Proof. This follows from Theorem K.9 applied twice:

$$\text{lk} \left(\sum_i a_i [C_i], \sum_j b_j [D_j] \right) = \sum_i a_i \cdot \text{lk} \left([C_i], \sum_j b_j [D_j] \right) \quad (\text{linearity in first arg.}) \quad (40)$$

$$= \sum_i a_i \sum_j b_j \cdot \text{lk}(C_i, D_j) \quad (\text{linearity in second arg.}) \quad (41)$$

$$= \sum_{i,j} a_i b_j \cdot \text{lk}(C_i, D_j) \quad (42)$$

$\square$

K.6. Generalization to Higher Dimensions

The homological framework above, developed for curves in $\mathbb{R}^3$, generalizes naturally to higher-dimensional manifolds (Hatcher, 2002). This generalization underlies our higher-dimensional linking experiments (Section 7.5).

Definition K.13 (Higher-dimensional linking number). Let M^m and N^n be disjoint, closed, oriented manifolds embedded in $\mathbb{R}^d$ where $d = m + n + 1$. The **linking number** $\text{lk}(M, N) \in \mathbb{Z}$ is the degree of the Gauss map $G : M \times N \rightarrow S^{d-1}$ defined by $G(x, y) = (x - y)/|x - y|$. Since $\dim(M \times N) = m + n = d - 1 = \dim(S^{d-1})$, the degree is well-defined (Hatcher, 2002, Chapter 2).

Remark K.14 (Alexander duality in higher dimensions). The homological interpretation extends via Alexander duality (Hatcher, 2002, Theorem 3.44): for $K \subset S^d$ compact and locally contractible, $\tilde{H}_k(K; \mathbb{Z}) \cong \tilde{H}^{d-k-1}(S^d \setminus K; \mathbb{Z})$. For an n -sphere $S^n \subset S^{2n+1}$, this yields $H_n(S^{2n+1} \setminus S^n; \mathbb{Z}) \cong \mathbb{Z}$, and the linking number of two disjoint n -spheres in $\mathbb{R}^{2n+1}$ measures how one represents a generator of this homology group.

Proposition K.15 (Bilinearity in higher dimensions). *For disjoint compact subsets $K_1, K_2 \subset \mathbb{R}^d$ with $d = m + n + 1$, the linking pairing $\text{lk} : H_m(K_1; \mathbb{Z}) \times H_n(K_2; \mathbb{Z}) \rightarrow \mathbb{Z}$ is bilinear. This follows from the same homological argument as Theorem K.9: linking number is defined on homology classes via Alexander duality (Hatcher, 2002, Theorem 3.44), and the resulting isomorphism to $\mathbb{Z}$ is a group homomorphism.*

For our experiments with $S^n \times S^n$ in $\mathbb{R}^{2n+1}$ (taking $m = n$, so $d = 2n + 1$), the higher-dimensional Hopf construction yields linked spheres with $\text{lk}(S^n, S^n) = \pm 1$, providing canonical test cases for topological obstructions in arbitrary dimensions.

K.7. Application to Cycle Basis Detection

The homological framework directly justifies our algorithmic approach.

Corollary K.16 (Sufficiency of basis cycle detection). *Let $G_{\mathcal{X}}, G_{\mathcal{Y}}$ be spatial graphs constructed from point clouds $\mathcal{X}, \mathcal{Y}$. Let $\mathcal{C}_{\mathcal{X}} = \{C_1, \dots, C_p\}$ and $\mathcal{C}_{\mathcal{Y}} = \{D_1, \dots, D_q\}$ be their fundamental cycle bases. Then $\mathcal{X}$ and $\mathcal{Y}$ exhibit detectable linking if and only if $\exists i, j : \text{lk}(C_i, D_j) \neq 0$.*

Proof. By Proposition K.4, the fundamental cycle basis generates $H_1(G; \mathbb{Z})$: any cycle C in $G_{\mathcal{X}}$ satisfies $[C] = \sum_i a_i [C_i]$ for some integers a_i . Similarly for cycles in $G_{\mathcal{Y}}$.

By Theorem K.12: $\text{lk}(C, D) = \text{lk}(\sum_i a_i [C_i], \sum_j b_j [D_j]) = \sum_{i,j} a_i b_j \cdot \text{lk}(C_i, D_j)$.

If $\text{lk}(C_i, D_j) = 0$ for all basis pairs (i, j) , then $\text{lk}(C, D) = 0$ for all cycle pairs. Conversely, if some $\text{lk}(C_i, D_j) \neq 0$, then that basis pair witnesses non-trivial linking. $\square$

This reduces the detection problem from searching over infinitely many possible cycles to a finite computation over $O(p \cdot q)$ basis cycle pairs, where p and q are the first Betti numbers of the respective graphs.

K.8. Complete Detection Algorithm

Algorithm 3 Link Detection for Point Cloud Data

Require: Point clouds $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^d$ (two classes)
Require: Parameters: k (neighbors), ϵ (distance threshold), projection dimension $d' = 3$
Ensure: Decision: linked or not; witness cycles if linked

- 1: **Projection:** Apply PCA to $\mathcal{X} \cup \mathcal{Y}$, project to $\mathbb{R}^3$
- 2: **Graph construction:** Build ϵ -filtered k -NN graphs $G_{\mathcal{X}}, G_{\mathcal{Y}}$
- 3: **Cycle extraction:** Compute fundamental cycle bases $\mathcal{C}_{\mathcal{X}}, \mathcal{C}_{\mathcal{Y}}$
- 4: **if** $\mathcal{C}_{\mathcal{X}} = \emptyset$ or $\mathcal{C}_{\mathcal{Y}} = \emptyset$ **then**
- 5: **return** (Not linked, no cycles found)
- 6: **end if**
- 7: **for each** $C \in \mathcal{C}_{\mathcal{X}}$ **do**
- 8: **for each** $D \in \mathcal{C}_{\mathcal{Y}}$ **do**
- 9: Compute $\ell \leftarrow \text{lk}(C, D)$ via Algorithm 2
- 10: **if** $\ell \neq 0$ **then**
- 11: **return** (Linked, witness: (C, D, ℓ))
- 12: **end if**
- 13: **end for**
- 14: **end for**
- 15: **return** (Not linked, all basis pairs checked)

K.9. Parameter Selection Guidelines

- **Number of neighbors k :** Typically $k \in [4, 20]$. Larger k creates denser graphs with more cycles but increases computational cost. Start with $k = 6$ and increase if insufficient cycles are found.
- **Distance threshold ϵ :** Set based on the empirical edge length distribution. Compute the pairwise distances for k -NN edges and use a percentile (e.g., median or 70th percentile). Too small ϵ fragments the graph; too large ϵ creates spurious long-range connections.
- **Mutual vs. standard k -NN:** Mutual k -NN is more robust to density variations but produces sparser graphs. Use standard k -NN for uniform-density data; mutual k -NN for heterogeneous data.

- **Minimum cycle length:** Filter cycles with fewer than 4 vertices, which are typically artifacts of graph construction noise.

K.10. Computational Complexity

- **k -NN graph construction:** $O(n \log n)$ using k -d trees or $O(n^2)$ brute force
- **Spanning forest:** $O(n + m)$ via BFS where $m = O(kn)$ edges
- **Cycle basis extraction:** $O(\beta \cdot n)$ where $\beta = m - n + c$ is the first Betti number
- **Linking number (per pair):** $O(L_1 \cdot L_2 \cdot N_{\text{sub}}^2)$ where L_i are cycle lengths
- **Total:** $O(n \log n + \beta_1 \cdot \beta_2 \cdot L^2)$ where L is average cycle length

For typical datasets with $n \sim 10^3$ – 10^4 points, $\beta \sim 10$ – 100 cycles, and $L \sim 10$ – 50 vertices per cycle, the algorithm runs in seconds to minutes.

L. CIFAR-10 Linking Experiments: Full Details

This section provides complete experimental details for the CIFAR-10 linking analysis (Section 7.6).

L.1. Data Preparation and Link Detection

Data augmentation. To increase point cloud density for reliable cycle detection, we apply $20\times$ augmentation to CIFAR-10 training data:

- Random horizontal flip (probability 0.5)
- Random crop with 4-pixel padding
- Color jitter (brightness, contrast, saturation: ± 0.2)

This yields 1,050,000 samples (105,000 per class) from the original 50,000 training images.

Dimensionality reduction. Images are flattened ($32 \times 32 \times 3 = 3072\text{D}$), normalized to $[0, 1]$, standardized (zero mean, unit variance per feature), then projected to 3D via PCA on the combined augmented dataset.

Link detection parameters.

- $k = 15$ nearest neighbors
- Mutual k -NN (edge only if both endpoints are mutual neighbors)
- Minimum cycle length: 30 vertices
- ε threshold: varied via binary search to find minimum linking threshold

Detected link statistics. At $\varepsilon = 0.0338$:

- Linking number: $\text{lk} = -1$ (Gauss integral: -1.0036)
- Bird cycle: 58 points (cycle index 21)
- Deer cycle: 40 points (cycle index 48)
- 3D bounding box diagonal: 15.555
- ε as fraction of range: 0.22%

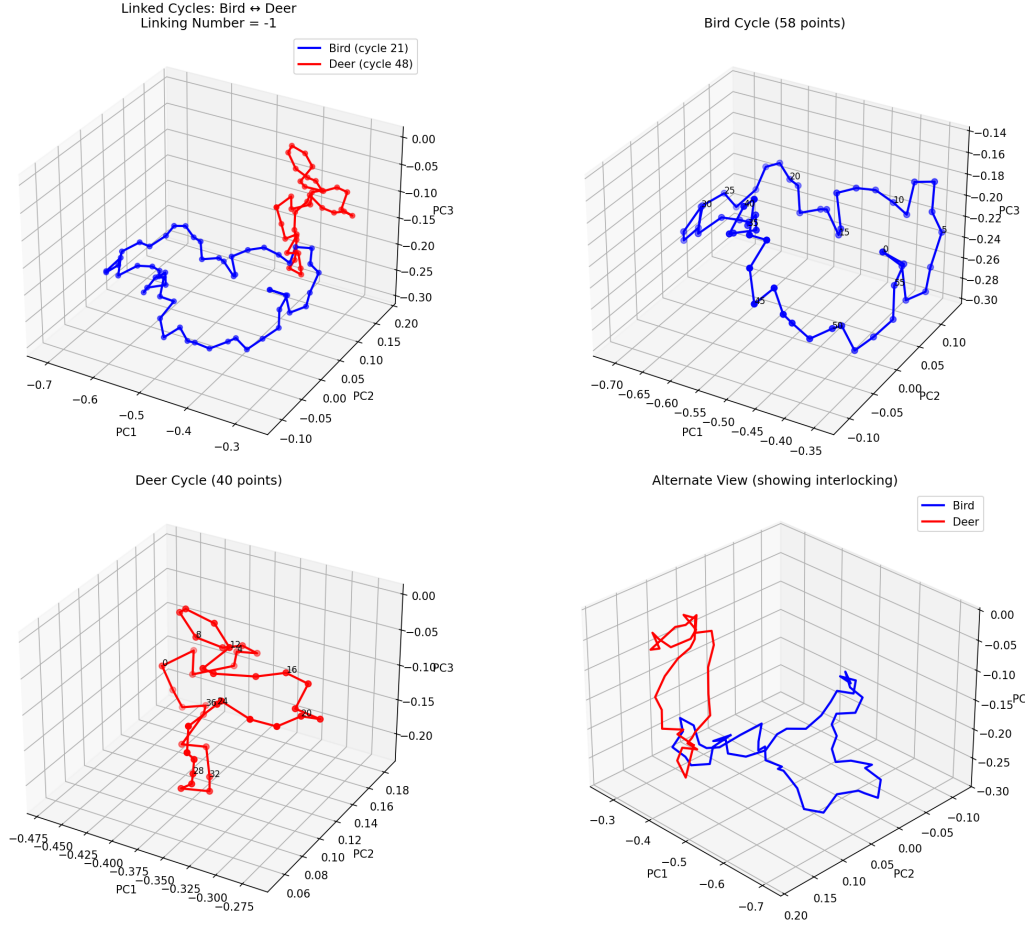


Figure 22. Linked cycles detected in CIFAR-10: bird (blue, 58 points) and deer (red, 40 points) classes form interlocked cycles with $lk = -1$ in 3D PCA space. The cycles were detected at $\varepsilon = 0.034$ (0.22% of the bounding box diagonal).

L.2. 10-Class Linking Consistency Matrix

Definition L.1 (Linking consistency). Given N independent runs of the linking detection algorithm on a class pair (X_i, X_j) , the *linking consistency* is the fraction of runs that detect non-trivial linking:

$$\text{consistency}(X_i, X_j) = \frac{1}{N} \sum_{r=1}^N \mathbf{1}[|lk_r(X_i, X_j)| \geq 1]$$

where lk_r denotes the linking number computed in run r . Consistency values near 1 indicate robust linking; values near 0 indicate the pair is consistently unlinked; intermediate values suggest sensitivity to algorithm parameters.

Sources of randomness. Although linking is a topological property of the underlying class manifolds, detecting it from finite point clouds involves stochastic steps that cause run-to-run variation:

1. **Data augmentation:** Each run regenerates the $20\times$ augmented dataset with a different random seed, applying random horizontal flips, crops, and color jitter. This produces a different point cloud realization of the same underlying image distribution.
2. **Graph structure:** Different point clouds yield different k -NN graphs, which in turn produce different fundamental cycle bases. A cycle that exists in one run’s graph may not exist in another’s.
3. **Cycle geometry:** Even when cycles exist, their precise vertex positions vary with the augmented samples. Two cycles

may thread through each other in one realization but not in another, depending on the specific 3D coordinates after PCA projection.

Linking consistency thus measures robustness to sampling variability: pairs with high consistency have manifold geometry that reliably produces interlocked cycles across different finite samples, while pairs with low consistency are either genuinely unlinked or have linking that is too fragile to detect reliably from finite data.

We ran the detection algorithm 11 times with different random seeds across all 45 class pairs. Table 14 shows the linking consistency for each pair.

Table 14. Linking consistency matrix for CIFAR-10 classes (fraction of 11 runs detecting $\text{lk} \neq 0$). Values ≥ 0.7 indicate strongly linked pairs; values ≤ 0.3 indicate unlinked pairs.

	air	auto	bird	cat	deer	dog	frog	horse	ship	truck
airplane	–	0.27	0.73	0.27	0.82	0.36	0.27	<u>0.18</u>	0.91	0.27
automobile		–	0.82	0.82	0.91	0.82	0.82	0.82	0.55	0.91
bird			–	0.91	0.82	0.91	0.91	0.82	0.45	0.73
cat				–	0.91	0.82	0.91	0.82	0.27	0.45
deer					–	0.91	0.82	0.91	0.45	0.27
dog						–	0.82	0.64	0.36	0.55
frog							–	0.82	<u>0.18</u>	0.55
horse								–	0.36	0.82
ship									–	0.82

Bold: strongly linked (≥ 0.7). Underlined: unlinked (≤ 0.3).

Semantic interpretation. Strongly linked pairs tend to share semantic features:

- **Animals:** deer-dog (91%), bird-cat (91%), cat-dog (82%), bird-deer (82%)
- **Vehicles:** automobile-truck (91%), ship-truck (82%)

Unlinked pairs are semantically dissimilar: frog-ship (18%), airplane-horse (18%).

L.3. CNN Architecture and Theoretical Connection

CNNs as special case of feedforward networks. A key insight enabling our theory to apply to CNNs is that any CNN layer with C channels and spatial resolution $H \times W$ can be viewed as a feedforward layer with width $C \cdot H \cdot W$. Specifically, if we flatten the feature maps at any layer, we obtain a vector in $\mathbb{R}^{C \cdot H \cdot W}$, and the convolution operation is simply a special case of a linear transformation (with weight sharing). Since our theoretical results (Theorems 3.7 and 6.2) apply to feedforward networks with bounded intermediate width, they also apply to CNNs where the *representation dimension* $C \cdot H \cdot W$ remains bounded.

For CIFAR-10 with input dimension $3 \times 32 \times 32 = 3072$, the topological obstruction applies to any CNN where all intermediate representations satisfy $C \cdot H \cdot W \leq 3072$.

Width-bounded design. To test whether narrow bottlenecks create topological barriers, we constrain all intermediate representations to ≤ 3072 D (matching CIFAR-10 input dimension):

Layer	Output Shape	Dimension
Input	$3 \times 32 \times 32$	3072
Conv block 1	$12 \times 16 \times 16$	3072
Conv block 2	$48 \times 8 \times 8$	3072
Conv block 3	$192 \times 4 \times 4$	3072
FC	3072	3072
Output	C (2 or 10)	–

Depth configurations.

- **L5:** 1 conv layer per resolution (5 total)
- **L8:** 2 conv layers per resolution (8 total)
- **L11:** 3 conv layers per resolution (11 total)

ConvBlock structure. Each `ConvBlock` consists of a 3×3 convolution followed by activation (no batch normalization). Each resolution stage contains 1–3 ConvBlocks depending on depth configuration:

- **L5:** 1 ConvBlock per stage with stride 2
- **L8:** 2 ConvBlocks per stage (first stride 1, second stride 2)
- **L11:** 3 ConvBlocks per stage (first two stride 1, last stride 2)

The channel count is chosen so that quadrupling channels while halving each spatial dimension preserves the representation dimension: $12 \times 16 \times 16 = 48 \times 8 \times 8 = 192 \times 4 \times 4 = 3072$.

Skip connections. When enabled, residual connections span each conv block: $x \mapsto x + \mathcal{G}(x)$ where $\mathcal{G}$ is the conv block transformation.

L.4. Training Protocol

- **Optimizer:** Adam with learning rate 10^{-3}
- **Epochs:** 100 maximum
- **Early stopping:** Patience 15 epochs on validation accuracy
- **Batch size:** 128
- **Loss:** Cross-entropy
- **Data split:** Standard CIFAR-10 train (50K) / test (10K)

L.5. Full Binary Classification Results

Table 15 shows all 42 baseline configurations for both binary tasks.

L.6. Effect of Skip Connections

With skip connections, the non-monotonic advantage disappears. This is consistent with theory: skip connections enable $|x| = x + 2 \operatorname{ReLU}(-x)$, allowing monotonic networks to synthesize non-monotonic behavior.

L.7. Interpretation: Why the Effect is Small

The 1–2% gap in CIFAR-10 (vs 8–15% in synthetic data) has several explanations:

1. Local vs global linking. In synthetic Hopf link data, 100% of points participate in the linked structure. In CIFAR-10, linking is *local*: the detected cycles involve ~ 100 points out of 105,000 per class (0.1%). The topological barrier affects only this small fraction.

2. Epsilon scale. The linking threshold $\varepsilon = 0.034$ is tiny relative to the data range (15.6 diagonal), indicating linking exists only at very fine scales. At coarser scales, the classes are separable by any activation.

Table 15. Full binary classification results: top 10 configurations per task at validation accuracy.

Linked pair: Deer vs Dog (91% linking consistency)					
Activation	Type	Depth	Skip	Val Acc	Test Acc
gelu	NONM	L8	no	90.80%	89.60%
mish	NONM	L8	no	90.80%	89.25%
mish	NONM	L11	no	90.70%	90.55%
swish	NONM	L8	no	90.10%	89.55%
leaky_relu	MONO	L11	no	89.60%	89.55%
leaky_relu	MONO	L8	no	89.60%	88.70%
relu	MONO	L8	no	89.60%	87.95%
elu	MONO	L8	no	89.10%	88.65%
relu	MONO	L11	no	89.10%	86.70%
gelu	NONM	L11	no	88.90%	87.85%

Unlinked pair: Frog vs Ship (18% linking consistency)					
Activation	Type	Depth	Skip	Val Acc	Test Acc
swish	NONM	L5	no	98.60%	97.20%
relu	MONO	L11	no	98.50%	97.75%
gelu	NONM	L5	no	98.40%	97.70%
relu	MONO	L5	no	98.40%	97.50%
gelu	NONM	L8	skip	98.30%	97.50%
leaky_relu	MONO	L11	no	98.30%	97.95%
leaky_relu	MONO	L5	no	98.20%	97.40%
mish	NONM	L11	no	98.10%	97.60%
mish	NONM	L5	no	98.10%	97.55%
mish	NONM	L11	skip	98.10%	97.55%

Table 16. Skip connection effect at L8 on deer-dog classification.

	No Skip	With Skip	Δ
Best NONM	90.80% (GELU)	88.10% (Swish)	−2.70%
Best MONO	89.60% (ReLU)	88.20% (ReLU)	−1.40%
NONM advantage	+1.20%	−0.10%	−

3. Capability vs learning. Non-monotonic activations *can* unlink in principle, but may not *learn* to do so due to optimization dynamics, insufficient depth, or competing objectives. The observed advantage is an empirical lower bound on the theoretical advantage.

4. 3D projection limitation. Linking is detected in 3D PCA space, not the original 3072D pixel space. The 3D linking is evidence of high-dimensional topological structure but not proof—some linking may be projection artifacts, and other high-dimensional linking may not be visible in 3D.

L.8. 10-Class Classification and Confusion-Linking Correlation

Beyond binary classification, we use the full 10-class CIFAR-10 task to demonstrate that **linking consistency reliably predicts between-class difficulty**. Dataset difficulty depends on many factors (image quality, resolution, class semantics, etc.), making cross-dataset comparisons challenging. However, *within* a single dataset like CIFAR-10, these confounding factors are controlled—all classes share the same image statistics, resolution, and collection methodology. This allows us to isolate the effect of topological structure: if two class pairs differ primarily in their linking consistency, any systematic difference in classification difficulty can be attributed to the geometric entanglement of their data manifolds.

Experimental setup. We trained 42 configurations ($7 \text{ activations} \times 3 \text{ depths} \times 2 \text{ skip settings}$) on 10-class CIFAR-10 using the same width-bounded CNN architecture. Each model produces a 10×10 confusion matrix, from which we extract

pairwise confusion rates for all 45 class pairs.

Correlation analysis. For each trained model, we compute the Spearman rank correlation between:

- **Linking consistency** (Table 14): fraction of runs detecting $lk \neq 0$ for each class pair
- **Symmetric confusion rate**: $\text{conf}(i, j) = P(\hat{y} = j | y = i) + P(\hat{y} = i | y = j)$

Table 17 shows the correlation results for all activations at depth L11 (no skip connections).

Table 17. Confusion-linking correlation for 10-class CIFAR-10 (L11, no skip). All correlations are statistically significant at $p < 0.01$.

Activation	Type	Test Acc	Spearman r	p -value	High-Link	Low-Link
ReLU	MONO	78.6%	0.436	0.0028	6.4%	2.1%
ELU	MONO	82.2%	0.486	0.0007	5.3%	1.9%
LeakyReLU	MONO	79.5%	0.478	0.0009	6.1%	2.1%
GELU	NON	80.3%	0.501	0.0005	6.0%	1.8%
Swish	NON	81.1%	0.493	0.0006	5.7%	1.9%
Mish	NON	80.6%	0.497	0.0005	5.9%	2.1%
MONO mean		–	0.467	–	6.0%	2.0%
NON mean		–	0.497	–	5.9%	1.9%

High-Link: mean confusion for pairs with linking consistency ≥ 0.7 (27 pairs). Low-Link: mean confusion for pairs with linking consistency ≤ 0.3 (8 pairs).

Key findings.

1. **Statistically significant correlation:** All models show Spearman $r \approx 0.44$ – 0.50 with $p < 0.003$. For $n = 45$ pairs, the critical value at $p < 0.01$ is $|r| > 0.38$; our observed correlations exceed this threshold.
2. **Medium-to-large effect size:** By Cohen’s conventions ($r = 0.1$: small, $r = 0.3$: medium, $r = 0.5$: large), our observed $r \approx 0.47$ represents a medium-to-large effect. The linking consistency matrix explains $\sim 22\%$ of variance in confusion rates ($r^2 \approx 0.22$).
3. **$3\times$ confusion ratio:** Class pairs with high linking consistency (≥ 0.7) have approximately $3\times$ higher confusion rates than pairs with low linking consistency (≤ 0.3): mean 6.0% vs 2.0%.
4. **Most confused pairs are strongly linked:** The five most confused class pairs under ReLU are:
 - cat–dog: 30.6% confusion (linking = 0.82)
 - bird–deer: 13.1% confusion (linking = 0.82)
 - bird–cat: 10.8% confusion (linking = 0.91)
 - cat–frog: 10.8% confusion (linking = 0.91)
 - automobile–truck: 10.4% confusion (linking = 0.91)

Consistency across depths. Table 18 shows that the correlation pattern is stable across network depths, consistent with the theoretical prediction that depth cannot overcome topological barriers.

Table 18. Confusion-linking correlation across depths (no skip). All $p < 0.006$.

Depth	MONO r	NON r	MONO ratio	NON ratio
L5	0.474	0.478	$2.43\times$	$2.81\times$
L8	0.444	0.493	$2.50\times$	$3.14\times$
L11	0.467	0.497	$2.89\times$	$3.08\times$

MONO: ReLU, ELU, LeakyReLU. NON: GELU, Swish, Mish. Ratio = (high-link confusion) / (low-link confusion).

Interpretation. The strong correlation between algorithmically-detected linking and classifier confusion provides evidence that **linking consistency is a reliable predictor of between-class difficulty**:

1. **Within-dataset comparison controls confounds:** By comparing class pairs within CIFAR-10, we hold constant factors like image resolution, collection methodology, and preprocessing. The remaining variation in difficulty can be attributed to intrinsic geometric structure.
2. **Linking captures geometric entanglement:** The linking detection algorithm identifies how “intertwined” class manifolds are in feature space—a complementary perspective to semantic similarity.
3. **Predictive across architectures:** The correlation is consistent across all activation functions and depths tested, suggesting linking consistency reflects a fundamental property of the data rather than an artifact of specific architectural choices.

Why non-monotonic activations do not help in 10-class classification. Unlike the binary classification setting where non-monotonic activations showed a modest advantage on linked pairs, the 10-class results show no differential benefit—in fact, non-monotonic activations exhibit slightly *higher* correlation with linking. Several factors explain this discrepancy:

1. **Complex loss trade-offs:** In 10-class classification, the network must balance confusion rates across all 45 class pairs simultaneously. Reducing confusion on one linked pair (e.g., cat–dog) may increase confusion on others due to shared decision boundaries. This multi-objective optimization dilutes any advantage from topological unlinking.
2. **Insufficient to overcome topology alone:** At the high dimensionality of CIFAR-10 (3072D input, width-3072 intermediate representations), topological obstructions are only one of many barriers to perfect classification. Generalization error, optimization landscape complexity, and limited model capacity all contribute to confusion, regardless of activation type.

Linking as a diagnostic tool for dataset difficulty. Despite the absence of a non-monotonic advantage, the linking consistency matrix remains a **statistically robust predictor** of classifier confusion ($r \approx 0.47$, $p < 0.003$). This predictive power is:

- **Consistent across activations:** All 6 activation functions show significant correlation.
- **Consistent across depths:** L5, L8, and L11 networks show similar correlation strength.
- **Practically meaningful:** Linked pairs have $\sim 3\times$ higher confusion than unlinked pairs.

This demonstrates that the linking detection algorithm captures genuine geometric structure in the data manifold that poses classification difficulty. Within a dataset, linking consistency can serve as a **diagnostic tool** for identifying hard-to-separate class pairs *before training*, potentially informing data collection priorities, class hierarchy design, or targeted augmentation strategies.

Comparison with inter-class distance metrics. A natural question is whether linking consistency simply measures inter-class distance. To address this, we compared linking consistency against two standard distance metrics computed in the original 3072D pixel space:

- **Average pairwise distance:** Mean Euclidean distance between all pairs of samples from two classes (subsampled to 500 samples per class for efficiency).
- **Minimum distance:** Minimum Euclidean distance between any two samples from different classes (proxy for margin).

Table 19 shows that linking consistency is a **stronger predictor** of confusion than either distance metric.

This comparison suggests that linking consistency captures more than just “how close” two classes are in feature space. The topological perspective—measuring how *intertwined* the class manifolds are rather than their average separation—provides additional signal about classification difficulty. Two classes may have similar average distances but very different linking structure depending on how their manifolds wrap around each other in the embedding space.

Table 19. Comparison of confusion predictors. Values are mean Spearman $|r|$ across all 42 trained models. Linking consistency outperforms both distance metrics.

Predictor	Mean $ r $	p -value	vs. Linking
Linking consistency	0.479	< 0.001	—
Average pairwise distance	0.412	< 0.001	−0.067
Minimum distance	0.356	< 0.001	−0.123

Limitations and future directions. The current linking detection algorithm is a **proof-of-concept** that projects high-dimensional data to 3D via PCA and detects binary linking (linked vs. unlinked). For practical applications, we envision several improvements:

1. **Quantitative linking numbers:** Compute actual linking numbers rather than binary detection, providing a continuous measure of entanglement strength.
2. **Multi-projection synthesis:** Use multiple random projections and translated affine subspaces to sample different “cross-sections” of the data manifold, then aggregate linking information across projections for a more robust approximation of higher-dimensional structure.
3. **Higher-dimensional linking:** Develop methods to compute linking invariants directly in high dimensions without projection, potentially using persistent homology or other topological data analysis techniques.

We expect these developments to strengthen the predictive power of topological metrics for dataset difficulty analysis.